

Order-Invariant Identification in SV-BVAR Models: Uncertainty, Financial Volatility, and GDP Tail Risk *

Manuel Gloria[†]

May 15, 2026

Abstract

Policy uncertainty and financial market stress jointly shape the distribution of future GDP growth, yet their interaction is poorly understood. Using a Bayesian VAR with stochastic volatility (SV-BVAR) estimated on UK data, this paper documents an asymmetric amplification mechanism: financial stress shocks are substantially more recessionary when policy uncertainty is high, whereas uncertainty shocks are not systematically more contractionary when financial stress is elevated. This asymmetry is clearly visible under order-invariant identification but largely muted under standard recursive Cholesky schemes. A 2025 counterfactual exercise confirms the policy relevance of this finding: persistent financial stress worsens GDP-at-Risk roughly two to three times more than the GDP median, indicating that tail risk can be materially understated when analysis focuses on central projections alone. Historical narrative checks and out-of-sample forecasting experiments provide additional support; forecasting performance remains broadly comparable across identification schemes.

JEL Classification: C11, C32, C53, E37, E44

Keywords: Bayesian VAR, stochastic volatility, policy uncertainty, financial conditions, GDP tail risk.

*The views expressed are those of the author and should not be taken to represent those of the Bank of England or its policy committees.

[†]Bank of England. Email: manuel.gloria@bankofengland.co.uk

1 Introduction

Policy uncertainty has become an increasingly important source of risk for financial stability. Episodes of heightened geopolitical tensions, trade fragmentation, fiscal stress and institutional uncertainty have contributed to a persistently elevated level of policy uncertainty, raising concerns about its potential impact on financial markets and the real economy.

Policy uncertainty is a financial stability concern for two reasons. First, when markets are less confident about the fiscal, monetary, or trade policy stance, macroeconomic outcomes become more dispersed. Higher uncertainty widens the distribution of expected GDP growth and increases downside risks, strengthening the case for resilience - for instance through more conservative stress-testing assumptions. Second, elevated policy uncertainty can itself act as a catalyst for financial market stress. Shifts in expectations about policy can trigger abrupt repricing episodes, testing structural vulnerabilities in market-based finance such as liquidity mismatches, leverage in non-bank financial intermediaries, and the procyclicality of margining practices. In this sense, policy uncertainty can both amplify macroeconomic tail risks and crystallise vulnerabilities already present in the financial system.

A key challenge is that policy uncertainty and financial market volatility do not always move together. While periods of elevated uncertainty are often associated with heightened market volatility, this relationship has been far from stable in recent years. Episodes of persistently high policy uncertainty have at times coincided with relatively subdued financial market volatility, suggesting that risks may build up beneath the surface. This divergence raises the concern that a sudden realignment of volatility with elevated uncertainty could lead to a sharp tightening of financial conditions, with potentially severe macroeconomic consequences. Studying policy uncertainty in isolation may therefore provide an incomplete picture of macro-financial risks; instead, it is essential to analyse uncertainty jointly with financial market volatility.

Because financial stability analysis is inherently concerned with adverse and extreme outcomes, an appropriate empirical framework must be able to capture nonlinearities and asymmetries in the transmission of shocks. In particular, the effects of uncertainty and financial stress are unlikely to be symmetric across the business cycle or across the distribution of future economic outcomes. Small deteriorations in financial conditions may have limited effects in normal times, while similar shocks can have disproportionately large effects when the economy is already fragile. This makes models that focus on average responses ill-suited for assessing financial stability risks.

For this reason, much of the financial stability literature has relied on quantile regressions (QR) to study macroeconomic tail risks, often referred to as Growth-at-Risk.

Quantile-based approaches have proven useful in documenting that downside risks to GDP growth are strongly related to financial conditions. However, recent work by [Carriero et al. \(2024\)](#) shows that Bayesian VARs with stochastic volatility (SV-BVARs) can provide an equally effective framework for analysing macroeconomic tail risks. By allowing volatility to evolve over time, stochastic-volatility BVARs generate asymmetric unconditional distributions and can capture downside risks in a unified and internally consistent way.

In addition to their strong performance in forecasting tail outcomes, SV-BVARs offer several practical advantages over QR approaches. First, unlike QR, which typically requires a second step to fit a full distribution after estimating individual quantiles, SV-BVARs directly model the entire predictive distribution of macroeconomic variables. Second, QR methods are prone to the issue of quantile crossing, where estimated quantiles may violate logical ordering, particularly in small samples or at extreme percentiles. Third, QR models must be specified separately for each variable and forecast horizon, whereas the VAR framework naturally accommodates multiple variables and dynamic interdependencies through its lag structure. These features make SV-BVARs a compelling alternative for assessing macro-financial tail risks in a coherent and flexible framework.

Despite these advantages, most SV-BVAR applications in macro-finance continue to rely on recursive (Cholesky-based) identification schemes to recover structural shocks. While widely used, this approach requires imposing an ordering of variables that can materially affect impulse responses and structural interpretation. This sensitivity is particularly problematic in financial stability applications, where models often include several highly correlated macro-financial variables and where the economic meaning of identified shocks is central. To address this issue, [Chan et al. \(2024\)](#) develop an algorithm for estimating order-invariant SV-BVARs, exploiting the heteroskedasticity introduced by stochastic volatility to identify structural shocks without recursive restrictions. Because order-invariant methods do not impose a triangular structure, the mapping between estimated shocks and economic categories is not automatic; I address this through a labelling strategy based on the Hungarian assignment algorithm (details in Section 3.5).

To the best of my knowledge, this is the first paper to analyse the joint role of policy uncertainty and financial market stress in shaping GDP tail risk using an order-invariant SV-BVAR. The central finding is that state dependence is asymmetric: financial stress shocks are substantially more recessionary when policy uncertainty is high, but uncertainty shocks are not systematically more contractionary in high-financial-stress regimes. This asymmetric amplification is clearly visible under order-invariant identification yet largely muted under recursive Cholesky schemes, and is robust to alterna-

tive orderings and to filtering out weakly identified posterior draws. Replacing policy uncertainty with geopolitical risk causes the amplification channel to disappear, as expected given the exogeneity of geopolitical events to domestic financial stress, confirming that the mechanism is specific to policy uncertainty. A 2025 counterfactual exercise reinforces the policy relevance of this finding: persistent financial stress worsens the 5th percentile of GDP roughly two to three times more than the median, confirming that tail risk can be materially understated when analysis focuses on central projections alone. Historical narrative checks across three benchmark UK episodes and out-of-sample forecasting experiments provide further supporting evidence; forecasting performance remains broadly comparable across identification schemes.

The rest of the paper is structured as follows: Section 2 reviews the relevant literature; Section 3 describes the model, data and identification; Section 4 presents the impulse response analysis; Section 5 reports the narrative check; Section 6 evaluates forecasting performance; Section 7 reports prior-sensitivity diagnostics; Section 8 presents a counterfactual policy application; and Section 9 concludes.

2 Literature Review

The paper relates to three strands of the macro-financial literature: (i) the literature on policy uncertainty and financial volatility, (ii) the Growth-at-Risk (GaR) literature on downside macroeconomic risks, and (iii) recent work on structural identification in VAR models, with particular emphasis on order-invariant identification schemes.

2.1 Policy Uncertainty, Financial Volatility, and Macroeconomic Fluctuations

A large empirical literature studies the macroeconomic effects of policy uncertainty and financial market volatility. While there is broad agreement that both uncertainty and financial shocks exert adverse effects on real activity, important questions remain regarding their relative importance, their interaction, the state-dependent and time-varying nature of their effects, and the most appropriate strategies for identifying and disentangling them.

[Caldara et al. \(2016\)](#) employs an SVAR identification strategy based on penalty functions and show that both financial and uncertainty shocks are an important source of macroeconomic disturbances and that the latter have especially severe effects when they coincide with tighter financial conditions. [Alessandri and Mumtaz \(2019\)](#) employ a regime switching VAR and find the real effects of uncertainty shocks to be six times

larger when a financial crisis is in place with respect to when financial markets function normally. [Angelini et al. \(2019\)](#) propose an identification approach via breaks in the unconditional volatility and similarly find that uncertainty's impact on real activity intensifies during periods of economic or financial turmoil. [Ludvigson et al. \(2021\)](#) introduce an identification strategy via "shock-based constraints" and show while higher macroeconomic uncertainty in recessions is often an endogenous response to output shocks, financial uncertainty is a likely source of output fluctuations. [Furlanetto et al. \(2019\)](#) disentangle financial shocks by origin and conclude that credit market shocks dominate uncertainty shocks, whose effects are modest and short-lived. [Pastor and Veronesi \(2017\)](#) provide evidence that the link between EPU and market volatility can weaken when political signals are imprecise, explaining episodes of high uncertainty but low volatility. [Choi and Yoon \(2022\)](#), using a century of US data, show that financial conditions - borrowing costs and credit availability - are crucial in propagating uncertainty shocks, though heightened uncertainty does not necessarily amplify financial shocks, suggesting asymmetric interactions. [Brianti \(2020\)](#) disentangle the two shocks based on the different responses of cash holdings and finds that uncertainty shocks explain about 20% of GDP forecast error variance, compared to 40% for financial shocks, highlighting their relative importance. [Ellington et al. \(2023\)](#), focusing on the UK, use a regime-switching model and document that uncertainty shocks have stronger spillover effects on financial markets when policy uncertainty is already high, and that they dampen GDP growth particularly under financial stress. Finally, [Pruser and Schlosser \(2017\)](#) underscore cross-country heterogeneity using a TVP-FAVAR model and distinguishing between a group of fragile countries and a group of stable countries, where the former are more strongly affected by EPU shocks.

The inflationary consequences of uncertainty shocks remain contested. [Alessandri and Mumtaz \(2019\)](#) find that uncertainty shocks are inflationary in normal times but deflationary during financial distress. [Meinen and Röhe \(2018\)](#), using SVAR models with sign restrictions, report ambiguous inflation responses in both the US and Euro area. [De Santis and Van Der Veken \(2021\)](#) use sign and narrative restrictions and show that financial shocks reduce both output and prices, whereas uncertainty shocks lower output but raise prices, as firms increase markups to self-insure against pricing risks. [Choi and Yoon \(2022\)](#) add further complexity by showing that incorporating historical data yields an inflationary effect of uncertainty shocks, suggesting that monetary policy may struggle to offset their impact.

2.2 Growth-at-Risk and Macroeconomic Tail Risk

While much of the uncertainty literature focuses on average macroeconomic responses, a parallel and rapidly growing literature studies macroeconomic downside risks through the lens of Growth-at-Risk. Following the seminal contribution of [Adrian et al. \(2019\)](#), which shows that lower quantiles of future GDP growth vary strongly with financial conditions while upper quantiles remain stable, GaR has become a central tool for analysing macro-financial vulnerabilities.

Subsequent work has expanded this framework in several directions. [Caldara et al. \(2021b\)](#) estimate a Markov-switching model in which the transition probabilities depend on real-time indicators capturing financial and macroeconomics conditions, and show that tail risk takes the form of transitions between regimes of high-growth-low-volatility and low-growth-high-volatility regimes. [Plagborg-Møller et al. \(2020\)](#) evaluate the in-sample and out-of-sample forecast performance of financial variables for GDP growth with both parametric and nonparametric models and find no predictors provide robust and precise advance warnings of tail risks.

While the GaR literature mostly focuses on the relationship between GDP and financial conditions, it touched on economic uncertainty as well. For example, [Jovanovic and Ma \(2022\)](#) estimate the distribution of future real growth of IP as a function of uncertainty measures and document that higher economic uncertainty is associated with a more dispersed and left-skewed future growth distribution. [Castellnuovo and Mori \(2022\)](#) demonstrate that uncertainty-based measures of skewness and tail risk depend sensitively on data frequency. Using quantile regressions and local projections, [Loria et al. \(2024\)](#) find that the left tail of GDP and industrial production reacts three to six times more strongly to financial and uncertainty shocks than the median. Similar conclusions are reached by [Hengge \(2019\)](#), who show that macroeconomic uncertainty predominantly affects the downside of the growth distribution, especially when combined with tight financial conditions.

While the GaR literature typically employ quantile regression methods, recent works show how tail risk can be captured in VAR framework as well. In fact, while the conditional distributions of VARs are typically symmetric, the unconditional distributions can be asymmetric if volatility is time-varying and therefore can be able to capture skewness and downside risk. [Carriero et al. \(2024\)](#) develop two versions of an SV-BVAR (one of which with a generalized factor structure) and show that both versions can perform as well as quantile regressions in capturing GDP tail risk. [Caldara et al. \(2021a\)](#) study the joint conditional distribution of GDP and corporate spreads in an SV-VAR where the means of endogenous variables influences their volatilities and vice versa and show that the interaction between mean and volatility is crucial to account

for variations in uncertainty and risk. [Karlsson et al. \(2023\)](#) extend the SV-BVAR to account for skewness in the stochastic volatilities and comparing the out of sample forecasting performances of different models find strong support with VAR models with heavy tails. [Kiss et al. \(2024\)](#) in a similar SV-BVAR with hyperbolic skew student t distribution for innovation show that shocks to EPU have a negative effect on inflation and industrial production. While these studies focus mainly on the US, the present papers applies a stochastic-volatility BVAR framework to the UK economy.

2.3 Structural Identification and Order-Invariant SV-BVARs

A third strand of the literature concerns the identification of structural shocks in SV-BVAR models. Beginning with the influential contributions of [Cogley and Sargent \(2005\)](#) and [Primiceri \(2005\)](#), recursive (Cholesky-based) identification schemes have been used extensively in empirical macroeconomic applications. However, such schemes rely on an assumed ordering of variables, which can materially affect estimated impulse responses and forecast performance. This sensitivity becomes more pronounced in models with stochastic volatility and in higher-dimensional systems.

Order-invariant identification approaches provide an alternative by avoiding arbitrary ordering assumptions. A prominent class of such methods exploits time variation in volatility to identify structural shocks through heteroskedasticity, following the seminal contribution of [Rigobon \(2003\)](#) and more recent work by [Bertsche and Braun \(2022\)](#) and [Lewis \(2021\)](#). In this framework, changes in the relative variances of structural shocks over time provide identifying information without imposing exclusion or sign restrictions.

Recent work extends these ideas to Bayesian VARs, with a focus on developing computationally efficient algorithms for estimating order-invariant SV-BVARs. This paper follows [Chan et al. \(2024\)](#), who propose a Gibbs sampling algorithm that draws the contemporaneous impact matrix sequentially while enforcing orthogonality constraints, thereby allowing for flexible identification without recursive restrictions. They show that their order-invariant approach leads to improved forecast performance.

[Davidson et al. \(2025\)](#) develop an algorithm for large, order-invariant stochastic volatility in mean VARs (SVMVARs) and show that adopting an order-invariant specification is critical to avoid overstating the impact of macroeconomic uncertainty. In contrast to their approach, this paper does not rely on a stochastic volatility in mean specification; instead, uncertainty enters the model as a separate observable variable, as described in the next section.

This project contributes to the literature by combining these three strands within a unified framework. To the best of my knowledge, the joint role of policy uncertainty

and financial market volatility in shaping macroeconomic tail risks has not yet been analysed using an order-invariant SV-BVAR framework. By doing so, the project aims to improve both the economic interpretation of identified shocks and the assessment of downside macroeconomic risks.

3 Model and Data

This section describes the empirical model, data and identification strategies used in the analysis. I estimate a Bayesian vector autoregression with stochastic volatility (SV-BVAR) and compare two alternative approaches for identifying structural shocks: a standard recursive (Cholesky) decomposition and an order-invariant identification scheme that follows [Chan et al. \(2024\)](#). Estimation is performed using a Gibbs sampler. Details of the algorithm and of the derivations for the conditional sampling steps under order-invariant identification are provided in Appendices [A1](#), [A2](#) and [A3](#).

The model includes the $n = 5$ variables described in Section [3.2](#) and $p = 12$ lags, corresponding to one year of monthly dynamics. Posterior inference is based on 20,000 Gibbs draws after a burn-in period of 1,000 iterations.

Let Y_t denote an $n \times 1$ vector of variables observed at time t . The reduced-form VAR of order p is

$$Y_t = c + A_1 Y_{t-1} + \cdots + A_p Y_{t-p} + u_t, \quad (1)$$

where c is a vector of intercepts, A_j are $n \times n$ coefficient matrices, and u_t denotes reduced-form residuals. Defining the regressor vector

$$X_t = (1, Y'_{t-1}, \dots, Y'_{t-p}), \quad (2)$$

and stacking coefficients into a matrix $A = (c, A_1, \dots, A_p)'$, the system can be written compactly as

$$Y_t = X_t A + u_t. \quad (3)$$

Structural shocks are recovered from reduced-form residuals through the contemporaneous matrix B_0 :

$$\varepsilon_t = B_0 u_t, \quad (4)$$

where structural shocks are normally distributed with zero mean and time-varying

covariance matrix Σ_t :

$$\varepsilon_t \sim N(0, \Sigma_t), \quad (5)$$

$$\Sigma_t = \text{diag}(e^{h_{1t}}, \dots, e^{h_{nt}}). \quad (6)$$

The vector of log-volatilities evolves as independent random walks:

$$h_{it} = h_{i,t-1} + \eta_{it}, \quad \eta_{it} \sim N(0, \sigma_{h,i}^2). \quad (7)$$

This stochastic volatility specification allows the variance of structural shocks to evolve over time, capturing the heteroskedastic dynamics that are commonly observed in macroeconomic and financial data. Periods of financial turmoil are typically characterised by large increases in volatility, while tranquil periods exhibit relatively stable fluctuations. In the context of this paper, modelling time-varying volatility is particularly important as our objective is to study the relationship between macro-financial variables and GDP tail risk. Standard VAR models with constant variance impose a fixed shock distribution and therefore cannot capture the clustering of extreme realizations that tends to occur during crises.

Allowing the volatility of structural shocks to vary over time enables the model to capture episodes in which macroeconomic disturbances become unusually large. When volatility rises, the predictive distribution of future GDP growth becomes wider and more asymmetric, increasing the probability of severe negative outcomes. As a result, stochastic volatility plays a key role in linking macro and financial shocks to the evolution of GDP tail risk.

3.1 Prior Distributions

Estimation is conducted in a Bayesian framework. The VAR coefficients are assigned Gaussian priors

$$\text{vec}(A) \sim N(\beta_0, V_\beta). \quad (8)$$

The prior mean is set to zero for all coefficients. The prior covariance matrix V_β is diagonal and introduces different degrees of tightness across parameters. In particular, intercepts are assigned relatively diffuse priors, while lag coefficients are more tightly centered around zero. This structure shrinks the dynamics of the system while allowing sufficient flexibility for the data to determine the posterior estimates. The

stochastic volatility innovations follow inverse-gamma priors:

$$\sigma_{h,i}^2 \sim IG(\nu_h, S_h). \quad (9)$$

These priors govern the persistence and variability of the volatility process. Priors for the contemporaneous impact matrix B_0 depend on the identification scheme and are discussed in sections 3.3 and 3.4.

3.2 Data

The model is estimated at a monthly frequency using data from January 1999 to December 2025. The five variables are: Economic Policy Uncertainty (EPU), Inflation (INF), real GDP growth (GDP), a monetary policy variable, and a financial stress index (FSI).

The EPU index is the UK version of the text-based measure developed in [Baker et al. \(2016\)](#). It quantifies policy-related uncertainty by tracking the frequency of newspaper articles that contain terms related to the economy, policy, and uncertainty. The methodology systematically scans a broad set of major newspapers, normalises counts over time, and aggregates them into a monthly index that captures shifts in political, fiscal, monetary and regulatory uncertainty. As such, it provides a timely proxy for changes in the policy environment that may influence macroeconomic expectations and financial conditions. The FSI is the Composite Index of Systemic Stress (CISS) developed in [Chavleishvili and Kremer \(2025\)](#), that aggregates stress across five segments - money, bond, financial equity, non-financial equity and forex - to provide a real-time measure of financial market stress.

The traditional monetary policy variable is replaced with the UK Goldman Sachs Financial Conditions Index (FCI), which captures a broader spectrum of financial variables beyond interest rates, including riskier asset prices. This substitution reflects growing evidence that monetary policy transmission occurs primarily through financial conditions rather than policy rates alone, and that the two do not necessarily follow the same path. The disconnect between interest rates and financial conditions was particularly evident in 2024, when historically high policy rates coincided with unexpectedly loose financial conditions driven by soaring equity markets and compressed corporate credit spreads.

[Caballero et al. \(2024\)](#), [Caballero and Simsek \(2025\)](#) further strengthen the case for FCI targeting as an independent objective of monetary policy. They argue that stabilising financial conditions reduces output fluctuations and that directly communicating the central bank's expected financial condition path eliminates misunderstandings

with markets, insulating financial conditions from noise. In contrast, communicating interest rates alone fails to achieve these benefits. Moreover, FCI gaps more accurately reflect the effective policy stance during periods of rapid monetary changes, making them a superior guide for policy calibration.

Beyond policy calibration, the FCI could also have a diagnostic role when assessed alongside CISS. For example, if financial market stress is elevated but financial conditions remain accommodative, this indicates that the system continues to support credit intermediation, liquidity provision, and payment services despite adverse conditions. In this way, the FCI offers insight into the resilience of financial intermediation under stress. Although some indicators between FCI and CISS may overlap, their low correlation (0.075) over the sample period confirms that they measure distinct phenomena.¹

3.3 Identification via Cholesky Decomposition

Identification under the Cholesky specification (CHOL) follows the triangular decomposition commonly used in stochastic-volatility VAR models (e.g. [Cogley and Sargent \(2005\)](#)). Let B_0 be equal to a lower-triangular matrix L with ones on the main diagonal, such as the structural shocks are mapped into the residuals via:²

$$\varepsilon_t = Lu_t \quad (10)$$

The covariance matrix of reduced-form innovations can therefore be written as

$$\text{Cov}(u_t) = L^{-1}D_tL^{-'} \quad (11)$$

where D_t is a diagonal matrix containing the time-varying variances of the structural shocks. This triangular structure implies a recursive identification in which variables ordered earlier in the system do not respond contemporaneously to those ordered later. While widely used in empirical macroeconomics, this approach introduces an ordering dependence that may affect the interpretation of impulse responses and structural shocks. In the empirical analysis, this specification therefore serves as a benchmark against which the order-invariant identification scheme is evaluated. In the empirical application the variables are ordered as follows:

¹INF and GDP are calculated as the annualized growth rate of the seasonally adjusted Consumer Price Index (CPI) and real GDP, respectively, using the transformation $x_t = (\log(x_t) - \log(x_{t-12})) * 100$. EPU is logged. $\log(\text{EPU})$, FCI, and FSI are included in the model in levels.

²Note that the structure of L implies that there are $m = n(n-1)/2$ free parameters to be estimated, corresponding to the number of off-diagonal elements in the lower triangular matrix.

$$Y_t = (EPU_t, INF_t, GDP_t, FCI_t, FSI_t)'. \quad (12)$$

This ordering implies that economic policy uncertainty is the most exogenous variable in the system, affecting all other variables contemporaneously but not responding to them within the same period. Inflation and output can respond immediately to uncertainty shocks but not to financial variables. Financial conditions and financial stress appear last in the ordering and therefore respond contemporaneously to all other shocks.

Although recursive identification is widely used due to its simplicity and computational convenience, the ordering above can be nonetheless problematic. For example, it can be argued that real variables do respond to financial variables in the same period. Or that inflation responds to GDP within the same period. This dependence, given that ordering is ultimately arbitrary, and different variable arrangements lead to different structural interpretations of shocks, is the main motivation for considering alternative identification approaches that do not rely on recursive structures.

To quantify how sensitive the Cholesky results are to the particular variable arrangement chosen, Section 4.1 reports results under two alternative orderings. The first places financial variables first:

$$Y_t^{\text{CHOL2}} = (FCI_t, FSI_t, INF_t, GDP_t, EPU_t)', \quad (13)$$

so that financial conditions and financial stress are now the most exogenous series, while uncertainty moves to the bottom of the recursive chain. The second ordering retains financial variables first but interleaves uncertainty before the macro block:

$$Y_t^{\text{CHOL3}} = (FCI_t, FSI_t, EPU_t, INF_t, GDP_t)'. \quad (14)$$

Both alternatives preserve the constraint that FCI precedes FSI, mirroring market practice in which broad financial conditions are viewed as driving narrower stress measures. Together with the baseline ordering in equation above, these three Cholesky specifications span a wide range of contemporaneous-impact assumptions and provide a transparent benchmark against which the order-invariant results can be evaluated.

3.4 Order-Invariant Identification

To avoid the dependence on variable ordering inherent in recursive identification, I also estimate the model using the order-invariant identification approach proposed

by [Chan et al. \(2024\)](#). In this framework, the contemporaneous impact matrix B_0 is treated as unrestricted, and the Gibbs sampling algorithm is constructed so that the posterior distribution does not depend on the ordering of variables in the VAR. Structural shocks are still defined as linear combinations of the reduced-form residuals, but B_0 is not constrained to be triangular.

The key philosophical advantage of this approach is that it leverages identification via heteroskedasticity: the presence of time-varying volatility introduces additional moment conditions that are not available in a homoskedastic VAR. As shown by [Rigobon \(2003\)](#), changes in volatility regimes provide independent variation that can be exploited to identify the contemporaneous relationships among variables. In a homoskedastic system, the contemporaneous impact matrix is only identified up to an orthogonal rotation, but heteroskedasticity provides extra information that allows for sharper identification.

However, it is important to note that identification via heteroskedasticity is only possible up to sign and permutation of the structural shocks. That is, while the method can recover the structural matrix up to a set of possible rotations, the exact ordering and sign of the shocks remain undetermined unless further restrictions are imposed. This indeterminacy reflects the fact that second-moment information alone cannot distinguish between certain equivalent representations of the system. In practice, researchers may impose additional sign or economic restrictions to achieve full identification (see Section 3.5), but the order-invariant approach itself does not rely on any particular variable ordering or recursive structure.

Conditional on B_0 , the VAR coefficients can be sampled equation-by-equation after transforming the system to eliminate cross-equation correlations and heteroskedasticity. This transformation premultiplies the system by B_0' and rescales the innovations by their time-varying standard deviations, so that the transformed system can be estimated using standard Bayesian regression techniques. The contemporaneous impact matrix is then updated conditional on the VAR coefficients and volatility states, with rows of B_0 drawn sequentially while enforcing orthogonality between rows ([Chan et al., 2024](#)). This ensures that structural shocks remain mutually uncorrelated while avoiding any recursive ordering restrictions.

The resulting Gibbs sampler produces draws from the joint posterior distribution of the VAR coefficients, the contemporaneous impact matrix, and the stochastic volatility states without imposing an arbitrary variable ordering. Technical details of the conditional sampling steps are provided in Appendices [A1](#), [A2](#) and [A3](#).

3.5 Post-Estimation Labelling of Structural Shocks

The labelling of structural shocks in order-invariant models is a nontrivial task, as the posterior distribution of B_0 is invariant to permutations and sign changes of the shocks. The strategy adopted here - assigning shocks to variables based on the largest contemporaneous impact using the Hungarian algorithm - is only one of many possible labelling approaches. The literature has not established a clear consensus on the best method; see, for example, [Lewis \(2021\)](#), who provides an overview of alternative strategies in his paper's Appendix.

Let $A_0^{(s)} = (B_0^{(s)})^{-1}$ denote the impact matrix obtained at posterior draw s , so that $u_t = A_0 \varepsilon_t$. In this representation, row i of A_0 gives the contemporaneous responses of variable i to all shocks, and column j gives the contemporaneous effects of shock j on all variables. The objective of the labelling procedure is to assign shocks to variables such that the magnitude of the diagonal entries of A_0 is maximized. The assignment problem can be written as a combinatorial optimization problem. Let $A = |A_0^{(s)}|$ denote the matrix of absolute contemporaneous impacts. The goal is to select one element in each row and each column so as to maximize the sum of the selected impacts. Formally, the optimal assignment solves

$$\max_{P \in \mathcal{P}} \sum_{i=1}^n A_{i,P(i)}, \quad (15)$$

where $\mathcal{P}$ denotes the set of all permutations of $\{1, \dots, n\}$. The optimal permutation is computed using the Hungarian algorithm, which solves the assignment problem efficiently in polynomial time. In practice, the algorithm is implemented using the `matchpairs` routine, which finds the optimal one-to-one matching between variables and shocks. The resulting permutation $P^{(s)}$ reorders the shock columns of the impact matrix,

$$\tilde{A}_0^{(s)} = A_0^{(s)} P^{(s)}, \quad (16)$$

so that each column corresponds to the shock label assigned to the associated variable. An important property of the Hungarian algorithm is that it always produces a valid permutation. Consequently, the labelling procedure succeeds for every posterior draw and the assignment stability is therefore 100%.

Although the assignment is always well defined, the strength of the assignment may vary across posterior draws. To assess how strongly each shock dominates the variable to which it is assigned, I compute a dominance margin (d): for variable i , d compares the absolute impact of its assigned shock with the largest impact generated

by any other shock:

$$d_i = \frac{|\tilde{a}_{ii}|}{\max(|\tilde{a}_{ii}|, \max_{j \neq i} |\tilde{a}_{ij}|)}. \quad (17)$$

The dominance margin lies in the interval $[0, 1]$. A value of $d_i = 1$ indicates that the assigned shock is the largest contributor to variable i , while values below unity indicate that another shock has a larger contemporaneous impact. The dominance margin is computed for each variable and for each posterior draw. Averaging across simulations therefore provides a measure of the overall strength of the assignment, shown in Table 1.

Table 1: Average dominance margin by variable

	EPU	INF	GDP	FCI	FSI
Average dominance margin	0.8226	0.9994	0.9984	0.9990	1.0000

The results show that the financial stress shock is always clearly identified, with an average dominance margin equal to one. INF, GDP and FCI also exhibit very high dominance margins, indicating strong identification. By contrast, the uncertainty shock exhibits the lowest dominance margin, indicating that other shocks occasionally generate contemporaneous effects on uncertainty that are comparable in magnitude.

To further quantify the strength of identification, I define a weak-dominance threshold. Following a conservative criterion, a draw is classified as weak if the dominance margin falls below 0.909, corresponding to the inverse of a 10% dominance gap. Formally, weak dominance occurs when

$$d_i < 0.909. \quad (18)$$

The fraction of posterior draws with weak dominance is reported in Table 2.

Table 2: Fraction of draws with weak dominance

	EPU	Inflation	GDP	FCI	FSI
Weak dominance share	0.6884	0.0029	0.0074	0.0046	0.0000

Across all simulations there are 14,067 weak assignments out of 100,000 checks. Weak dominance occurs almost exclusively for the uncertainty variable, while the remaining variables exhibit very strong dominance. Finally, I analyse cases in which the assigned shock is not the largest contributor to the corresponding variable. These cases occur when the dominance margin falls below unity. The total number of such violations is 16,910 out of 100,000 checks. Table 3 reports the violation count matrix,

where rows correspond to variables and columns indicate the shock that produces the largest contemporaneous impact.

Table 3: Dominance violation counts

	EPU	Inflation	GDP	FCI	FSI
EPU	0	0	0	0	16302
Inflation	0	0	0	0	147
GDP	0	0	0	0	245
FCI	0	0	0	0	216
FSI	0	0	0	0	0

The violation pattern is highly systematic. Whenever dominance fails, the shock that dominates is always the financial stress shock. In particular, when the uncertainty shock is weakly identified, the largest contemporaneous contribution to uncertainty originates from the financial stress shock.

Overall, the diagnostics indicate that the labelling procedure is extremely stable. The Hungarian assignment successfully produces a valid mapping between shocks and variables for every posterior draw. Financial stress shocks are always sharply identified, while the uncertainty shock exhibits weaker dominance because financial stress shocks occasionally exert a larger contemporaneous effect on uncertainty itself. This result highlights the close interaction between financial stress and policy uncertainty, suggesting that episodes of elevated financial stress may be an important driver of uncertainty fluctuations in the data, and confirming that disentangling uncertainty and financial shocks is not straightforward.

From a methodological perspective, this finding also provides additional support for moving beyond recursive identification schemes. Under the Cholesky ordering considered in Section 3.3, uncertainty was assumed not to respond contemporaneously to financial shocks. The dominance diagnostics indicate that such a restriction may be overly strong, as financial stress shocks frequently contribute substantially to contemporaneous movements in uncertainty.

4 Generalized Impulse Response Functions

In standard VAR models with homoskedastic shocks, impulse response functions (IRFs) describe the dynamic propagation of a structural shock through the system under the assumption that the distribution of shocks is constant over time. In this setting, the response to a one-standard-deviation shock is time-invariant and independent of the state of the economy. In the present framework, however, the VAR is estimated with stochastic volatility, implying that the variance of the structural shocks evolves over

time. As a consequence, the transmission of shocks becomes state-dependent. The same structural innovation may generate different responses depending on the contemporaneous volatility environment and on the interaction with future volatility realizations.

Because of this feature, conventional IRFs computed under fixed volatility assumptions are not appropriate. Instead, I follow the approach of [Koop et al. \(1996\)](#) and compute *Generalized Impulse Response Functions* (GIRFs), which account for the nonlinear and state-dependent propagation of shocks in models with time-varying volatility. Formally, the GIRF at horizon h for a shock δ is defined as

$$GIRF(h, \delta, \Omega_{t-1}) = E[Y_{t+h} \mid \delta_t = \delta, \Omega_{t-1}] - E[Y_{t+h} \mid \Omega_{t-1}], \quad (19)$$

where Ω_{t-1} denotes the information set available at time $t - 1$. The GIRF therefore measures the expected difference between two simulated paths of the system - one in which a shock is introduced and one in which it is not - while integrating over the future distribution of shocks and volatility innovations.

Because impulse responses are state-dependent in this framework (through both volatilities and the last p observations), I implement two complementary exercises that both normalize deterministic shocks using date-specific volatility targets. The first one computes **date-conditioned GIRFs**, initialized at a single date: the end of the sample (31 December 2025). This exercise answers a point-in-time question: what is the expected propagation of each structural shock if it hits the system in that specific state. The second exercise computes **regime-conditioned GIRFs**. Instead of conditioning on one date, it applies the same volatility-based normalization date-by-date and then averages date-specific GIRFs over low, median, and high regimes defined by quantile thresholds. Specifically, high regimes are dates above the 80th percentile (p80), low regimes are dates below the 20th percentile (p20), and median regimes are dates in the p40–p60 band. This averaging exercise is more robust than the single-date experiment because the result is not driven by one idiosyncratic endpoint realization of the latent volatility state and recent data history.³

Starting with the first exercise, my focus is on EPU and FSI shocks (Figures 1 and 2, where solid lines report median GIRFs and shaded areas report 68% credible bands); the other date-conditioned GIRFs are reported in Appendix A5. The date-conditioned evidence indicates that OINV and CHOL imply materially different contemporaneous transmission channels. First, the inflation response to an EPU shock changes sign across identification schemes: it is positive under CHOL but negative under OINV.

³A detailed explanation of how GIRFs are calculated, including shock normalization and volatility conditioning, is provided in Appendix A4.

Second, the FSI response to an EPU shock is close to zero under OINV but large and positive under CHOL. This pattern is consistent with the possibility that placing EPU first and FSI last in the recursive ordering overstates the measured pass-through from uncertainty to financial stress. Third, the response of EPU to an FSI shock is near zero under CHOL but positive and large under OINV, suggesting that the recursive exclusion of contemporaneous feedback from financial stress to uncertainty is too restrictive in this application.

The date-conditioned GDP responses also provide a useful distinction between central dynamics and uncertainty around the median path. For an FSI shock, the median GDP response is broadly similar under CHOL and OINV, indicating comparable central-path propagation across identification schemes. However, the lower edge of the 68% band is generally more negative under OINV than under CHOL at short and medium horizons, indicating stronger downside risk even when median effects are close. In addition, under both identification schemes, the GDP response to an uncertainty shock remains larger than the GDP response to a financial-stress shock. This ranking is robust across the updated date-conditioned charts and reinforces the finding that uncertainty disturbances are quantitatively important for real activity in this sample.

Taken together, these differences suggest that the order-invariant identification captures economically meaningful interactions that are muted or constrained under the recursive Cholesky ordering, especially for contemporaneous uncertainty-stress feedback and for downside GDP tail responses.

Figure 1: Date-conditioned generalized impulse response functions: Economic Policy Uncertainty (EPU) shock

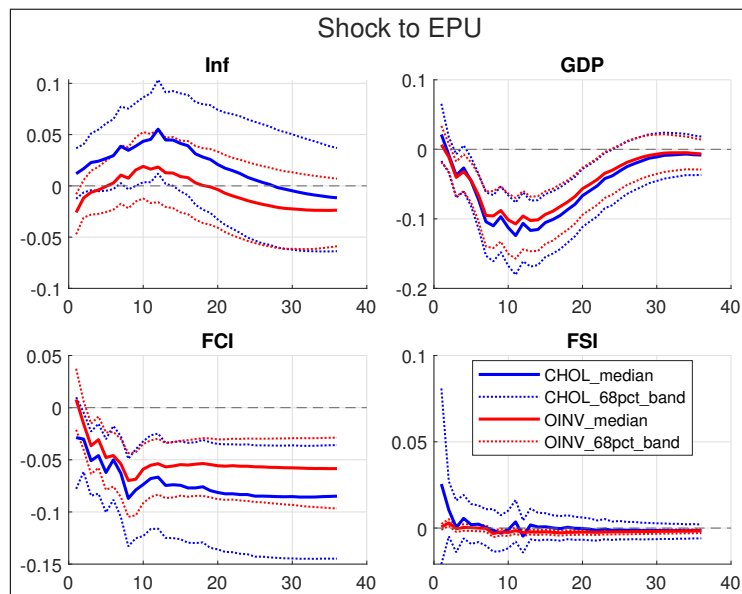
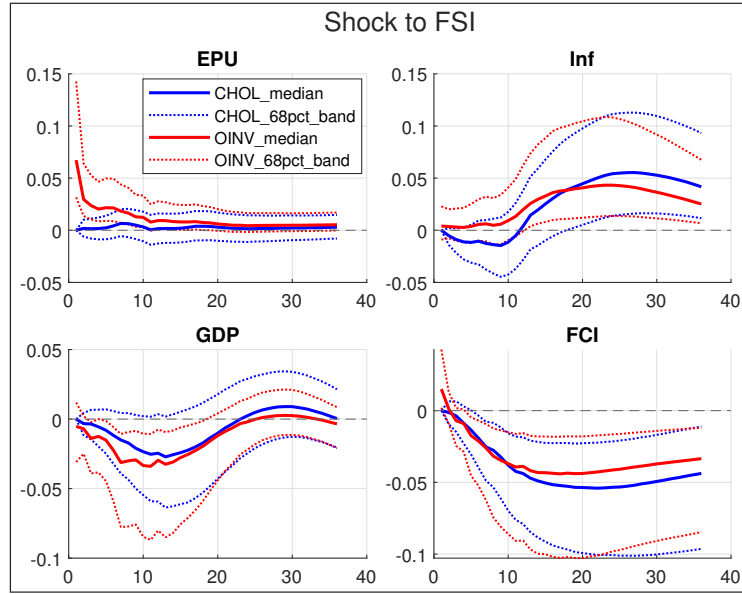
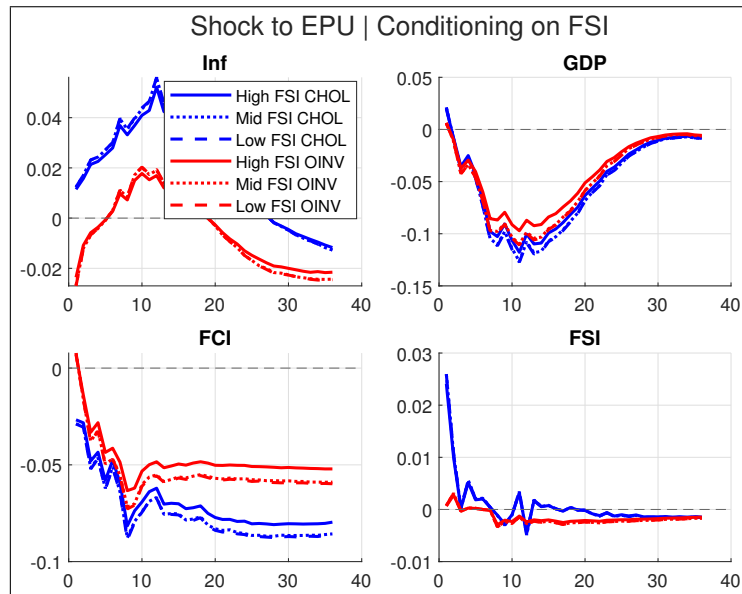


Figure 2: Date-conditioned generalized impulse response functions: Financial Stress Index (FSI) shock



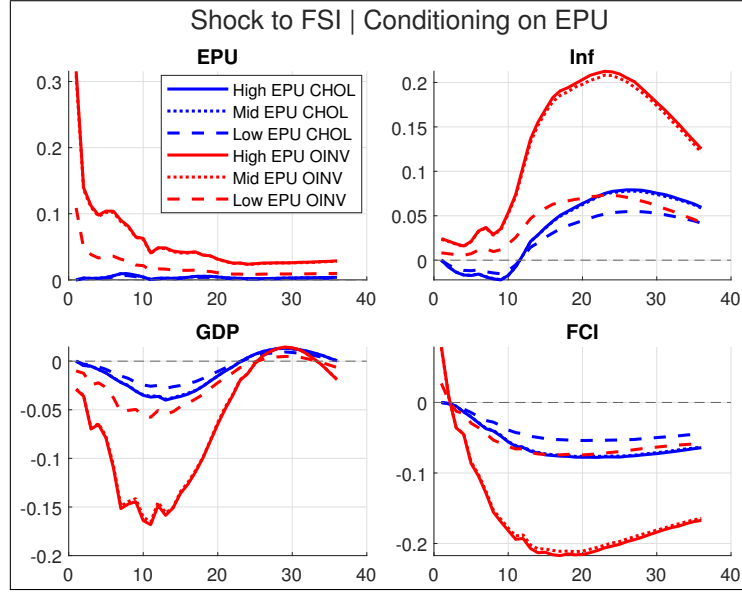
Figures 3 and 4 report the key regime-conditioned comparisons: an EPU shock conditional on low/median/high FSI and an FSI shock conditional on low/median/high EPU. Remaining regime-conditioned GIRFs are reported in Appendices A6 and A7.⁴

Figure 3: Regime-conditioned GIRFs conditional on FSI regime: EPU shock



⁴An analysis of the statistical significance of the gap between the high and low regimes is are reported in Appendices A8 and A9, specifically Figures 27 to 31 and Figures 32 to 36.

Figure 4: Regime-conditioned GIRFs conditional on EPU regime: FSI shock



The regime-conditioned GIRFs provide the strongest new evidence and show that state dependence is not symmetric across shock types.

First, consider an uncertainty (EPU) shock conditional on the financial-stress regime (Figure 3). For GDP, differences between high-FSI and low-FSI states remain modest and do not indicate systematic amplification in high-stress states. Under OINV, the peak GDP response is -0.0971 in high-FSI states, -0.1101 in the median-FSI regime, and -0.1109 in low-FSI states; cumulative 12-month effects are -0.6990, -0.7925, and -0.7978, respectively. CHOL shows the same ordering (high closer to median/low): peaks are -0.1178 (high), -0.1265 (median), and -0.1274 (low), with cumulative effects -0.7758, -0.8343, and -0.8411. High-minus-low diagnostics also remain small in economic terms, although OINV shows stronger statistical evidence than CHOL: the OINV cumulative high-minus-low difference is 0.0914 with 68% interval [0.0330, 0.1734], while CHOL implies 0.0545 with interval [0.0076, 0.1200]. Horizon-wise evidence points in the same direction: the 68% interval excludes zero at 10/12 horizons under OINV and 7/12 horizons under CHOL. Economically, this suggests that uncertainty shocks are contractionary in all regimes, but their marginal GDP effect does not rise materially with financial stress in this sample. Visual high-minus-low profiles are shown in Appendix A8.

The same weak regime dependence appears when the response variable is FSI itself. For an EPU shock conditional on FSI, OINV produces responses that are essentially flat across regimes (high-FSI peak 0.0028 and cum12 -0.0075; median-FSI peak 0.0031 and cum12 -0.0086; low-FSI peak 0.0031 and cum12 -0.0087). CHOL produces larger positive FSI responses in levels (high-FSI peak 0.0241 and cum12 0.0373;

median-FSI peak 0.0260 and cum12 0.0402; low-FSI peak 0.0259 and cum12 0.0403), but the high-versus-low regime gap is still very small. The cumulative high-minus-low difference is 0.0008 under OINV (68% interval [-0.0004, 0.0028]) and -0.0016 under CHOL (68% interval [-0.0154, 0.0072]). Horizon-wise exclusion counts are limited (5/12 for OINV; 0/12 for CHOL), which supports a cautious reading: this channel may be present but is weak in economic magnitude and not robustly separated from zero across horizons. Appendix A8 reports the full high-minus-low bands.

Second, consider a financial-stress (FSI) shock conditional on the uncertainty regime (Figure 4). Here the asymmetry is large and the median regime is close to the high regime under OINV. For the EPU response under OINV, the high- and median-EPU peaks are 0.3154 and 0.3091, both far above the low-EPU peak of 0.1088; cumulative effects are 1.2605 (high), 1.2380 (median), and 0.4335 (low). For GDP under OINV, the high- and median-EPU contractions are similarly close (peak -0.1682 and -0.1655; cumulative -1.3125 and -1.2874), both much larger than low-EPU responses (peak -0.0577; cumulative -0.4504). Quantitatively, amplification is much stronger when moving from low to median uncertainty than from median to high uncertainty: for FSI→EPU, the cumulative increase from low to median is about +0.8045 versus only +0.0225 from median to high; for FSI→GDP, the additional contraction is about -0.8370 from low to median versus -0.0251 from median to high.

Interpreting EPU regimes historically, the low-uncertainty threshold is $p_{20} = 71$, close to the pre-GFC average EPU level (81.5 over 1998 to August 2007). The median band (p_{40} – $p_{60} = 107.55$ to 133.22) maps to periods such as January 2021 to August 2022 (average 113.6, between COVID-19 and the LDI episode) and April 2023 to July 2024 (average 123.2, after the Russia/energy-crisis peak and before the later tariff-related peak under the Trump administration). By contrast, p_{80} is 183.46, while extreme episodes are much larger (Brexit: 558; LDI: 400; Trump tariffs: 498). This helps rationalize the results: moving from pre-GFC-type low uncertainty to the structurally higher uncertainty of recent years already generates most of the amplification, while moving further into the extreme upper tail adds comparatively less. High-minus-low 68% diagnostics confirm strong amplification: for OINV, cum12 differences are 0.8316 [0.4877, 1.3588] for FSI→EPU and -0.8529 [-1.6558, -0.3325] for FSI→GDP, with 11/12 and 9/12 horizons excluding zero, respectively. Under CHOL, high-minus-low differences are much weaker and 68% intervals include zero at all horizons for both channels. This pattern is consistent with an amplification channel in which high uncertainty worsens confidence and financing conditions, making stress shocks substantially more persistent and recessionary. The horizon-by-horizon high-minus-low bands for these channels are reported in Appendix A9.

Overall, the regime-conditioned evidence indicates that the economically relevant

nonlinearity is the amplification of financial-stress shocks in high-uncertainty regimes, rather than the mirror channel from uncertainty shocks in high-stress regimes. This asymmetric state dependence is clearly visible under OINV and largely muted under CHOL. Combined with the date-conditioned evidence, the results reinforce the view that identification matters for structural interpretation: recursive restrictions can compress contemporaneous uncertainty-stress interactions and understate downside amplification mechanisms that are visible under order-invariant identification.

For completeness, Appendix [A10](#) reports forecast error variance decompositions conditioned on the same date and regime states. Because the SV-BVAR FEVD is time-varying, with shares mechanically driven by relative volatility levels rather than structural propagation differences, GIRFs remain the more informative tool for assessing regime-dependent transmission in this framework.

4.1 Robustness to Alternative Cholesky Orderings

The regime-conditioned results in the previous subsection rely on a single Cholesky ordering (CHOL1: *EPU, INF, GDP, FCI, FSI*). To assess whether the key findings are driven by this particular recursive structure, I re-estimate the model under two alternative orderings introduced in Section [3.3](#): CHOL2 (*FCI, FSI, INF, GDP, EPU*) and CHOL3 (*FCI, FSI, EPU, INF, GDP*). These alternatives place financial variables first in the recursive chain, reversing the baseline assumption that real and uncertainty variables are contemporaneously exogenous to financial conditions. Full regime-conditioned GIRF plots for CHOL2 and CHOL3 are reported in Appendices [A11](#) and [A12](#); here I focus on the summary statistics that matter most for the paper’s main conclusions.

Table [4](#) reports peak responses and twelve-month cumulative responses under high and low regimes, together with the regime difference ($\Delta = \text{High} - \text{Low}$), for the four key shock–response–regime combinations.

Table 4: Regime-conditioned GIRF summary: peak and cumulative (12-month) responses under four identification schemes. ΔPeak and ΔCum12 denote the high-minus-low regime difference.

Experiment	Model	Peak _{Hi}	Cum12 _{Hi}	Peak _{Lo}	Cum12 _{Lo}	ΔPeak	ΔCum12
<i>FSI shock $\rightarrow$ GDP EPU regime</i>							
	CHOL1	−0.040	−0.234	−0.028	−0.163	−0.012	−0.072
	CHOL2	−0.045	0.465	−0.022	0.222	−0.023	0.243
	CHOL3	−0.030	−0.247	−0.018	−0.148	−0.012	−0.099
	OINV	−0.168	−1.313	−0.058	−0.450	−0.111	−0.862
<i>FSI shock $\rightarrow$ EPU EPU regime</i>							
	CHOL1	0.009	0.050	0.007	0.035	0.003	0.016
	CHOL2	0.021	0.137	0.010	0.065	0.011	0.071
	CHOL3	0.020	0.151	0.012	0.091	0.008	0.060
	OINV	0.315	1.260	0.109	0.433	0.207	0.827
<i>EPU shock $\rightarrow$ GDP FSI regime</i>							
	CHOL1	−0.118	−0.776	−0.127	−0.841	0.010	0.065
	CHOL2	−0.063	−0.416	−0.068	−0.450	0.005	0.034
	CHOL3	−0.111	−0.870	−0.119	−0.930	0.008	0.061
	OINV	−0.097	−0.699	−0.111	−0.798	0.014	0.099
<i>EPU shock $\rightarrow$ FSI FSI regime</i>							
	CHOL1	0.024	0.037	0.026	0.040	−0.002	−0.003
	CHOL2	0.004	−0.014	0.005	−0.015	0.000	0.001
	CHOL3	0.003	−0.021	0.004	−0.022	0.000	0.002
	OINV	0.003	−0.008	0.003	−0.009	0.000	0.001

The table reveals a clear pattern. For the channel that constitutes the paper’s main finding—the amplification of FSI shocks in high-uncertainty regimes—the order-invariant identification produces regime asymmetries that are an order of magnitude larger than any Cholesky specification. The OINV cumulative GDP response gap ($\Delta\text{Cum12} = -0.86$) dwarfs the corresponding Cholesky gaps, which range from -0.10 (CHOL3) to $+0.24$ (CHOL2). The sign reversal under CHOL2 is notable: because this ordering treats financial stress as contemporaneously exogenous, the recursive structure absorbs the uncertainty-stress interaction into the reduced-form residuals rather than the structural shock, leading to a misleading positive gap. A similar pattern holds for the FSI $\rightarrow$ EPU channel, where the OINV difference ($+0.83$) is around an order of magnitude larger than the largest Cholesky figure (about $+0.07$ under CHOL2).

Figure 5 overlays the median GIRFs of GDP to an FSI shock under the four identification schemes, separately for the high- and low-EPU regimes. The OINV response (solid line) is substantially larger and more persistent than any Cholesky variant, especially in the high-uncertainty state, confirming visually the quantitative gap docu-

mented in the table. Figure 6 shows the corresponding overlay for the EPU response to an FSI shock; the same dominance of the OINV scheme is evident. For completeness and symmetry, Figures 7 and 8 report the mirror channel: GDP and FSI responses to an EPU shock conditional on the FSI regime.

Figure 5: Overlay of median GIRFs: GDP response to an FSI shock, conditional on EPU regime. Solid = OINV, dashed/dotted = Cholesky variants.

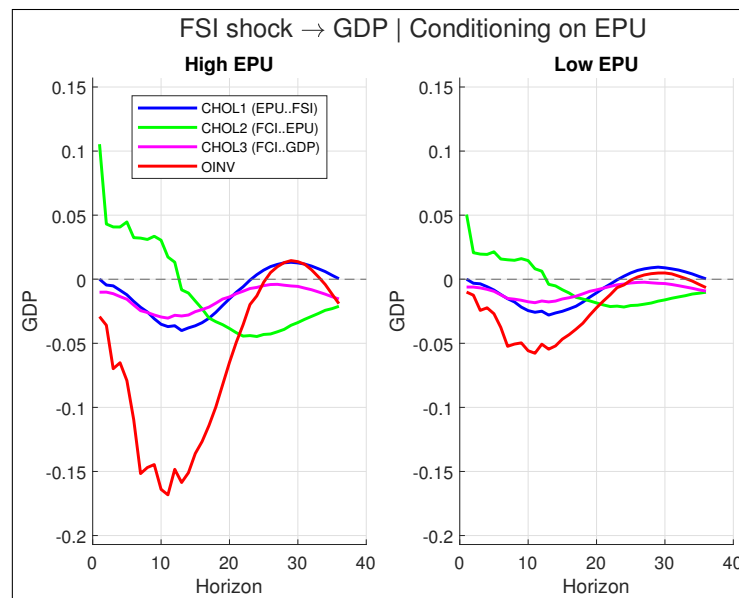


Figure 6: Overlay of median GIRFs: EPU response to an FSI shock, conditional on EPU regime. Solid = OINV, dashed/dotted = Cholesky variants.

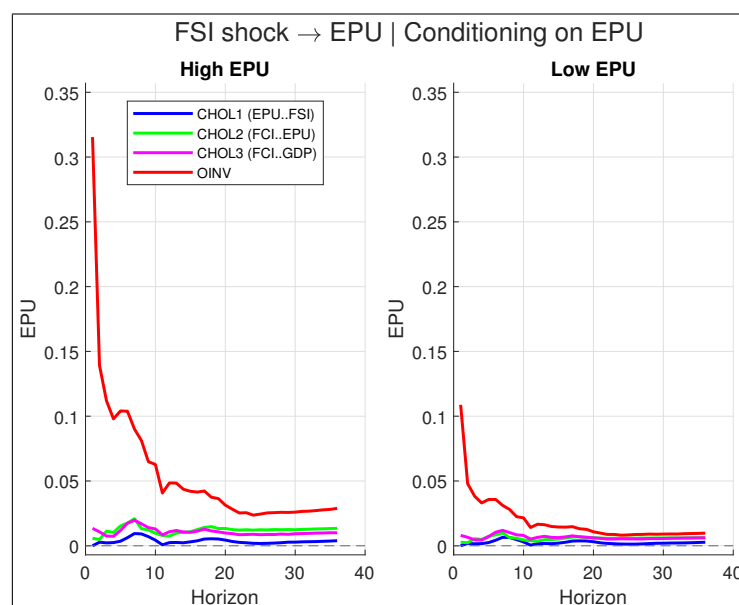


Figure 7: Overlay of median GIRFs: GDP response to an EPU shock, conditional on FSI regime. Solid = OINV, dashed/dotted = Cholesky variants.

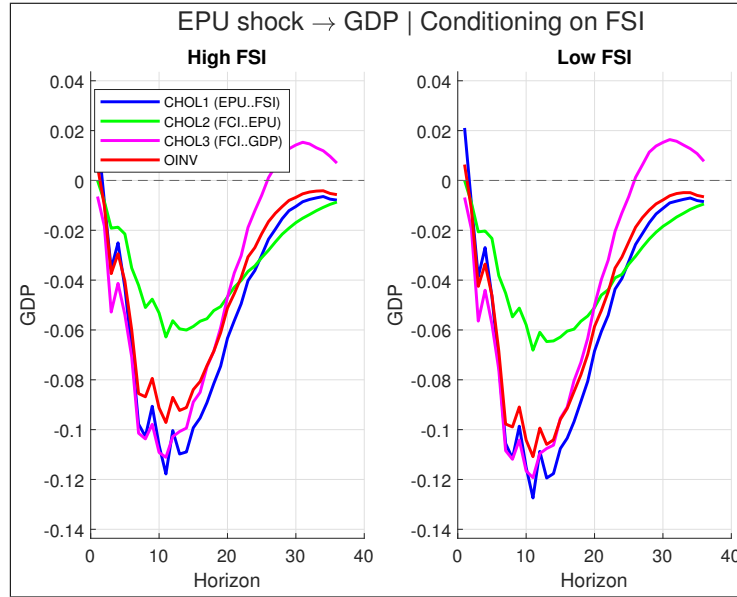
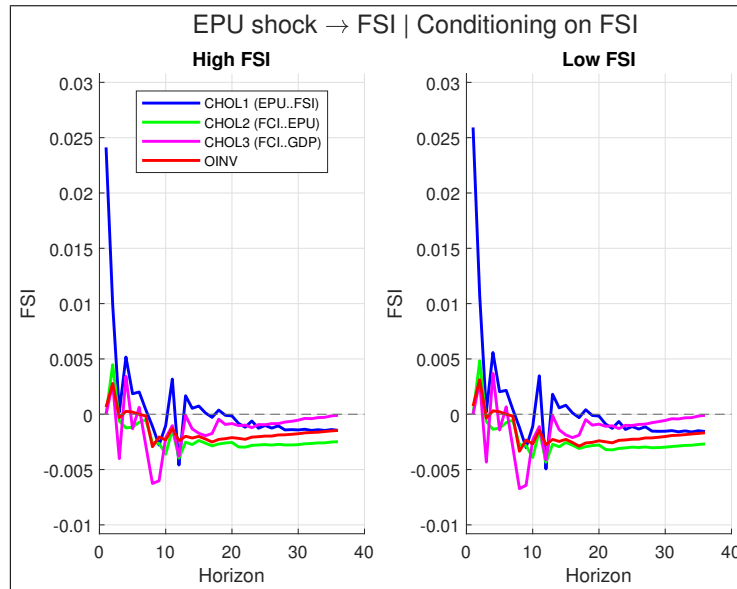


Figure 8: Overlay of median GIRFs: FSI response to an EPU shock, conditional on FSI regime. Solid = OINV, dashed/dotted = Cholesky variants.



By contrast, the EPU→GDP and EPU→FSI channels—shown in Figures 7 and 8, and where Section 4 found little regime dependence—remain muted across all four identification schemes once the sign convention is interpreted correctly. For EPU→GDP, high-state responses are slightly less negative than low-state responses (cum12 differences between about +0.03 and +0.10). For EPU→FSI, cumulative high-minus-low differences are near zero in all schemes (around −0.003 to +0.002). High-minus-low 68% intervals in the extended diagnostics are also weak for these channels, especially

outside OINV EPU→GDP where evidence is moderate rather than strong. This confirms that the absence of strong amplification in the uncertainty-to-real-activity channel is a robust feature of the data, not an artefact of a particular ordering.

Taken together, the robustness exercise strengthens the paper’s two core conclusions. First, the amplification of financial-stress shocks in high-uncertainty regimes is a genuine structural feature that is visible only when the contemporaneous impact matrix is left unrestricted; all three Cholesky orderings compress this asymmetry by at least an order of magnitude. Second, the absence of strong regime dependence for uncertainty shocks is ordering-invariant, ruling out the possibility that the baseline Cholesky specification was inadvertently suppressing an economically important channel.

4.2 Robustness to Weak-Dominance Draws

As documented in Section 3.5, the impact-dominance labelling procedure assigns each posterior draw a dominance margin for every variable, measuring the extent to which the assigned structural shock is the single largest contributor to that variable’s contemporaneous impact. Draws where this margin falls below a threshold of $1/1.1 \approx 0.909$ are flagged as weakly identified. For the EPU shock, approximately 69% of posterior draws are flagged as weak—a consequence of the fact that the FSI shock frequently competes with the EPU shock for dominance over the EPU variable.

This raises a natural question: is the asymmetric state dependence documented above driven by the well-identified minority of posterior draws, or does it require the weakly identified majority? If the amplification of financial-stress shocks in high-uncertainty regimes survives when the analysis is restricted to the roughly 31% of draws where EPU identification is strong, the finding is robust to identification uncertainty. If it does not, the result must be interpreted with greater caution.

To answer this question, I exploit the fact that the regime-conditioned GIRFs computed in Section 4 retain the full posterior-draw dimension. The baseline GIRF arrays have shape $(n \times H \times N_{\text{draws}} \times n)$, where N_{draws} is the number of thinned posterior draws used for computational tractability (every fifth draw from the MCMC chain). The dominance margin diagnostics produced during the labelling step of Section 3.5 are thinned with the same stride, so that each position in the GIRF draw dimension maps one-to-one to its dominance classification. Draws where the EPU dominance margin exceeds the 0.909 threshold are retained; the remaining draws are discarded. The filtered GIRF arrays are obtained by extracting the strong-draw positions from the baseline arrays. No re-simulation is required: the filtered set is an exact partition of the baseline draws, sharing the same random innovations.

Table 5 reports the results for the four key shock–response–regime combinations.

Table 5: Weak-dominance robustness: high-minus-low regime differences under baseline (all thinned draws) and EPU-strong filter. Peak and cumulative (12-month) median differences, with zero-exclusion flag (excl0 = 1 if 68% band excludes zero at peak horizon).

Experiment	Filter	ΔPeak	h^*	ΔCum12	excl0
<i>EPU shock $\rightarrow$ GDP FSI regime</i>					
	Baseline ($n = 4,000$)	−0.0005	1	+0.090	0
	EPU-strong ($n = 1,217$)	−0.0006	1	+0.099	0
<i>FSI shock $\rightarrow$ EPU EPU regime</i>					
	Baseline ($n = 4,000$)	+0.205	1	+0.821	1
	EPU-strong ($n = 1,217$)	+0.168	1	+0.689	1
<i>FSI shock $\rightarrow$ GDP EPU regime</i>					
	Baseline ($n = 4,000$)	−0.110	11	−0.852	1
	EPU-strong ($n = 1,217$)	−0.092	11	−0.744	1
<i>EPU shock $\rightarrow$ FSI FSI regime</i>					
	Baseline ($n = 4,000$)	+0.0003	8	+0.0009	1
	EPU-strong ($n = 1,217$)	+0.0004	8	+0.0008	1

The results are reassuring. For the paper’s main finding—the amplification of FSI shocks in high-uncertainty regimes—the EPU-strong subset preserves the sign, timing, and approximate magnitude of the baseline regime asymmetry. The FSI→GDP cumulative gap narrows modestly from −0.852 (baseline) to −0.744 (EPU-strong), and the FSI→EPU gap from +0.821 to +0.689, but both remain large and the 68% credible bands continue to exclude zero at the peak horizon. The roughly 15–18% attenuation is consistent with the well-identified draws being drawn from a posterior region with somewhat smaller contemporaneous impacts, rather than suggesting that the asymmetry depends on identification ambiguity. For the EPU→GDP and EPU→FSI channels, where Section 4 found little regime dependence, the filtered results are essentially unchanged.

Appendix A13 provides the full set of comparison figures, overlaying baseline and EPU-strong high-minus-low difference bands for all shock–response pairs under both FSI and EPU conditioning.

4.3 Robustness to Alternative Measures of Uncertainty

While it is plausible that financial stress can have an impact on policy uncertainty, as policymakers may respond to crisis, it is much less plausible that financial stress would cause a spike in other measures of uncertainty. For example, geopolitical risk

is largely driven by exogenous international events. To address this, I repeat the estimation and the regime-conditioned analysis using the Geopolitical Risk Index (GPR) as an alternative measure of uncertainty. Geopolitical risk is constructed to capture exogenous international events and, by design, should not be directly caused by domestic financial stress. If the amplification channel is specific to policy uncertainty, the FSI→GPR effect should be absent or much weaker.

First, I verify that the weak dominance issue identified for EPU in Section 3.5 disappears when using GPR (Tables 6, 7 and 8). In the GPR model, shock assignment is consistently clear, with no evidence of weak dominance for the GPR variable. Whereas 69% of EPU draws exhibited weak dominance, none of the GPR draws do. This supports the interpretation that GPR is more exogenous and less entangled with domestic financial shocks than EPU.

Table 6: Average dominance margin by variable

	GPR	INF	GDP	FCI	FSI
Average dominance margin	1.0000	1.0000	0.9987	0.9986	1.0000

Table 7: Fraction of draws with weak dominance

	GPR	Inflation	GDP	FCI	FSI
Weak dominance share	0	0.0002	0.0063	0.0066	0.0000

Table 8: Dominance violation counts

	GPR	Inflation	GDP	FCI	FSI
GPR	0	0	0	0	0
Inflation	0	0	0	0	11
GDP	0	0	0	0	228
FCI	0	0	0	0	327
FSI	0	0	0	0	0

Second, I repeat the regime-conditioned GIRF analysis for the GPR model. The key comparisons are the responses of GPR and EPU to an FSI shock conditional on the GPR/EPU regime. If the amplification channel is specific to policy uncertainty, the FSI→GPR response should show little or no regime dependence, while the FSI→EPU response should show strong amplification in high-EPU regimes. The results are shown in Figures 9 and 10.

Figure 9: Regime-conditioned responses of EPU and GPR to an FSI shock. Shaded areas show 68% credible intervals for high and low uncertainty regimes.

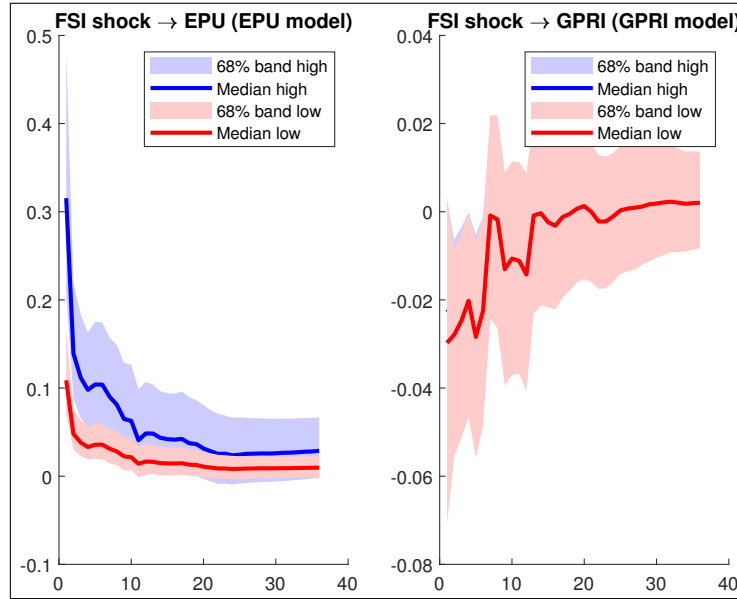
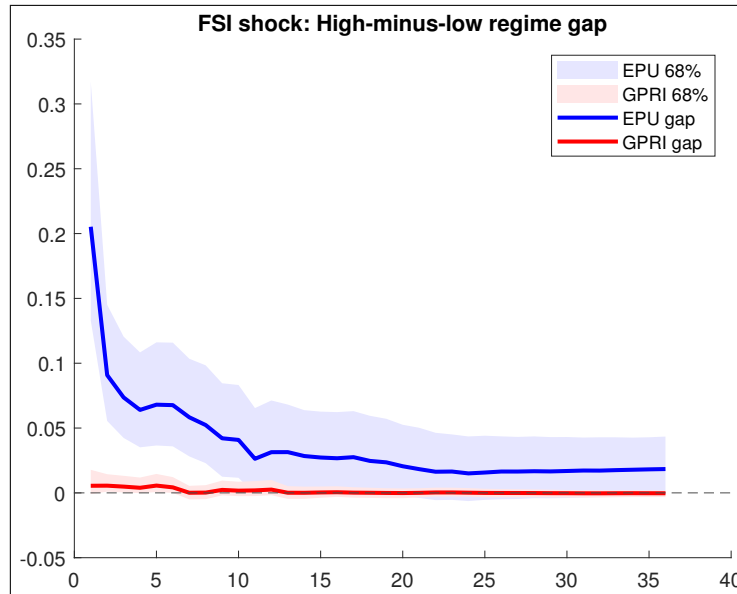


Figure 10: High-minus-low regime gap for the response of EPU and GPR to an FSI shock. Shaded areas show 68% credible intervals.



The response of the GPR to an FSI shock is small and negative, with the 68% credible interval including zero for most of the horizon. Furthermore, while the regime gap is large and statistically significant for EPU, it is negligible for GPR. These results confirm that the amplification of financial stress shocks is specific to policy uncertainty and does not generalize to alternative, more exogenous measures of uncertainty such as GPR. This finding strengthens the interpretation that the FSI→EPU amplification channel reflects genuine macro-financial feedback, rather than a generic feature of un-

certainty proxies or a spurious correlation. As expected, financial stress shocks do not amplify geopolitical risk, supporting the exogeneity of GPR and the robustness of the main results.

Results are also confirmed in an additional robustness check, reported in see Appendix A14 (Tables 18–20 and Figures 71–72), where we test a joint 6-variable specification that includes both EPU and GPRI. The Appendix confirms results of this subsection.

5 Historical Narrative Check of Structural Shocks

GIRFs show the response of each variable to structural shocks, but they do not by themselves guarantee that labelled shocks are economically plausible in key historical episodes. A complementary check is therefore to ask whether estimated structural shocks align with the narrative that policymakers and markets associate with major events. If the labelling strategy is informative, episode-level shock rankings should be coherent and reasonably robust across specifications.

I implement a compact narrative exercise using three UK episodes: Brexit referendum (June 2016), COVID onset (March 2020), and the gilt crisis (September 2022). For each model I reconstruct structural shocks from posterior objects, standardize each shock to unit variance, and compute episode windows of ± 3 months around the event date. Within each window, shocks are ranked by their peak absolute standardized realization. I report results for four specifications: OINV and the three recursive orderings specified in Section 4.1, denoted CHOL 1, CHOL 2, and CHOL 3. The benchmark priors are straightforward. For Brexit, an uncertainty shock should dominate. For COVID onset, a real-activity shock should dominate. For the gilt crisis, a financial-stress shock should dominate. These are deliberately coarse expectations, intended as a plausibility test rather than a strict identification restriction.

Figures 11, 12, and 13 show standardized structural shocks around each episode for all four models. Table 9 reports the top-ranked shock by episode and model with the signed peak value. Because the ranking metric is based on peak magnitude, interpretation should also use sign and joint shock configuration in the figures.

Figure 11: Historical narrative check around the Brexit referendum: standardized structural shocks across OINV and recursive CHOL 1–3. Vertical dashed lines mark event dates.

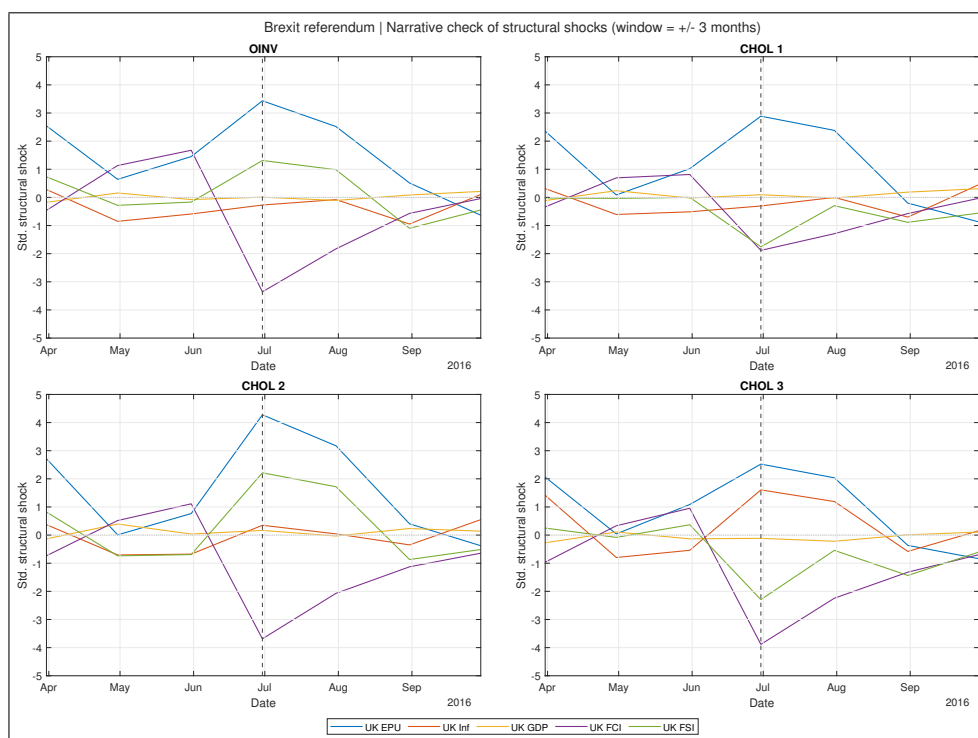


Figure 12: Historical narrative check around COVID onset: standardized structural shocks across OINV and recursive CHOL 1–3. Vertical dashed lines mark event dates.

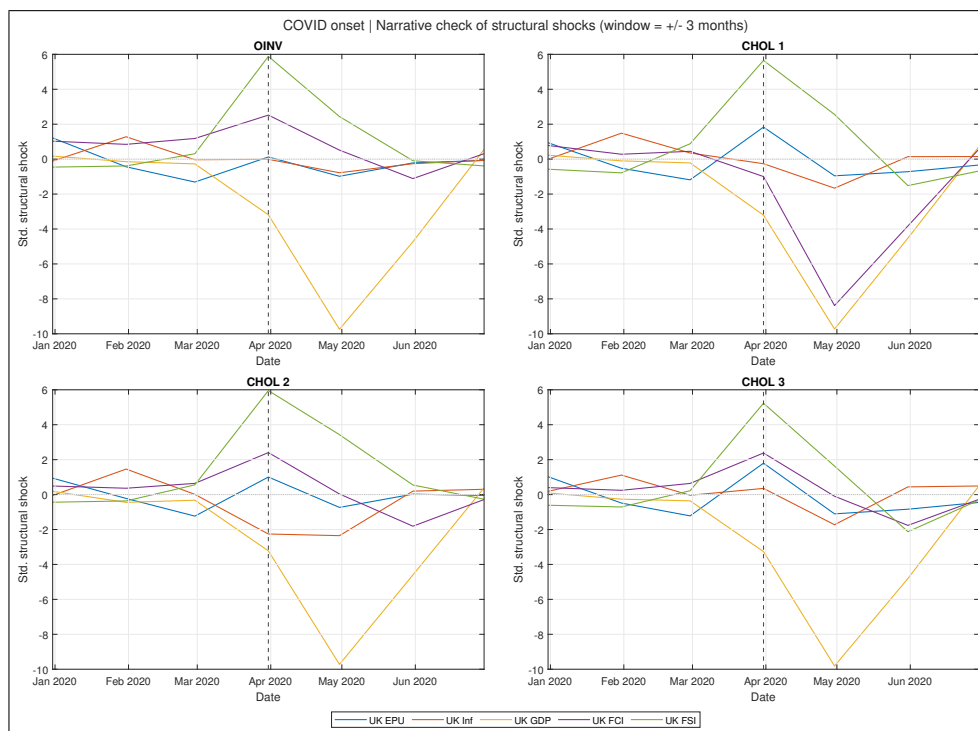


Figure 13: Historical narrative check around the September 2022 gilt crisis: standardized structural shocks across OINV and recursive CHOL 1–3. Vertical dashed lines mark event dates.

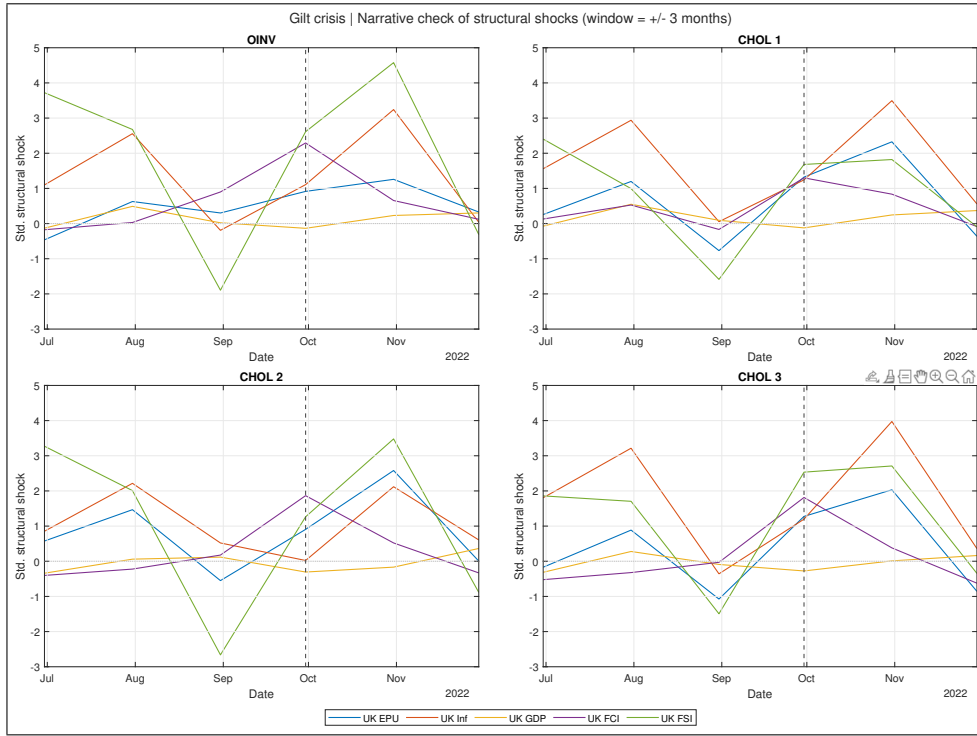


Table 9: Top-ranked structural shock by episode and identification scheme (signed peak standardized value in parentheses)

Episode	OINV	CHOL 1	CHOL 2	CHOL 3
Brexit referendum	UK EPU (+3.43)	UK EPU (+2.89)	UK EPU (+4.28)	UK FCI (-3.88)
COVID onset	UK GDP (-9.74)	UK GDP (-9.73)	UK GDP (-9.72)	UK GDP (-9.81)
Gilt crisis	UK FSI (+4.58)	UK Inf (+3.50)	UK FSI (+3.48)	UK Inf (+3.98)

Overall, the four specifications are similar and all recover key elements of each episode. That said, two differences remain relevant. First, OINV still matches all three benchmark priors: EPU dominates in Brexit, GDP dominates in COVID, and FSI dominates in the gilt episode. Second, recursive results remain ordering-sensitive, especially in Brexit and the gilt crisis, where the leading shock and associated sign pattern vary across CHOL orderings.

Sign-aware interpretation adds an important nuance. For Brexit, all models show a strong positive uncertainty shock and financial-conditions loosening; however, CHOL 1 and CHOL 3 imply financial-stress loosening, and CHOL 3 additionally shows a relatively large positive inflation shock, while OINV and CHOL 2 are more in line with a stress-based interpretation. For COVID onset, all models identify a very large negative GDP shock and a strong financial-stress tightening, with differences concentrated

in secondary shocks (for example, the sign of financial-conditions adjustments). For the gilt episode, all models capture heightened financial stress, but OINV shows the strongest FSI signal, whereas CHOL 1 and CHOL 3 assign relatively more weight to inflation. In this sense, OINV looks clearly stronger than CHOL 1 and CHOL 3, while OINV and CHOL 2 appear more comparable.

There are three reasons why CHOL 1–3 do not align equally well with the narrative benchmark in this application. First, recursive identification imposes triangular contemporaneous restrictions that force shock attribution to depend on ordering, even when the underlying reduced-form co-movements are unchanged. Second, the Brexit and gilt episodes combine policy-news shocks, financial repricing, and inflation dynamics within a short horizon; with that overlap, alternative recursive orderings naturally reassign the same movement across labels. Third, a magnitude-only ranking can obscure direction and composition issues: a model may match the dominant label while still implying a less plausible signed mix of secondary shocks. Taken together with Section 4, the narrative evidence is best read as supportive rather than decisive: OINV tends to produce the most coherent episode interpretation, especially relative to CHOL 1 and CHOL 3, but gaps are not uniformly large and CHOL 2 often delivers a similar narrative picture. This does not imply that every episode can be reduced to a single shock, nor that narrative agreement proves causal identification.

6 Forecasting Performance

The previous two sections showed that the impact-dominance identification (OINV) produces impulse responses and episode-level structural shock patterns that appear more economically plausible than those obtained under recursive Cholesky orderings. However, the ultimate usefulness of an empirical model also depends on its predictive performance. In this section I therefore compare the forecasting properties of the two identification schemes.

My focus is on forecasting *GDP downside risk*. In particular, I evaluate the ability of the model to predict the lower tail of the predictive distribution of GDP growth. This objective is motivated by the literature on macroeconomic tail risks and by the results of [Carriero et al. \(2024\)](#), who show that stochastic-volatility Bayesian VARs estimated under recursive identification can produce tail forecasts that are competitive with quantile regression approaches. Given this evidence, we ask whether the alternative identification strategy proposed in this paper provides additional gains in forecasting performance. In particular, we examine whether replacing the recursive Cholesky decomposition with the impact-dominance identification improves the model’s ability to capture GDP tail risk.

I evaluate forecasting performance using a recursive out-of-sample experiment. At each recursion the model is estimated using an expanding window of observations and multi-step-ahead predictive distributions are generated. The initial estimation window contains 180 observations, corresponding to approximately fifteen years of monthly data. Forecasts are then generated up to a horizon of 36 months ahead. Given a total sample size of $T = 336$ observations, the number of recursive forecast origins is therefore

$$x = T - (180 + 36 - 1) = 121.$$

For each of these 121 forecast origins the model is re-estimated and a full predictive distribution for the next 36 months is generated. The same forecasting experiment is implemented for both identification schemes, allowing for a direct comparison of their predictive performance. Forecasts are generated using posterior simulation. For each MCMC draw of the model parameters, a future path of the VAR is simulated conditional on the most recent observed values of the data, as explained in [Appendix A15](#).

Because the predictive distribution is time-varying and asymmetric, forecast quality is evaluated with multiple complementary metrics. I use quantile scores to assess lower-tail accuracy, RMSE to assess point-forecast accuracy at the center of the distribution, and CRPS to assess full-distribution accuracy. This combination helps identify whether any ranking difference between CHOL and OINV is tail-specific or reflects broader predictive performance.

For tail accuracy, I compute quantile scores at $\tau \in \{0.01, 0.05, 0.10\}$:

$$QS_{t,h}^{\tau} = (\tau - \mathbf{1}\{y_{t+h} < q_{t,h}^{\tau}\})(y_{t+h} - q_{t,h}^{\tau}),$$

where $q_{t,h}^{\tau}$ is the predicted τ -quantile of GDP growth at horizon h , y_{t+h} is the realization, and $\mathbf{1}\{\cdot\}$ is the indicator function. Lower values indicate better tail forecast accuracy.

For point-forecast accuracy, I compute RMSE using the median forecast $m_{t,h}$:

$$\text{RMSE} = \sqrt{\frac{1}{NT} \sum_{t=1}^N \sum_{h=1}^T (y_{t+h} - m_{t,h})^2},$$

where N is the number of recursive forecast origins and T is the forecast horizon length.

For full-distribution accuracy, I compute the continuous ranked probability score

(CRPS):

$$\text{CRPS}_{t,h} = \int_{-\infty}^{\infty} (F_{t,h}(u) - \mathbf{1}\{y_{t+h} \leq u\})^2 du,$$

where $F_{t,h}(u)$ is the predictive CDF. Lower CRPS values indicate better overall density forecasts. All metrics are computed horizon-by-horizon for each recursion and then averaged across the 121 recursive forecast origins for each identification scheme.

6.1 Results

Table 10 reports the full metric set introduced above, averaged over all recursive forecast origins and all horizons. Three points stand out. First, CHOL is marginally better on lower-tail quantile scores at 1%, 5%, and 10%. Second, OINV is marginally better on RMSE (computed from median forecasts) and on CRPS. Third, the magnitudes are small in all cases, so neither identification dominates uniformly.

Table 10: Forecast-performance summary: CHOL vs OINV (averages over all origins and horizons)

Metric	CHOL	OINV	Preferred
QS (1%)	1.4145	1.4331	CHOL (lower is better)
QS (5%)	1.7622	1.7803	CHOL (lower is better)
QS (10%)	1.8985	1.9025	CHOL (lower is better)
RMSE (median forecast)	6.6544	6.5693	OINV (lower is better)
CRPS	3.3781	3.2964	OINV (lower is better)
Coverage at 1%	0.0849	0.0852	CHOL (closer to 0.01)
Coverage at 5%	0.0962	0.0978	CHOL (closer to 0.05)
Coverage at 10%	0.1148	0.1217	CHOL (closer to 0.10)

Figure 80 in Appendix A16 shows the horizon-by-horizon QS profiles at 1%, 5%, and 10%. In line with Table 10, CHOL tends to score slightly better in lower-tail quantile loss, but differences are modest and horizon-dependent. For example, over the first ten horizons the two models are often nearly indistinguishable, with occasional rank reversals at specific horizons.

The interpretation is that different metrics emphasize different parts of the predictive distribution. Quantile scores at low τ values focus on specific lower-tail cutoffs, where CHOL remains slightly sharper in this sample. By contrast, RMSE and CRPS integrate more central and full-distribution features, where OINV has a small advantage. This metric sensitivity is economically plausible and not contradictory: a model can fit a specific tail quantile slightly better while another performs slightly better on integrated distributional accuracy.

Calibration diagnostics reinforce that practical differences are limited. Figure 81 reports empirical coverage for the 1%, 5%, and 10% lower quantiles. Both specifications exhibit excess exceedances relative to nominal levels (for example, empirical 1% coverage around 8.5%), indicating that downside tails are underpredicted in absolute terms; however, CHOL is marginally closer to nominal coverage on average.

Overall, the expanded scorecard points to small, metric-dependent differences rather than a single dominant identification scheme in forecasting. CHOL retains a slight edge on tail-quantile accuracy and calibration proximity, while OINV retains a slight edge on RMSE and CRPS. The practical implication remains that identification choice matters much more for structural interpretation than for aggregate forecast performance in this experiment. Taken together with the impulse-response evidence in Section 4 and the episode-level narrative evidence in Section 5, these results suggest that the impact-dominance identification can produce economically more plausible structural dynamics without materially deteriorating the forecasting performance of the model.

6.2 Robustness Check: Alternative Cholesky Orderings

To assess whether these forecasting conclusions depend on the baseline recursive ordering, I run a reduced-origin robustness exercise with four models: CHOL1, OINV, CHOL2, and CHOL3. To keep computation tractable, I evaluate every sixth recursive origin and enforce a like-for-like comparison by restricting CHOL1 and OINV to the same reduced-origin subset used for CHOL2 and CHOL3.

Table 11 reports the reduced-origin summary. CHOL2 performs strongly in this short-run sample and attains the lowest quantile scores and CRPS. OINV remains competitive and still improves over CHOL1 on RMSE and CRPS. CHOL3 shows an abnormally high *mean* CRPS, but this is not representative of its typical behavior: its median CRPS is close to the other models, and Appendix A17 shows the mean is driven by a narrow outlier cluster (one origin at long horizons).

Table 11: Reduced-origin forecasting robustness (every 6th origin): four-model comparison

Model	QS (1%)	QS (5%)	RMSE (median)	Mean CRPS
CHOL1	1.3644	1.6919	6.4730	3.3364
OINV	1.3928	1.7234	6.3595	3.2457
CHOL2	1.2865	1.6103	6.3546	3.2069
CHOL3	1.3983	1.7232	6.4388	20.0192

The corresponding horizon-by-horizon robustness profiles lead to the same con-

clusion: forecast rankings remain metric-dependent and broadly similar across identification choices, while structural conclusions in Section 4 are considerably sharper and more robust.

7 Priors Diagnostics

A natural concern in comparing CHOL and OINV is that apparent differences in structural results might be driven by prior choices rather than by identification itself. This section addresses that concern directly by running a prior-sensitivity exercise around the baseline calibration and checking whether conclusions are robust. The rationale is straightforward - in this SV-BVAR, posterior sampling is iterative: VAR coefficients, stochastic volatilities, and structural parameters are sampled conditionally on each other. As a result, changing structural priors can in principle propagate to reduced-form objects. A dedicated diagnostic is therefore needed to verify that the main CHOL-OINV differences are not artifacts of prior tightness.

The sensitivity design is intentionally simple. All common components are held fixed across runs (data sample, lag length, MCMC size, and non-structural priors) and only identification-specific prior tightness is varied: (i) for CHOL, the precision on free lower-triangular impact parameters is scaled; (ii) for OINV, the variance on elements of B_0 is scaled. A light grid is used for both schemes, with prior scales $\{0.5, 1, 2\}$, and for each run we report summary moments of posterior coefficients, stochastic volatility, and impact-matrix structure. We report two standardized distance diagnostics: one for reduced-form VAR coefficients (Table 13) and one for stochastic-volatility states (Table 14). For coefficients, the distance metric is

$$z_j = \frac{\bar{\alpha}_j^{CHOL} - \bar{\alpha}_j^{OINV}}{\sqrt{\mathbb{V}(\alpha_j^{CHOL}) + \mathbb{V}(\alpha_j^{OINV})}}.$$

Here, j indexes one element of the stacked coefficient vector; $\bar{\alpha}_j^{CHOL}$ and $\bar{\alpha}_j^{OINV}$ are posterior means of that element under each identification; and $\mathbb{V}(\alpha_j^{CHOL})$ and $\mathbb{V}(\alpha_j^{OINV})$ are posterior variances of the same element. The numerator is therefore the CHOL-OINV mean gap for coefficient j , while the denominator scales that gap by a pooled posterior standard deviation. This normalization makes the metric unit-free and directly comparable across coefficients with different scales. In particular, $|z_j| = 0.1$ means the two posterior means differ by one-tenth of a posterior standard deviation, while $|z_j| = 1$ means a one-standard-deviation gap. For volatility states, Table 14 ap-

plies the same construction to each element of h , i.e.

$$z_j^h = \frac{\bar{h}_j^{CHOL} - \bar{h}_j^{OINV}}{\sqrt{\mathbb{V}(h_j^{CHOL}) + \mathbb{V}(h_j^{OINV})}}.$$

Table 12: Prior-sensitivity summary by identification and prior scale

Ident.	Prior scale	mean($ \alpha $)	mean(sd(α))	mean(h)	Median $ B_{0,ij} , i \neq j$	Median $ B_{0,ij} , i > j$
CHOL	0.5	0.067785	0.080713	-2.5282	4.5627×10^{-7}	0.081281
CHOL	1.0	0.067788	0.080682	-2.5288	2.2640×10^{-7}	0.081402
CHOL	2.0	0.067821	0.080703	-2.5281	1.4201×10^{-8}	0.081184
OINV	0.5	0.066204	0.070893	-2.0186	0.077446	0.041144
OINV	1.0	0.066286	0.070829	-1.8152	0.087065	0.048029
OINV	2.0	0.066495	0.070729	-1.6236	0.096536	0.055995

Table 12 delivers two main messages. First, reduced-form coefficient summaries are highly stable within each identification across prior scales, which indicates that moderate changes in structural-prior tightness do not materially shift posterior VAR dynamics. Second, structural summaries move in expected ways but remain interpretable once constraints are accounted for. The column “Median $|B_{0,ij}|, i \neq j$ ” uses all off-diagonal entries and is therefore not directly comparable between CHOL and OINV because CHOL imposes many exact zeros by construction. For strict cross-scheme comparability, the last column reports only lower-triangular off-diagonal elements ($i > j$), which are free under CHOL and unrestricted under OINV after labelling. On this matched set, CHOL medians are around 0.081 across scales, while OINV medians increase from about 0.041 to 0.056 as prior scale rises.

Table 13: Coefficient-distance diagnostics for reduced-form VAR coefficients (α): CHOL vs OINV

Prior scale	Share($ z_j < 0.25$)	Share($ z_j < 0.50$)	Median($ z_j $)
0.5	0.80656	0.93770	0.094576
1.0	0.80656	0.93770	0.095644
2.0	0.80000	0.93770	0.097203

Table 13 confirms that CHOL and OINV remain close in reduced-form terms over the full prior grid. The thresholds $|z_j| < 0.25$ and $|z_j| < 0.5$ are heuristic benchmarks for “very small” and “small” standardized differences, not hard decision cutoffs. In this application, median $|z_j| \approx 0.095$ means that the typical CHOL–OINV difference in VAR coefficients is about one-tenth of a posterior standard deviation, which is quanti-

tatively small. This is consistent with the distributional shares: around 80% of coefficients satisfy $|z_j| < 0.25$ and about 94% satisfy $|z_j| < 0.5$ at all prior scales.

Table 14: Volatility-distance diagnostics for stochastic-volatility states (h): CHOL vs OINV

Prior scale	Share($ z_j^h < 0.25$)	Share($ z_j^h < 0.50$)	Median($ z_j^h $)
0.5	0.07099	0.22346	1.014511
1.0	0.06790	0.22099	1.153837
2.0	0.09753	0.25000	1.221456

Table 14 complements Table 13 by showing that CHOL and OINV are not close in volatility-state space. Across all prior scales, only about 7%–10% of volatility-state elements satisfy $|z_j^h| < 0.25$, about 22%–25% satisfy $|z_j^h| < 0.5$, and median $|z_j^h|$ is around 1.0–1.22. Hence, unlike reduced-form coefficients, stochastic-volatility states differ materially across identification schemes in standardized terms.

The combination of Tables 13 and 14 is economically interpretable rather than contradictory. Reduced-form coefficients describe average dynamic propagation in the observation equation and are estimated very similarly under CHOL and OINV, which is why their standardized gaps are small and stable. By contrast, the h processes are latent states and therefore identification-sensitive decomposition objects: changing contemporaneous restrictions (triangular under CHOL versus unrestricted under OINV, with post-labelling) can reallocate second-moment variation between the impact matrix and volatility states, even when reduced-form dynamics remain close. In that sense, large $|z_j^h|$ values indicate materially different latent variance decomposition across identification schemes, not necessarily large discrepancies in observables or forecasting accuracy.

The results are therefore nuanced but coherent. Prior perturbations leave reduced-form coefficient summaries in Table 12 and coefficient-distance diagnostics in Table 13 broadly unchanged, while structural summaries and latent volatility decomposition respond in expected identification-specific ways. Taken together with Section 6, the evidence suggests that identification affects structural interpretation and latent variance attribution more than it affects aggregate predictive performance metrics.

Taken together, these diagnostics support the interpretation that the main CHOL–OINV differences documented in this paper are primarily identification-driven, not an artifact of a particular prior calibration.

8 A Counterfactual Application for 2025 Tail Risk

An important practical advantage of VAR-based frameworks is that they can be used not only for historical decomposition and forecasting, but also for scenario analysis through conditional counterfactual paths. This flexibility is particularly useful in the present nonlinear SV-BVAR setting, because it allows us to assess whether responses at the center of the predictive distribution differ from responses in the lower tail. In policy terms, this means we can evaluate not only what happens to expected growth, but also how downside risk evolves under alternative financial conditions.

This application is motivated by one of the central empirical puzzles discussed in this paper: over 2025, policy uncertainty and financial stress did not move together in a persistent way. Following the April 2025 policy-uncertainty shock ("Liberation Day"), financial stress increased only briefly and retraced soon after. We therefore ask a counterfactual question: how would UK GDP tail risk in 2025 have evolved if financial volatility had remained substantially higher after the April shock?

Methodologically, we implement conditional forecasts from end-March 2025 onward and keep the realized paths of *INF*, *FCI*, and *EPU* fixed over April–December 2025. We then vary only the path of *FSI* and compare the resulting GDP predictive distributions. The outcomes of interest are the GDP median and GDP-at-Risk, measured as the 5th percentile of the predictive distribution. This setup isolates the effect of alternative financial-stress trajectories while holding the remaining conditioning information unchanged. We consider three alternative *FSI* paths over April–December 2025. In the first, *FSI* rises to the April 2025 level (0.18) and, instead of normalizing from May onward, remains at that elevated level for the rest of the year. In the second, more severe scenario, *FSI* rises to the October 2022 gilt-crisis level (approximately 0.63–0.64) and remains there throughout the period. In the third, intermediate scenario, *FSI* rises to 0.64 in April and then declines linearly to 0.18 by December.

Table 15 reports the quantitative comparison at a single policy-relevant horizon, December 2025 (nine months ahead), relative to the baseline conditional forecast with realized *FSI*. Figures 14–15 provide the visual counterpart. Under baseline conditioning, the December GDP median is 1.1769 and the 5th percentile is 0.0663. In the *FSI* = 0.18 flat scenario, the median declines to 1.1242 ($\Delta = -0.0527$), while the 5th percentile declines to -0.0047 ($\Delta = -0.0710$). In the intermediate *FSI* : 0.64 $\rightarrow$ 0.18 scenario, the median is 1.0342 ($\Delta = -0.1427$) and the 5th percentile is -0.2006 ($\Delta = -0.2669$). In the severe *FSI* = 0.64 flat scenario, the median is 1.0007 ($\Delta = -0.1762$), while the 5th percentile falls to -0.3331 ($\Delta = -0.3994$). In all three cases, the absolute deterioration in the 5th percentile is larger than the median deterioration, with implied tail-to-median delta ratios of about 1.35, 1.87, and 2.27, respectively. Quantitatively, this confirms that

tighter financial conditions primarily amplify downside tail risk rather than producing a proportionate shift in central GDP outcomes.

Table 15: Quantitative GDP effects at December 2025 (9-month horizon)

Scenario	GDP median	GDP 5th pct	Δ median vs base	Δ 5th pct vs base	$ \Delta \text{ 5th pct} / \Delta \text{ median} $
Baseline (realized <i>FSI</i>)	1.1769	0.0663	–	–	–
<i>FSI</i> = 0.18 flat	1.1242	-0.0047	-0.0527	-0.0710	1.35
<i>FSI</i> : 0.64 $\rightarrow$ 0.18	1.0342	-0.2006	-0.1427	-0.2669	1.87
<i>FSI</i> = 0.64 flat	1.0007	-0.3331	-0.1762	-0.3994	2.27

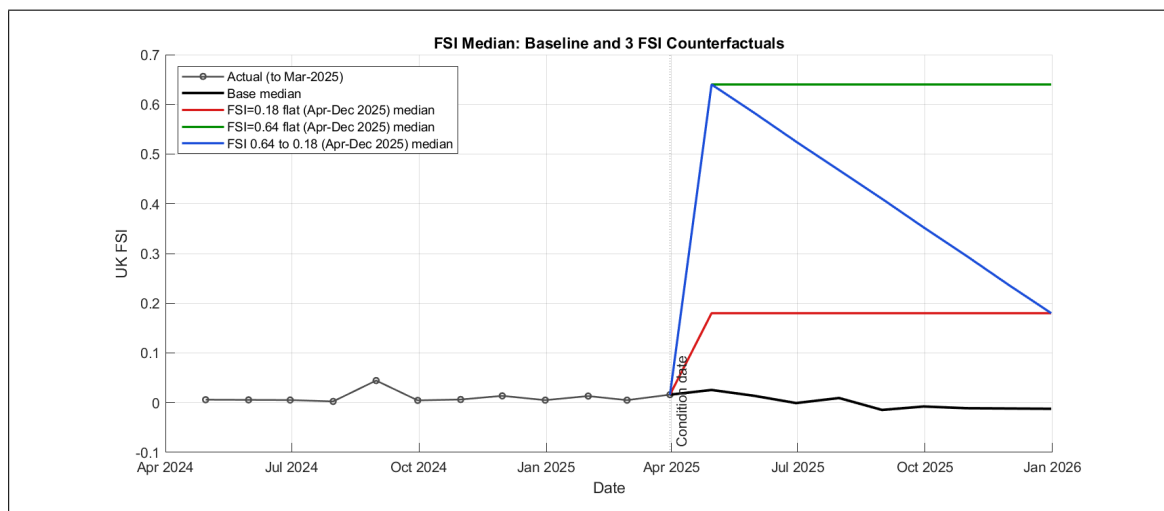


Figure 14: Alternative *FSI* conditioning paths used in the 2025 counterfactual exercise: 0.18 flat, 0.63–0.64 flat, and 0.64 declining linearly to 0.18 by December.

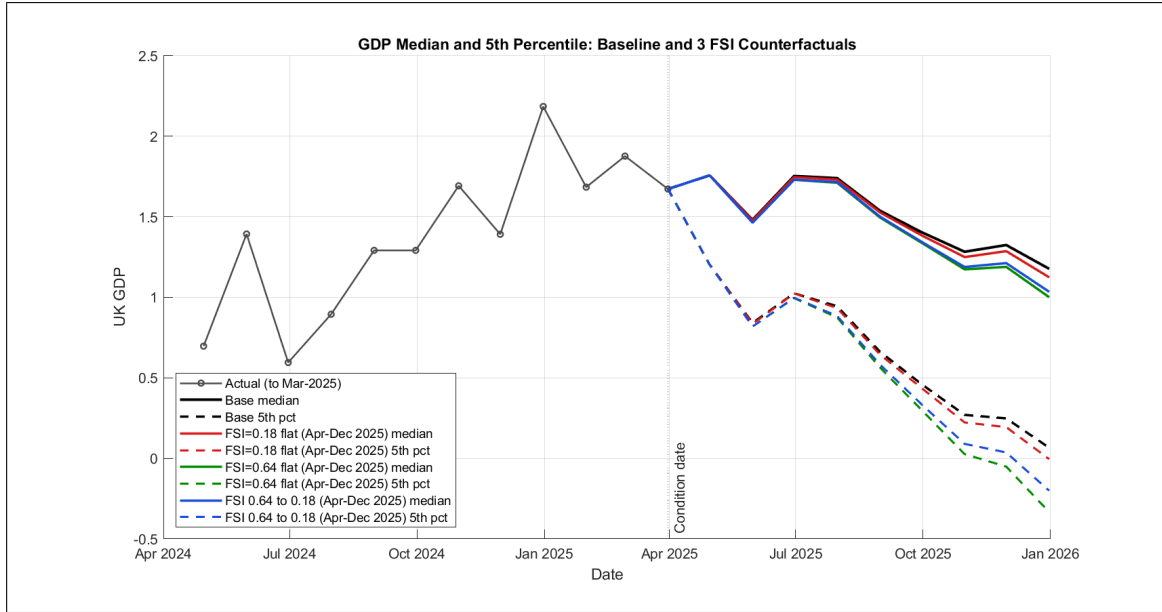


Figure 15: Counterfactual GDP forecasts from end-March 2025 under alternative financial-stress paths. The figure compares baseline conditioning with three *FSI* scenarios and reports both the median and 5th percentile (GDP-at-Risk).

The policy relevance of this exercise is direct. If financial stress had remained elevated after the April 2025 uncertainty shock, downside GDP risk by year-end would have been materially worse than suggested by central projections alone. This implies that risk monitoring frameworks should place explicit weight on tail metrics, not only on median forecasts, because financial conditions can disproportionately worsen left-tail outcomes even when central activity appears comparatively resilient.

9 Conclusions

This paper has studied how policy uncertainty and financial market stress jointly shape UK macroeconomic tail risks, using a Bayesian VAR with stochastic volatility and contrasting recursive Cholesky identification with the order-invariant approach of [Chan et al. \(2024\)](#). The central finding is that state dependence is asymmetric: financial stress shocks are substantially more recessionary when policy uncertainty is high, but uncertainty shocks are not systematically more contractionary in high-financial-stress regimes. Under order-invariant identification, the cumulative GDP contraction from a financial-stress shock roughly triples when moving from a low- to a high-uncertainty regime, while all three alternative Cholesky orderings compress this amplification by at least an order of magnitude. The result is robust to filtering out weakly identified posterior draws. Replacing policy uncertainty with geopolitical risk causes the amplification channel to disappear, confirming that the mechanism reflects genuine

policy-uncertainty feedback rather than a generic uncertainty effect.

Two complementary pieces of evidence support this structural interpretation. First, a historical narrative check across three benchmark episodes - Brexit, COVID onset, and the September 2022 gilt crisis - shows that the order-invariant specification matches all three episode priors, whereas recursive orderings are sensitive to variable arrangement, especially for Brexit and the gilt crisis. The narrative evidence is supportive but not decisive: it corroborates rather than independently establishes the identification ranking. Second, an out-of-sample forecasting experiment demonstrates that the order-invariant model performs comparably to the Cholesky-identified model in predicting GDP downside risk: quantile scores, coverage rates, RMSE, and CRPS all show small, metric-dependent differences. Identification choice therefore matters substantially more for structural interpretation than for aggregate predictive accuracy.

The counterfactual application in Section 8 translates the structural findings into a policy-relevant scenario framework. Under persistent gilt-crisis-level financial stress throughout 2025, the 5th percentile of GDP deteriorates by approximately 0.40 percentage points relative to baseline, compared with only 0.18 percentage points for the median, a tail-to-median amplification ratio of about 2.3. This confirms that financial-stability deterioration can be materially understated when policy analysis focuses on central projections rather than on tail risk metrics. More broadly, the results underscore that policymakers should monitor uncertainty and financial stress jointly, not in isolation: stress-testing and risk-assessment frameworks that ignore their interaction risk underestimating the severity of adverse macro-financial scenarios.

In summary, order-invariant SV-BVARs uncover an economically meaningful amplification channel - from financial stress through elevated uncertainty to GDP downside risk - that recursive identification compresses. By relaxing restrictive ordering assumptions while retaining a transparent labelling strategy, the approach provides richer structural insights that are directly relevant for financial-stability assessment and macroprudential policy.

References

- Adrian, T. et al. (2019). Vulnerable growth. *American Economic Review*, 109(4):1263–1289.
- Alessandri, P. and Mumtaz, H. (2019). Financial uncertainty and macroeconomic activity. *Journal of Monetary Economics*, 101:31–46.
- Angelini, E. et al. (2019). Uncertainty shocks and the business cycle: Evidence from a break in volatility. *Journal of Economic Dynamics and Control*, 104:103–120.
- Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *The quarterly journal of economics*, 131(4):1593–1636.
- Bertsche, D. and Braun, R. (2022). Identification of structural vector autoregressions by stochastic volatility. *Journal of Business & Economic Statistics*, 40(1):328–341.
- Brianti, M. (2020). Uncertainty shocks and the macroeconomy: Cash holdings as an identification device. *Journal of Monetary Economics*, 114:20–36.
- Caballero, R. J., Caravello, T., and Simsek, A. (2024). Financial conditions targeting. *NBER Working Paper*, (w33206).
- Caballero, R. J. and Simsek, A. (2025). Fci-plot: Central bank communication through financial conditions. Technical report, National Bureau of Economic Research.
- Caldara, D. et al. (2016). The economic effects of uncertainty shocks: Evidence from a svar. *American Economic Journal: Macroeconomics*, 8(4):1–39.
- Caldara, D. et al. (2021a). Mean-volatility interaction in sv-vars. *Journal of Monetary Economics*, 101:31–46.
- Caldara, D. et al. (2021b). Tail risk regimes in macroeconomic forecasting. *Journal of Applied Econometrics*, 36(7):857–876.
- Carriero, A., Clark, T. E., and Marcellino, M. (2024). Capturing macro-economic tail risks with bayesian vector autoregressions. *Journal of Money, Credit and Banking*, 56(5):1099–1127.
- Castelnuovo, E. and Mori, A. (2022). Mixed-frequency models and uncertainty. *International Journal of Forecasting*, 38(2):456–470.
- Chan, J. C., Koop, G., and Yu, X. (2024). Large order-invariant bayesian vars with stochastic volatility. *Journal of Business & Economic Statistics*, 42(2):825–837.

- Chavleishvili, S. and Kremer, M. (2025). Ciss of death: measuring financial crises in real time. *Review of Finance*, 29(3):685–710.
- Choi, S. and Yoon, C. (2022). Uncertainty shocks and financial conditions: Evidence from a century of us data. *Journal of Financial Economics*, 146(2):456–478.
- Cogley, T. and Sargent, T. J. (2005). Drifts and volatilities: Monetary policies and outcomes in the post WWII U.S. *Review of Economic Dynamics*, 8(2):262–302.
- Davidson, S. N., Hou, C., and Koop, G. (2025). Investigating economic uncertainty using stochastic volatility in mean vars: the importance of model size, order-invariance and classification. *Journal of Business & Economic Statistics*, 43(4):992–1007.
- De Santis, R. and Van Der Veken, W. (2021). Financial and uncertainty shocks: Effects on output and prices. *European Economic Review*, 134:103–120.
- Ellington, M. et al. (2023). Uncertainty shocks and financial markets: Evidence from the uk. *Economic Journal*, 133(650):1234–1256.
- Furlanetto, F. et al. (2019). Disentangling credit and uncertainty shocks in the us economy. *Journal of Monetary Economics*, 101:42–58.
- Hengge, M. (2019). Macroeconomic uncertainty and the tails of growth. *Journal of Economic Dynamics and Control*, 104:103–120.
- Jovanovic, B. and Ma, Y. (2022). Uncertainty and the distribution of future growth. *Journal of Economic Dynamics and Control*, 127:104–120.
- Karlsson, S. et al. (2023). Skewness in stochastic volatility bvars. *Journal of Applied Econometrics*, 38(2):456–470.
- Kim, S., Shephard, N., and Chib, S. (1998). Stochastic volatility: likelihood inference and comparison with arch models. *The review of economic studies*, 65(3):361–393.
- Kiss, G. et al. (2024). Hyperbolic skew student t innovations in sv-bvars. *Journal of Business and Economic Statistics*, 42(2):825–837.
- Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in non-linear multivariate models. *Journal of econometrics*, 74(1):119–147.
- Lewis, D. J. (2021). A survey of identification and labelling strategies in structural vars. *Journal of Economic Surveys*, 35(5):1234–1260.
- Loria, F. et al. (2024). Quantile regression and tail risk. *Journal of Applied Econometrics*, 39(1):1–25.

- Ludvigson, S. C. et al. (2021). Shock-based identification of the effects of uncertainty shocks. *American Economic Review*, 111(3):903–947.
- Meinen, P. and Röhe, O. (2018). Economic uncertainty and inflation: Evidence from svar models with sign restrictions. *Journal of International Money and Finance*, 87:1–15.
- Pastor, L. and Veronesi, P. (2017). Political uncertainty and risk premia. *Journal of Financial Economics*, 125(3):365–393.
- Plagborg-Møller, M. et al. (2020). Tail risk in macroeconomic forecasting. *Review of Economics and Statistics*, 102(4):699–714.
- Primiceri, G. E. (2005). Time varying structural vector autoregressions and monetary policy. *The Review of economic studies*, 72(3):821–852.
- Pruser, C. and Schlosser, A. (2017). Cross-country heterogeneity in the effects of economic policy uncertainty shocks. *Journal of International Money and Finance*, 70:1–25.
- Rigobon, R. (2003). Identification through heteroskedasticity. *Review of Economics and Statistics*, 85(4):777–792.

Appendix

A1 Gibbs Sampling Algorithm

Posterior inference is obtained through a Gibbs sampling algorithm that iterates over four blocks of parameters:

1. **VAR coefficients.** Conditional on the current draw of B_0 and the stochastic volatility path, the VAR coefficients are sampled from their Gaussian posterior distribution. Under Cholesky identification, the coefficients are drawn jointly from the system form of the VAR: the lower-triangular impact matrix is embedded across all time periods, and the likelihood is weighted by time-varying precisions $\exp(-h_{it})$. This yields a Gaussian posterior with precision matrix equal to prior precision plus an SV-weighted data term, and the draw is obtained using a Cholesky factorization of that precision matrix. Under the order-invariant approach, coefficients are sampled equation-by-equation after the transformed-system step described in Appendix A2.
2. **Stochastic volatility.** Given structural shocks ε_t , the log-volatility paths h_{it} are sampled using the auxiliary mixture sampler Kim et al. (1998), which first linearizes the state-space by squaring and taking the logarithm of the original system. A seven component Gaussian mixture is then used to approximate the resulting $\log\chi^2$ distribution of the transformed error term.
3. **Identification parameters.** Under Cholesky identification, the free elements of the lower-triangular impact matrix are stacked in a vector a and sampled from a Gaussian conditional posterior. The likelihood contribution is constructed from reduced-form residuals using a stacked design matrix that maps each free element of a into the structural system, again weighted by the stochastic-volatility precisions $\exp(-h_{it})$. The resulting posterior precision is the sum of prior precision and the SV-weighted quadratic form from this design matrix, and sampling is implemented via Cholesky factorization. Under the order-invariant scheme, rows of B_0 are drawn sequentially with orthogonality constraints, as detailed in Appendix A3.
4. **Volatility innovation variances.** The innovation variances of the stochastic volatility process $\sigma_{h,i}^2$ are sampled from their inverse-gamma conditional posterior distribution.

After discarding the burn-in period, the retained posterior draws are used to compute impulse responses, predictive distributions, and tail-risk measures.

A2 Conditional Sampling of VAR Coefficients under OINV Identification

This section describes the conditional Gibbs step used to sample the VAR coefficients under order-invariant (OINV) identification in the presence of stochastic volatility and contemporaneously correlated shocks.

Consider the reduced-form VAR

$$Y_t = X_t A + U_t, \quad t = 1, \dots, T, \quad (20)$$

where $Y_t \in \mathbb{R}^n$, $X_t \in \mathbb{R}^k$, and $A \in \mathbb{R}^{k \times n}$. Reduced-form residuals satisfy

$$U_t = B_0^{-1} \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}\left(0, \text{diag}\left(e^{h_t}\right)\right), \quad (21)$$

with B_0 an unrestricted contemporaneous impact matrix and $h_t = (h_{t1}, \dots, h_{tn})'$ stochastic log-volatilities.

Let a_i denote column i of A , and let e_i be the i -th canonical basis vector in $\mathbb{R}^n$. The coefficient matrix admits the decomposition

$$A = \sum_{j=1}^n a_j e_j', \quad (22)$$

so that

$$XA = \sum_{j=1}^n X a_j e_j'. \quad (23)$$

Conditioning on all coefficients except a_i yields

$$Y = XA_{-i} + Xa_i e_i' + U, \quad (24)$$

where $A_{-i} = \sum_{j \neq i} a_j e_j'$. Define

$$U^{(-i)} \equiv Y - XA_{-i} = Xa_i e_i' + U. \quad (25)$$

This isolates a linear regression in a_i , but with residuals that are contemporaneously correlated and heteroskedastic.

Premultiplication by B_0' transforms reduced-form residuals into structural shocks,

$$U_t B_0' = \varepsilon_t', \quad (26)$$

eliminating cross-equation correlation. Let

$$\Lambda = \text{diag}\left(\text{vec}\left(e^{h/2}\right)\right), \quad (27)$$

where $\text{vec}(\cdot)$ stacks all time-equation pairs. Dividing by Λ standardizes the innovations to unit variance.

Applying both transformations gives

$$\Lambda^{-1} \text{vec}\left(U^{(-i)} B_0'\right) = \Lambda^{-1} (B_0(:, i) \otimes X) a_i + \eta, \quad \eta \sim \mathcal{N}(0, I). \quad (28)$$

Define

$$y_i \equiv \Lambda^{-1} \text{vec}\left(U^{(-i)} B_0'\right), \quad (29)$$

$$W_i \equiv \Lambda^{-1} (B_0(:, i) \otimes X). \quad (30)$$

The conditional likelihood is therefore

$$y_i = W_i a_i + \eta, \quad \eta \sim \mathcal{N}(0, I). \quad (31)$$

Assuming a Gaussian prior

$$a_i \sim \mathcal{N}(0, V_\beta), \quad (32)$$

the conditional posterior is conjugate:

$$a_i \mid \cdot \sim \mathcal{N}\left(K_i^{-1} W_i' y_i, K_i^{-1}\right), \quad K_i = W_i' W_i + V_\beta^{-1}. \quad (33)$$

A Gibbs sweep samples A equation by equation. For each $i = 1, \dots, n$, construct $U^{(-i)}$ by setting $a_i = 0$, rotate by B_0' , standardize by Λ , and draw a_i from its Gaussian conditional posterior. This yields an exact draw from the joint posterior of A conditional on (B_0, h) .

A3 Conditional Sampling of Structural Parameters under OINV Identification

This section describes the conditional sampling of the contemporaneous impact matrix B_0 under order-invariant identification. The key restriction is that rows of B_0 are drawn sequentially and must be orthogonal to all previously drawn rows.

Let ϕ_i denote row i of B_0 . Order-invariance requires

$$\phi_i \phi_j' = 0 \quad \text{for } j < i. \quad (34)$$

At the same time, ϕ_i must follow its Gaussian posterior distribution implied by the likelihood and prior.

Ignoring orthogonality constraints, suppose the conditional posterior for ϕ_i is

$$\phi_i \sim \mathcal{N}\left(m_i, \frac{1}{T}\Sigma_i\right). \quad (35)$$

Let C_i be the Cholesky factor of the precision matrix,

$$\Sigma_i^{-1} = C_i' C_i. \quad (36)$$

Define the transformed variable

$$\tilde{\phi}_i = C_i \phi_i, \quad (37)$$

so that

$$\tilde{\phi}_i \sim \mathcal{N}\left(C_i m_i, \frac{1}{T}I\right). \quad (38)$$

This rotation renders the posterior spherical, simplifying the imposition of linear constraints.

Let

$$\Gamma_{-i} = \begin{bmatrix} \phi_1 \\ \vdots \\ \phi_{i-1} \end{bmatrix}. \quad (39)$$

Orthogonality requires

$$\Gamma_{-i} \phi_i = 0. \quad (40)$$

Equivalently, ϕ_i must lie in the null space of Γ_{-i} , so that

$$\phi_i = \Gamma_{-i}^\perp \alpha, \quad (41)$$

where the columns of $\Gamma_{-i}^\perp$ form a basis for $\text{null}(\Gamma_{-i})$.

Transforming this restriction into the spherical Gaussian space yields

$$\tilde{\phi}_i = C_i \Gamma_{-i}^\perp \alpha. \quad (42)$$

Let v_1 be a unit vector spanning the image of this null space,

$$v_1 = \frac{C_i \Gamma_{-i}^\perp}{\|C_i \Gamma_{-i}^\perp\|}, \quad (43)$$

and complete it to an orthonormal basis

$$V = [v_1 \ v_2], \quad V'V = I. \quad (44)$$

Express the transformed variable as

$$\tilde{\phi}_i = V\zeta, \quad (45)$$

where ζ collects coordinates along constrained and unconstrained directions. The posterior mean in this basis is

$$\hat{\zeta}_j = v_j' C_i m_i. \quad (46)$$

Sampling proceeds as follows. Along the constrained direction v_1 , draw

$$\zeta_1 \sim \mathcal{N}(\hat{\zeta}_1, 1/T), \quad (47)$$

which ensures orthogonality. Along the remaining directions,

$$\zeta_j = \hat{\zeta}_j + \frac{1}{\sqrt{T}} \varepsilon_j, \quad \varepsilon_j \sim \mathcal{N}(0, 1), \quad j > 1. \quad (48)$$

Mapping back to the original parameterization gives

$$\phi_i = C_i^{-T} V \zeta. \quad (49)$$

Finally, a sign normalization

$$\phi_i \leftarrow \phi_i \cdot \text{sign}(\phi_{ii}) \quad (50)$$

ensures a positive diagonal element.

This procedure produces exact draws from the Gaussian posterior of B_0 conditional on previously drawn rows, while enforcing the order-invariant orthogonality restrictions.

A4 Computation of GIRFs

This appendix documents both the date-conditioned GIRFs and the regime-conditioned GIRFs used in the second set of charts.

For each posterior draw s , GIRFs are obtained by simulating two paths over horizon H : a baseline path and a shocked path. The GIRF at horizon h is their difference:

$$GIRF_s(h) = Y_{t+h}^{(\text{shock})} - Y_{t+h}^{(\text{baseline})}.$$

In both paths, the same sequences of structural innovations and volatility innovations are used. This common-random-numbers design isolates the causal effect of the deterministic shock.

Shock Normalization and Volatility Conditioning. In each simulation, the deterministic component of the structural shock is injected only at $t = 1$ in the shocked path. The algorithm rescales that deterministic component so that the impact on the shocked variable has the desired magnitude in economically interpretable units. Let j denote the shocked variable, let B_0 be the contemporaneous matrix, and let

$$\kappa_j = (B_0^{-1})_{jj}$$

be the own-impact coefficient of structural shock j on variable j . If τ_j denotes the target impact, the deterministic structural shock is set to

$$\tilde{\varepsilon}_{j,1} = \frac{\tau_j}{\kappa_j}.$$

Hence the first-period impact on variable j is normalized to τ_j by construction. The two GIRF exercises differ in how τ_j is chosen:

1. *Date-conditioned GIRFs.* The simulation function is called with `shock_target` set equal to $\sigma_{j,1}$ at the end-of-sample conditioning date t^* . Therefore:

$$\tau_j(s) = \sigma_{j,1}(t^*, s),$$

where $\sigma_{j,1}(t^*, s)$ is the first-horizon standard deviation of shocked variable j in the date-specific simulated volatility path for posterior draw s . In this exercise, conditioning comes from a single initialization state (last p observations and initial latent volatility), and shock normalization is tied to that date's volatility.

2. *Regime-conditioned GIRFs.* The simulation function is called with `shock_target`

set equal to $\sigma_{j,1}$ for each conditioning date t in a regime set R . Therefore

$$\tau_j(t, s) = \sigma_{j,1}(t, s),$$

so deterministic shock size co-moves with local volatility conditions for each conditioning date t and posterior draw s , and the resulting date-specific GIRFs are then averaged across all $t \in R$.

This distinction is deliberate. Both exercises are conditioned GIRFs because both condition on the information set at the chosen date(s), including recent lags and the initial latent volatility state, and both use volatility-based shock normalization. The difference is that date-conditioned GIRFs apply this normalization at one date only, whereas regime-conditioned GIRFs apply it date-by-date over a set of dates and then average across that set.

Date-Conditioned GIRFs (End-of-Sample State). For the first set of GIRFs, conditioning uses the end-of-sample state. The simulation is initialized from the last p observed values of the data and from the last estimated volatility state. Future log-volatilities are simulated via the random-walk law of motion, transformed to standard deviations, and used to scale structural shocks along the simulation horizon.

Regime-Conditioned GIRFs (Average Across Dates). To compute regime-conditioned GIRFs we first define regimes. Regimes are defined on valid conditioning dates ($t = p + 1, \dots, T$) using threshold splits:

$$\mathcal{H}^{FSI} = \{t : FSI_t \geq q_{0.80}(FSI)\}, \quad \mathcal{L}^{FSI} = \{t : FSI_t \leq q_{0.20}(FSI)\},$$

$$\mathcal{H}^{EPU} = \{t : EPU_t \geq q_{0.80}(EPU)\}, \quad \mathcal{L}^{EPU} = \{t : EPU_t \leq q_{0.20}(EPU)\}.$$

For the median-state analysis, I additionally define

$$\mathcal{M}^{FSI} = \{t : q_{0.40}(FSI) \leq FSI_t \leq q_{0.60}(FSI)\}, \quad \mathcal{M}^{EPU} = \{t : q_{0.40}(EPU) \leq EPU_t \leq q_{0.60}(EPU)\}.$$

For each posterior draw s , each shock j , and each conditioning date t in a regime set:

1. Initialize from date-specific conditions: $Y_{t-p+1:t}$ and the aligned volatility state used by the simulation code.
2. Simulate a future volatility path and corresponding standard deviations.
3. Simulate baseline and shocked paths using identical random innovations.

4. Compute date-specific GIRF differences and then average across all dates in the regime for that draw and shock.

Formally, for regime R :

$$\overline{GIRF}_{s,j}^R(h) = \frac{1}{|R|} \sum_{t \in R} \left(Y_{t+h,s,j}^{(\text{shock})} - Y_{t+h,s,j}^{(\text{baseline})} \right).$$

Reported regime-conditioned GIRFs (median and tail bands) are obtained from the distribution of $\overline{GIRF}_{s,j}^R(h)$ across posterior draws. This yields six state-dependent objects for each shock-response pair: high/median/low FSI and high/median/low EPU.

A5 Date-Conditioned GIRFs (Additional Shocks)

Figures 1 and 2 are reported in Section 4. This appendix reports the remaining date-conditioned GIRFs.

Figure 16: Date-conditioned generalized impulse response functions: Inflation shock

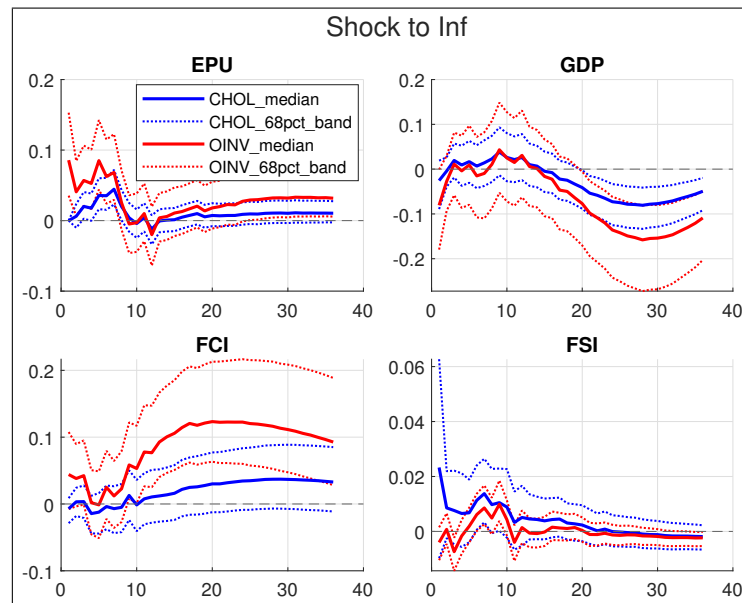


Figure 17: Date-conditioned generalized impulse response functions: GDP shock

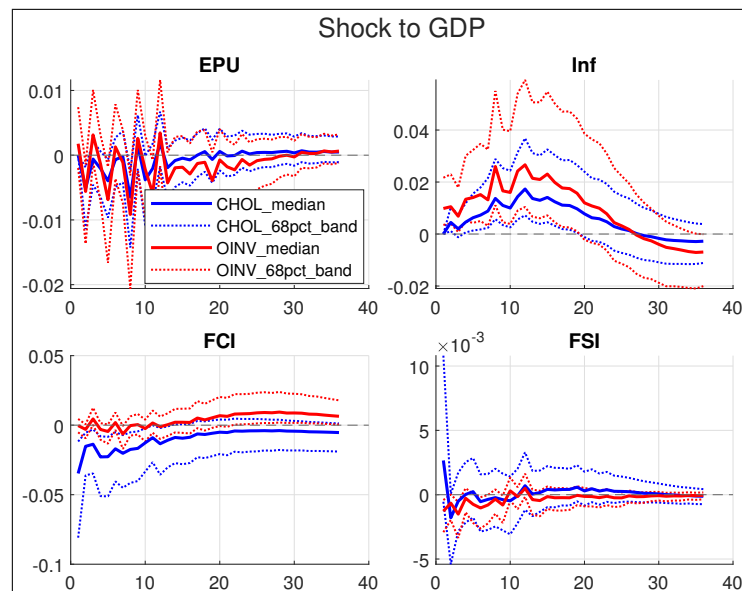
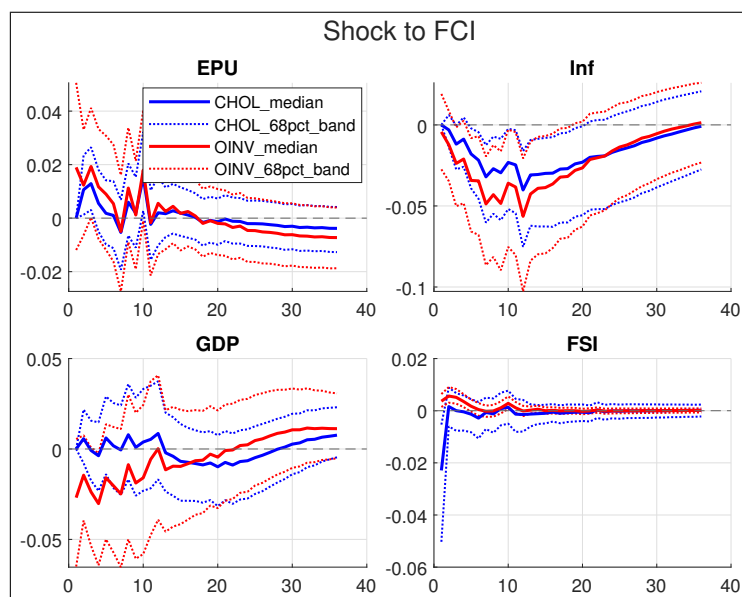


Figure 18: Date-conditioned generalized impulse response functions: Financial Conditions Index (FCI) shock



A6 Regime-Conditioned GIRFs by FSI Regime (Additional Shocks)

Figure 3 is reported in Section 4. This appendix reports the remaining shocks conditional on FSI regimes.

Figure 19: Regime-conditioned GIRFs conditional on FSI regime: Inflation shock

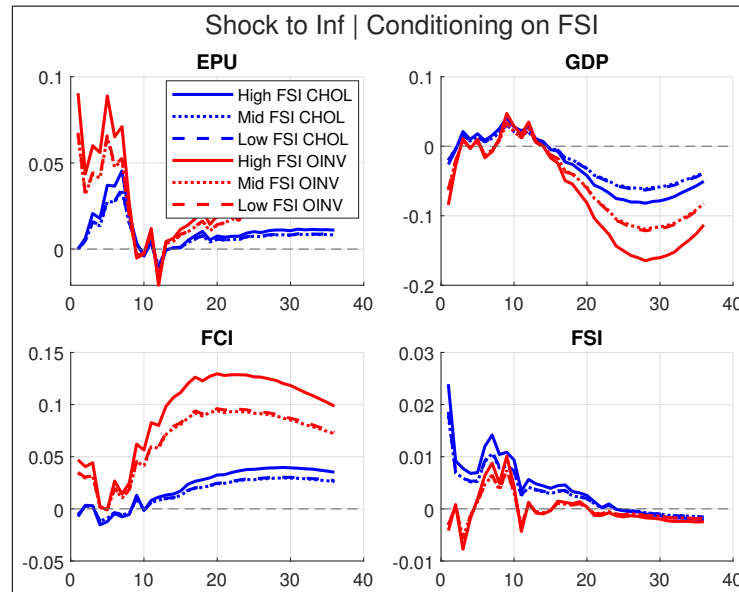


Figure 20: Regime-conditioned GIRFs conditional on FSI regime: GDP shock

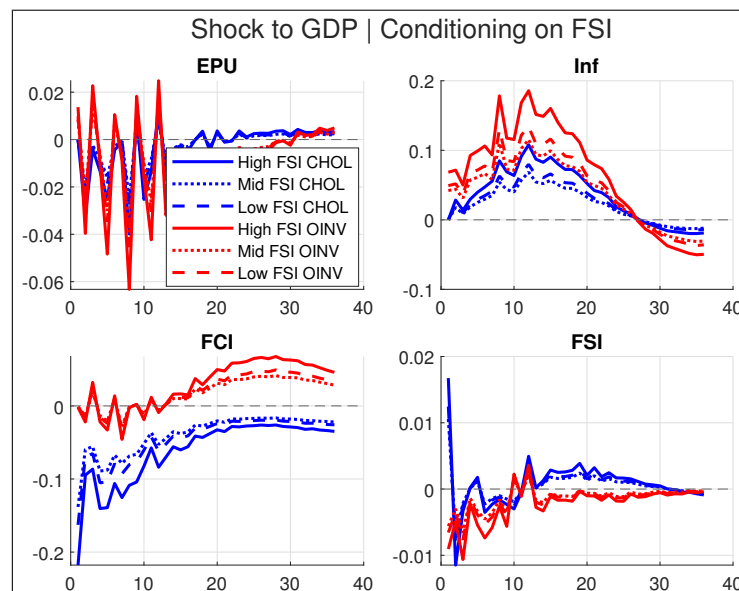


Figure 21: Regime-conditioned GIRFs conditional on FSI regime: Financial Conditions Index (FCI) shock

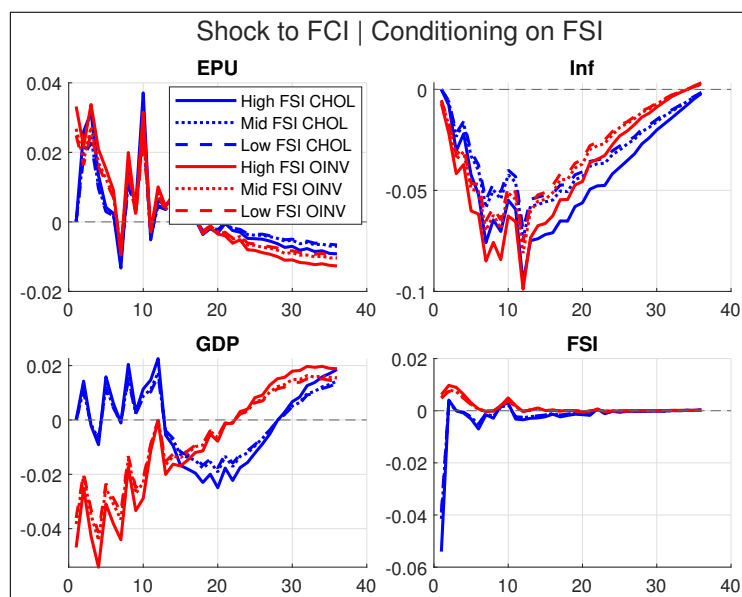
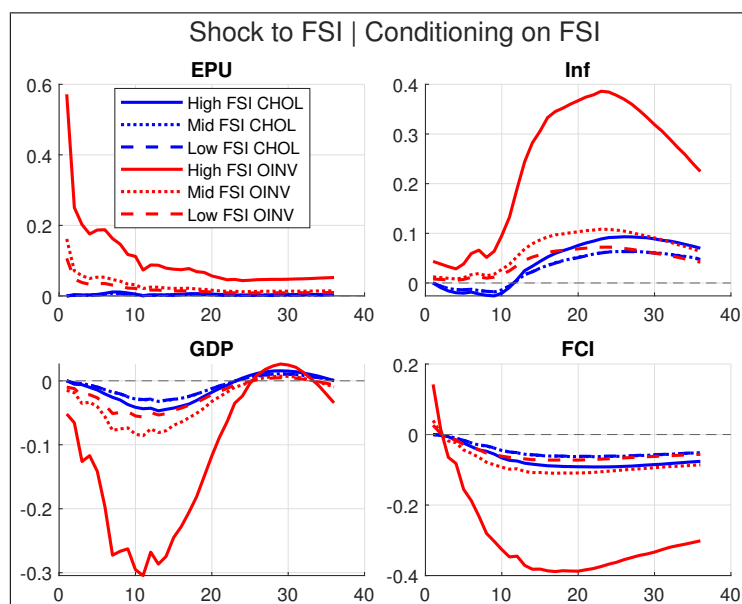


Figure 22: Regime-conditioned GIRFs conditional on FSI regime: Financial Stress Index (FSI) shock



A7 Regime-Conditioned GIRFs by EPU Regime (Additional Shocks)

Figure 4 is reported in Section 4. This appendix reports the remaining shocks conditional on EPU regimes.

Figure 23: Regime-conditioned GIRFs conditional on EPU regime: EPU shock

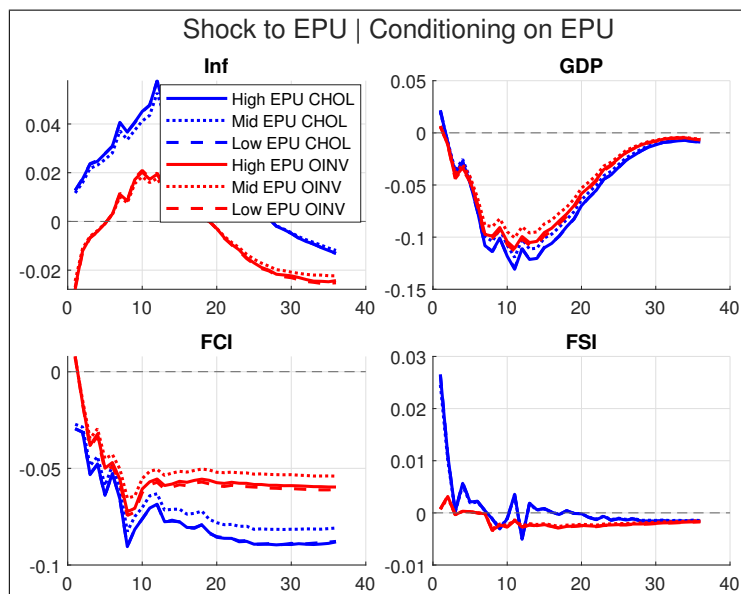


Figure 24: Regime-conditioned GIRFs conditional on EPU regime: Inflation shock

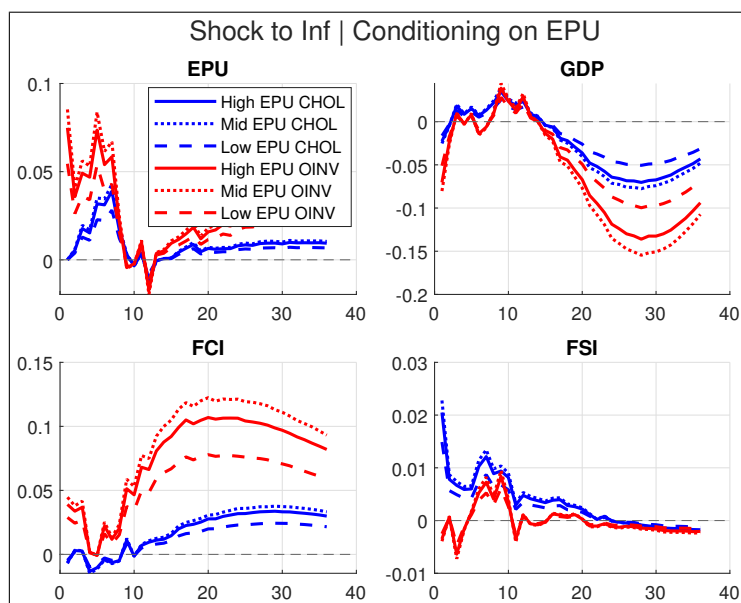


Figure 25: Regime-conditioned GIRFs conditional on EPU regime: GDP shock

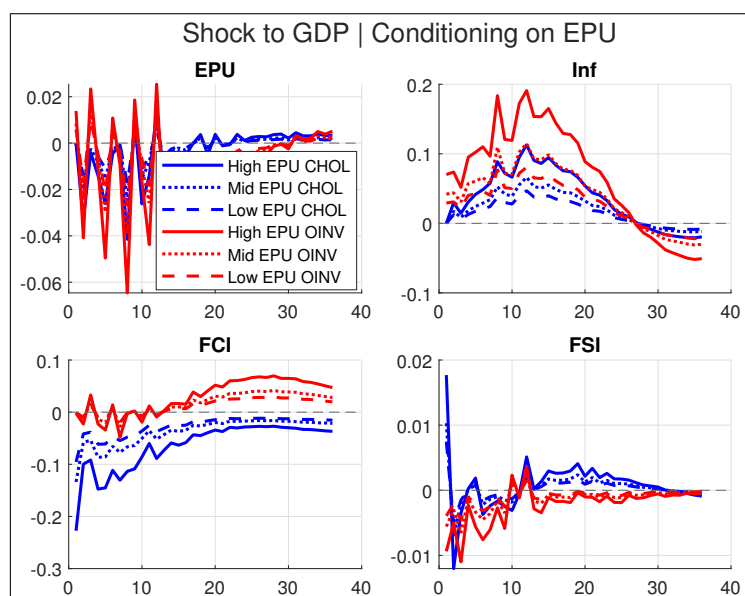
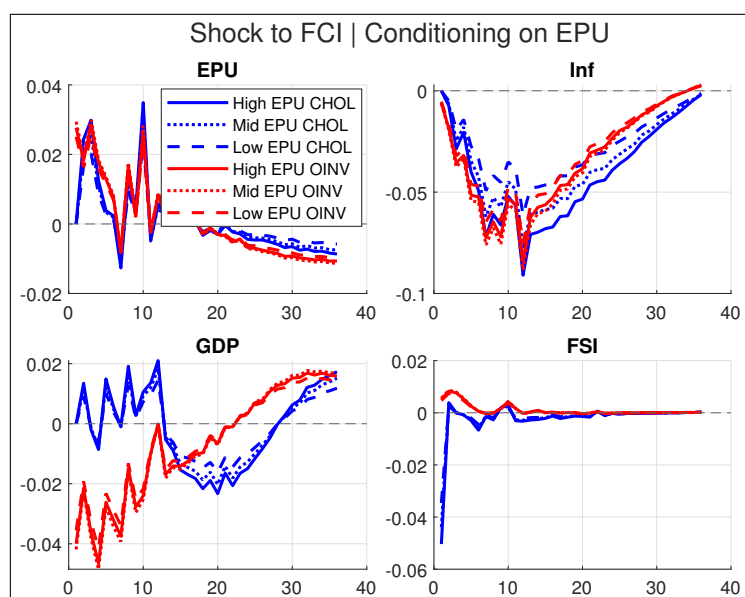


Figure 26: Regime-conditioned GIRFs conditional on EPU regime: Financial Conditions Index (FCI) shock



A8 High-Minus-Low GIRF Differences (Conditioning on FSI)

This appendix reports the high-minus-low regime differences for regime-conditioned GIRFs when conditioning on FSI (high FSI minus low FSI), with 68% credible bands.

Figure 27: High-minus-low GIRF difference conditional on FSI regime: EPU shock

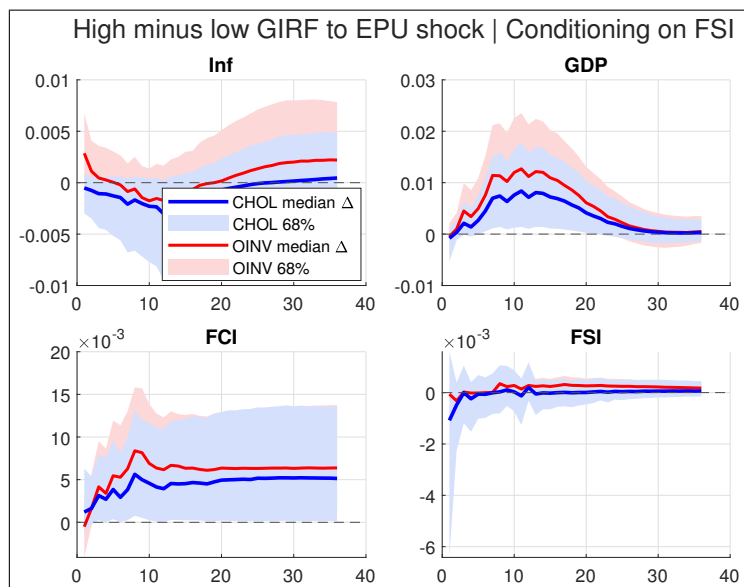


Figure 28: High-minus-low GIRF difference conditional on FSI regime: Inflation shock

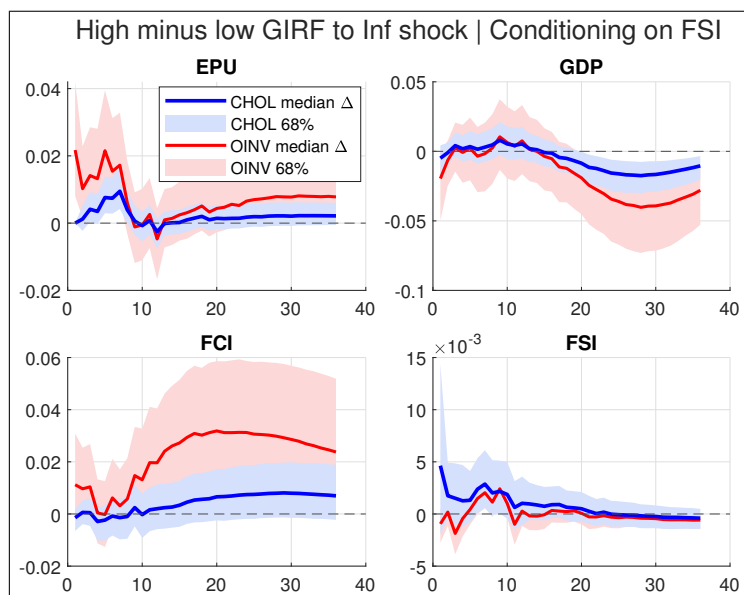


Figure 29: High-minus-low GIRF difference conditional on FSI regime: GDP shock

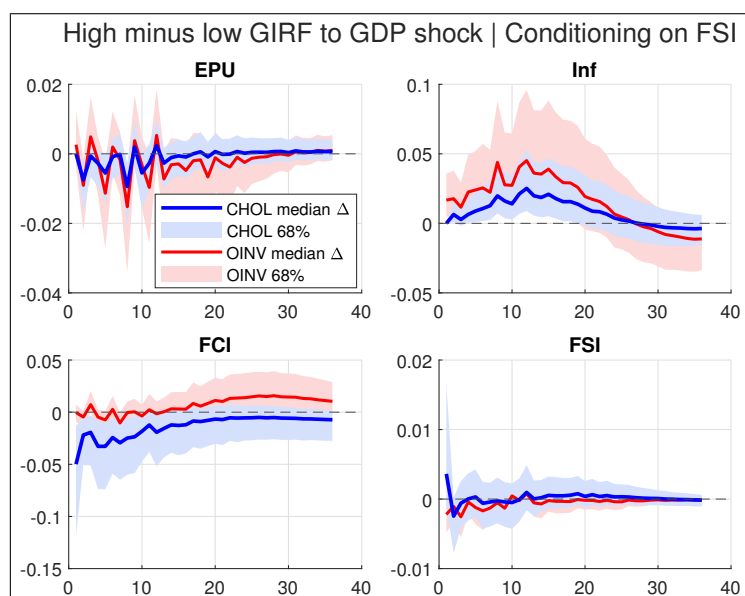


Figure 30: High-minus-low GIRF difference conditional on FSI regime: Financial Conditions Index (FCI) shock

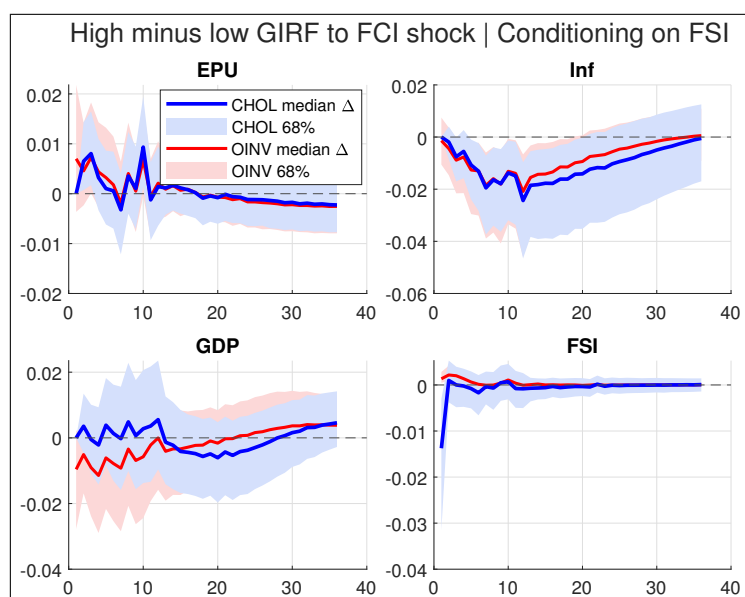
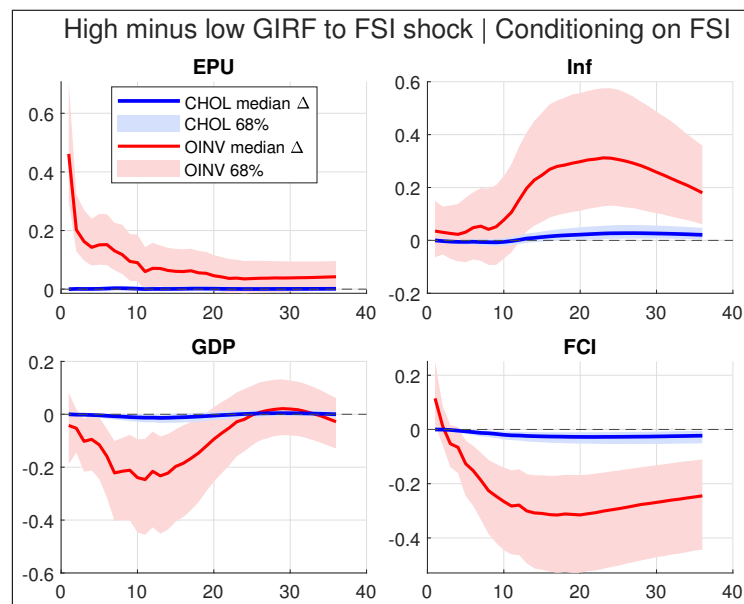


Figure 31: High-minus-low GIRF difference conditional on FSI regime: Financial Stress Index (FSI) shock



A9 High-Minus-Low GIRF Differences (Conditioning on EPU)

This appendix reports the high-minus-low regime differences for regime-conditioned GIRFs when conditioning on EPU (high EPU minus low EPU), with 68% credible bands.

Figure 32: High-minus-low GIRF difference conditional on EPU regime: EPU shock

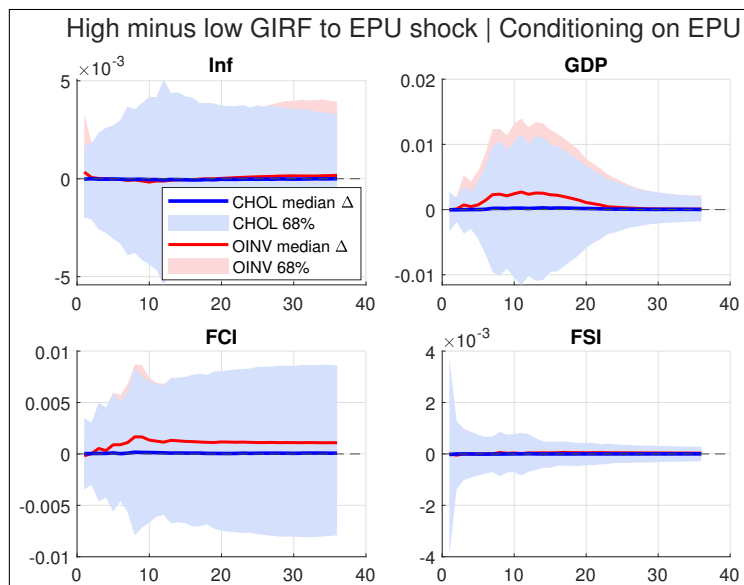


Figure 33: High-minus-low GIRF difference conditional on EPU regime: Inflation shock

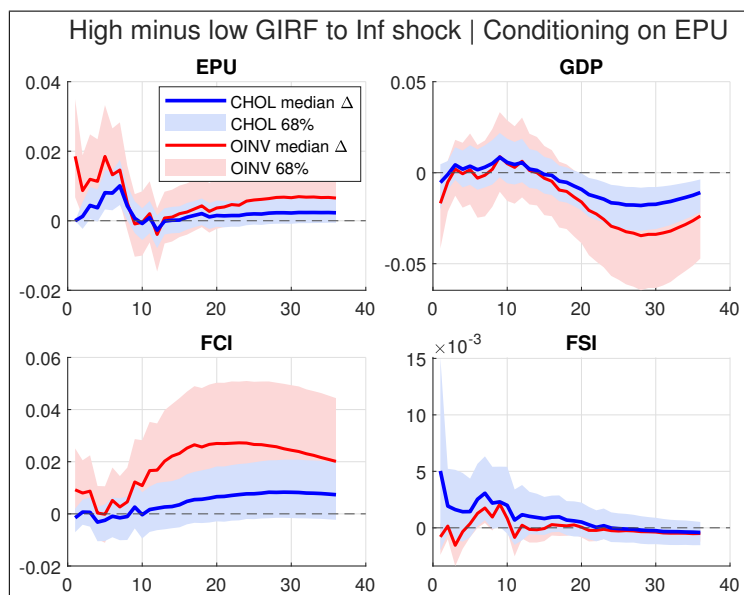


Figure 34: High-minus-low GIRF difference conditional on EPU regime: GDP shock

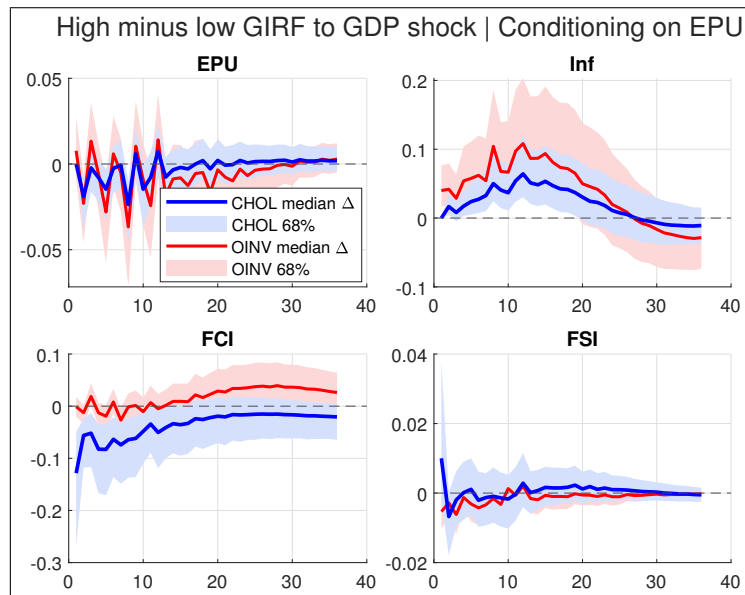


Figure 35: High-minus-low GIRF difference conditional on EPU regime: Financial Conditions Index (FCI) shock

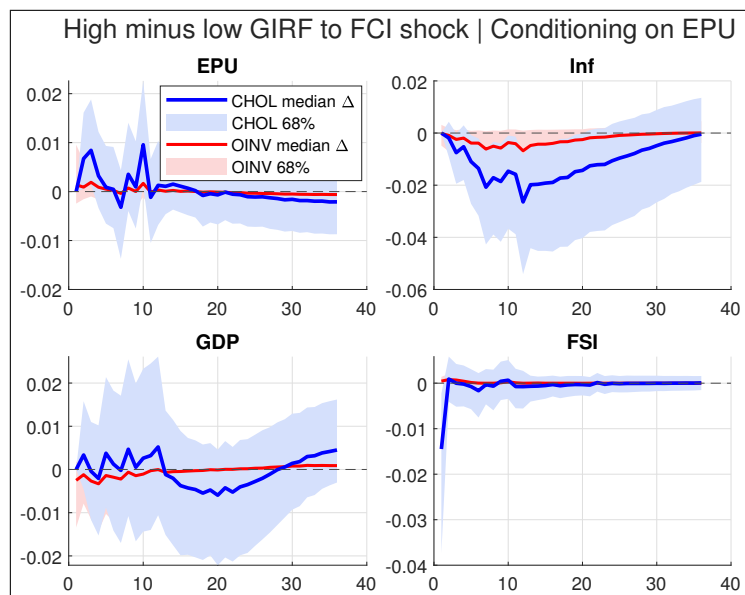
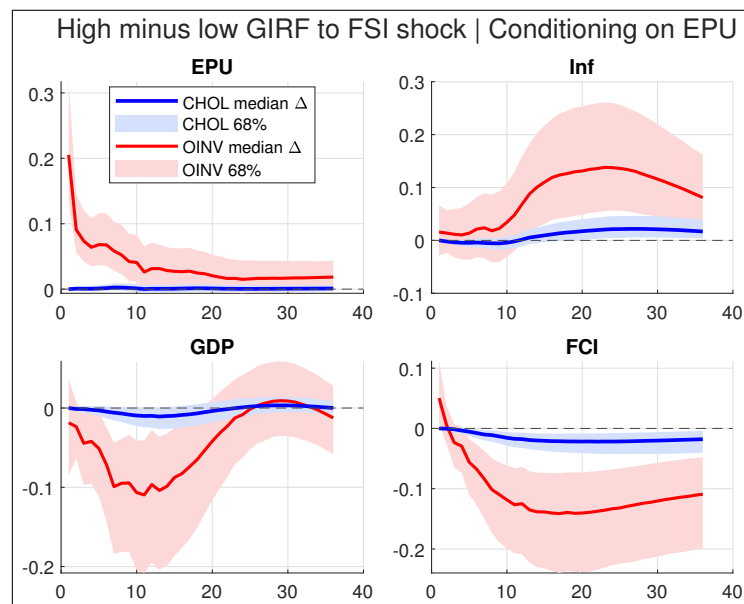


Figure 36: High-minus-low GIRF difference conditional on EPU regime: Financial Stress Index (FSI) shock



A10 Forecast Error Variance Decomposition

In a standard constant-volatility VAR, the forecast error variance decomposition (FEVD) assigns a fixed share of each variable's h -step forecast error variance to each structural shock. Because volatility is constant, these shares are time-invariant and depend only on the structural MA coefficients. In the SV-BVAR estimated in this paper, however, the variance of structural shocks evolves over time, making the FEVD inherently state-dependent.

Let Φ_s denote the reduced-form MA coefficient matrix at lag s , obtained recursively from the companion form of the VAR. The h -step forecast error for variable i can be decomposed as

$$e_{i,t+h} = \sum_{s=0}^{h-1} \sum_{j=1}^n \left(\Phi_s B_0^{-1} \right)_{ij} \varepsilon_{j,t+h-s}.$$

Since the structural shocks are orthogonal with time-varying variances $e^{h_{j,t}}$, the contribution of shock j to the h -step forecast error variance of variable i , conditioned on date t , is

$$\text{FEVD}_{i,j,t}(h) = \frac{e^{h_{j,t}} \sum_{s=0}^{h-1} \left(\Phi_s B_0^{-1} \right)_{ij}^2}{\sum_{k=1}^n e^{h_{k,t}} \sum_{s=0}^{h-1} \left(\Phi_s B_0^{-1} \right)_{ik}^2}. \quad (51)$$

The second equality exploits the random-walk property of the log-volatility process: under a random walk, $E[e^{h_{j,t+s}} | h_{j,t}] \approx e^{h_{j,t}}$ for all $s > 0$, so the volatility weight factors out of the horizon summation.

Equation (51) shows that the SV-conditioned FEVD is a reweighted version of the standard time-invariant FEVD, where each shock's cumulative squared impulse-response contribution is scaled by its current variance level $e^{h_{j,t}}$. This has two implications. First, the FEVD is date-dependent: the same structural propagation channels produce different decompositions depending on the relative volatility state. Second, regime comparisons based on FEVD shares are driven by *relative* volatility levels across shocks, not by differences in structural transmission.

Date-conditioned FEVD. Tables 16 and 17 report the posterior-mean FEVD at the end-of-sample conditioning date (December 2025) for CHOL and OINV, at horizon $h = 12$. FEVDs are computed at each MCMC draw using (51) and averaged across draws. The posterior mean preserves the sum-to-one constraint by linearity.

Several patterns emerge. Under both identification schemes, GDP, inflation, and FCI are largely self-explained (own-shock shares above 77%). FEVDs stabilize by $h = 12$ and remain unchanged at $h = 36$, consistent with the random-walk volatility

specification. Under OINV, the FSI shock explains approximately 9% of EPU’s forecast error variance, a channel that is mechanically excluded under CHOL due to the recursive ordering. This is consistent with the contemporaneous FSI→EPU feedback documented in the GIRF analysis.

Table 16: Date-conditioned FEVD at $h = 12$ — CHOL (December 2025)

	EPU	Inf	GDP	FCI	FSI
EPU	0.791	0.028	0.161	0.018	0.001
Inf	0.015	0.900	0.073	0.012	0.001
GDP	0.017	0.023	0.960	0.000	0.000
FCI	0.035	0.010	0.178	0.777	0.000
FSI	0.159	0.088	0.195	0.152	0.407

Table 17: Date-conditioned FEVD at $h = 12$ — OINV (December 2025)

	EPU	Inf	GDP	FCI	FSI
EPU	0.670	0.071	0.142	0.027	0.090
Inf	0.024	0.876	0.076	0.016	0.007
GDP	0.007	0.022	0.955	0.012	0.004
FCI	0.020	0.026	0.081	0.858	0.015
FSI	0.113	0.116	0.302	0.091	0.378

Regime-conditioned FEVD. Figure 37 through 40 report stacked-area plots of regime-averaged FEVDs, computed by averaging (51) across all dates in each regime set (high/low FSI, high/low EPU) and across posterior draws.

A recurring pattern in the regime-conditioned FEVD is that the share attributed to a given shock appears *smaller* precisely when that shock’s volatility is elevated. For example, the EPU shock’s contribution to FSI’s forecast error variance is higher in low-EPU regimes (0.223 under CHOL, 0.185 under OINV) than in high-EPU regimes (0.159 and 0.113, respectively). This is counterintuitive at first glance, but follows directly from the composition effect inherent in (51): when conditioning on high-EPU dates, FSI volatility is also elevated (due to the positive association between EPU and FSI), and the expansion of FSI’s own-shock share mechanically compresses all other shocks’ shares – including EPU’s. The effect operates symmetrically: every non-dominant shock gains share in calm regimes, not because the structural channel strengthens, but because the dominant shock’s volatility retreats.

This composition effect means that regime differences in FEVD shares reflect *relative* volatility reweighting rather than changes in structural transmission. The cumulative squared MA coefficients $C_{ij}(h) = \sum_s (\Phi_s B_0^{-1})_{ij}^2$ - which encode the structural

propagation mechanism - are common across regimes; only the $e^{h_{j,t}}$ weights differ. Consequently, regime-conditioned FEVDs are informative about which shocks are relatively more volatile in a given state, but they do not capture whether the *absolute* transmission of a shock is amplified or attenuated.

For this reason, the GIRF analysis in Section 4 provides a more direct and informative characterization of regime-dependent transmission. GIRFs measure the absolute difference between shocked and baseline paths, capturing both the size of the shock (via state-dependent volatility normalization) and the nonlinear propagation through the VAR dynamics and initial conditions. The FEVD, by contrast, is a zero-sum relative-share object that cannot distinguish between a genuine amplification of a transmission channel and a mechanical redistribution of shares driven by the volatility state.

Figure 37: Regime-conditioned FEVD: CHOL, conditioning on FSI

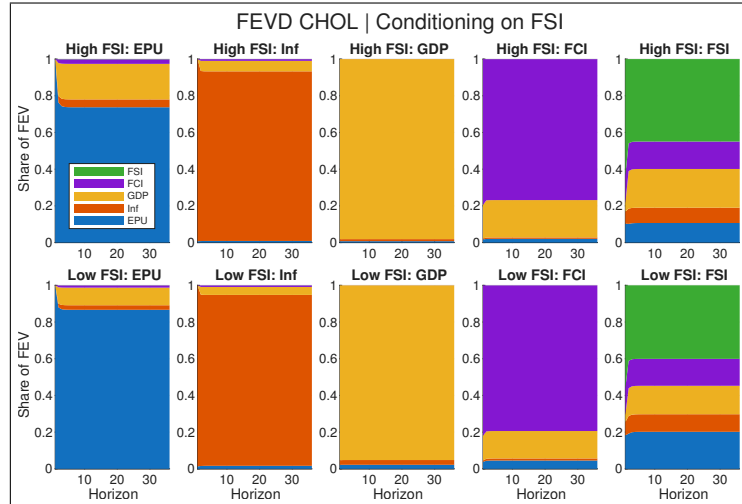


Figure 38: Regime-conditioned FEVD: OINV, conditioning on FSI

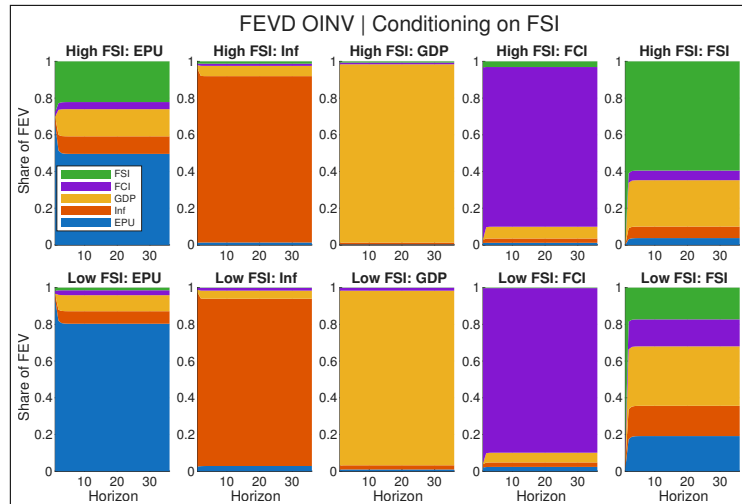


Figure 39: Regime-conditioned FEVD: CHOL, conditioning on EPU

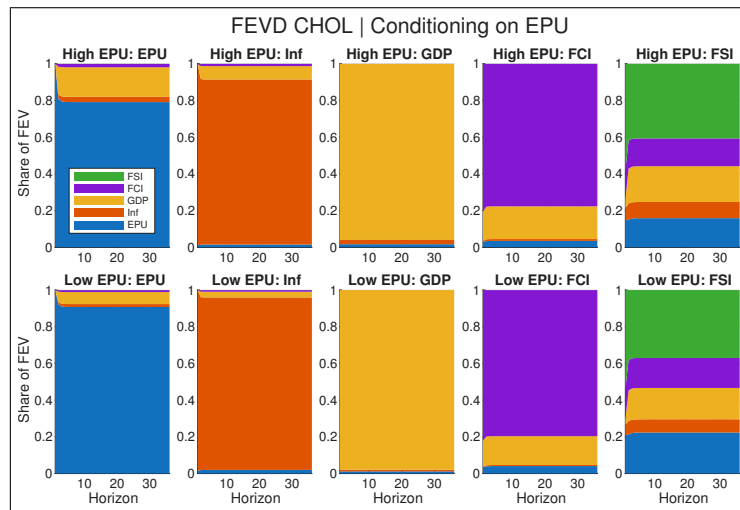
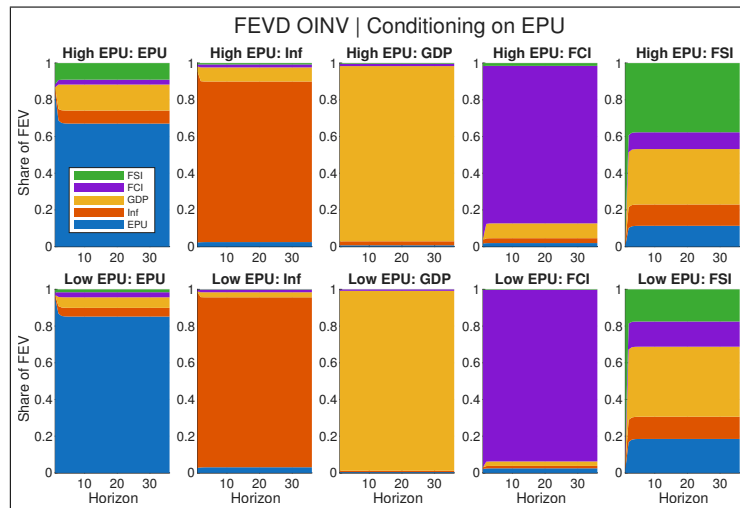


Figure 40: Regime-conditioned FEVD: OINV, conditioning on EPU



A11 Regime-Conditioned GIRFs under CHOL2

This appendix reports the full set of regime-conditioned generalised impulse response functions estimated under the second Cholesky ordering, CHOL2: (FCI , FSI , INF , GDP , EPU). In this specification financial variables appear first in the recursive chain and are therefore contemporaneously exogenous to all other variables. The figures are directly comparable to those reported for the baseline ordering (CHOL1) and the order-invariant identification (OINV) in Section 4.

Figure 41: CHOL2 – GIRF of each variable to a one-standard-deviation EPU shock, conditional on FSI regime.

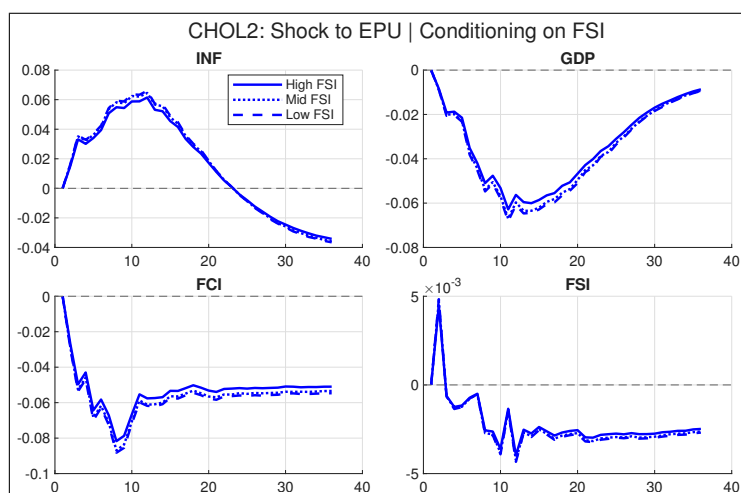


Figure 42: CHOL2 – GIRF of each variable to a one-standard-deviation EPU shock, conditional on EPU regime.

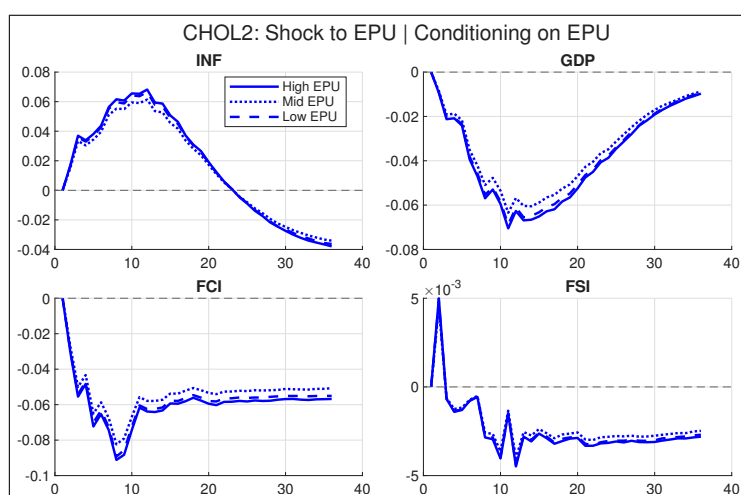


Figure 43: CHOL2 – GIRF of each variable to a one-standard-deviation INF shock, conditional on FSI regime.

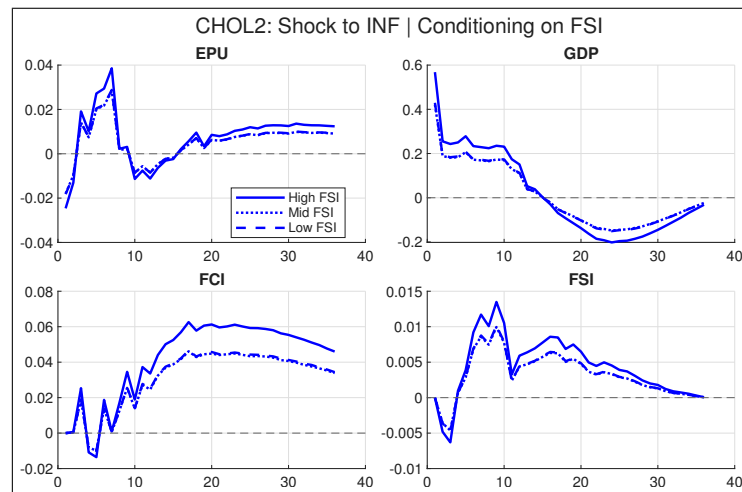


Figure 44: CHOL2 – GIRF of each variable to a one-standard-deviation INF shock, conditional on EPU regime.

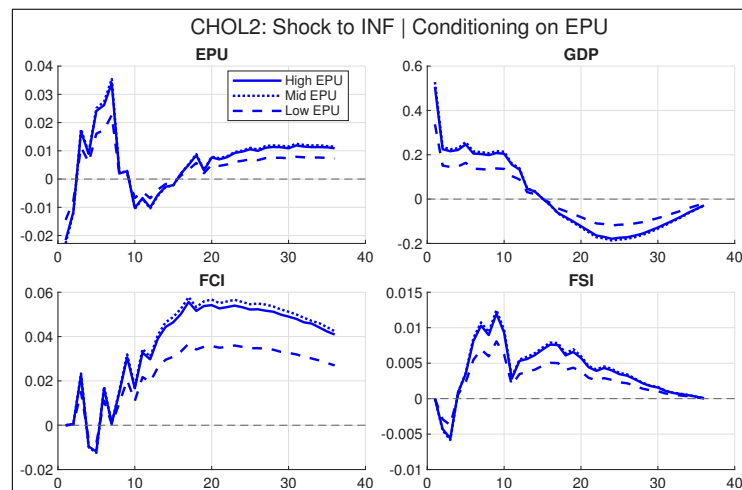


Figure 45: CHOL2 – GIRF of each variable to a one-standard-deviation GDP shock, conditional on FSI regime.

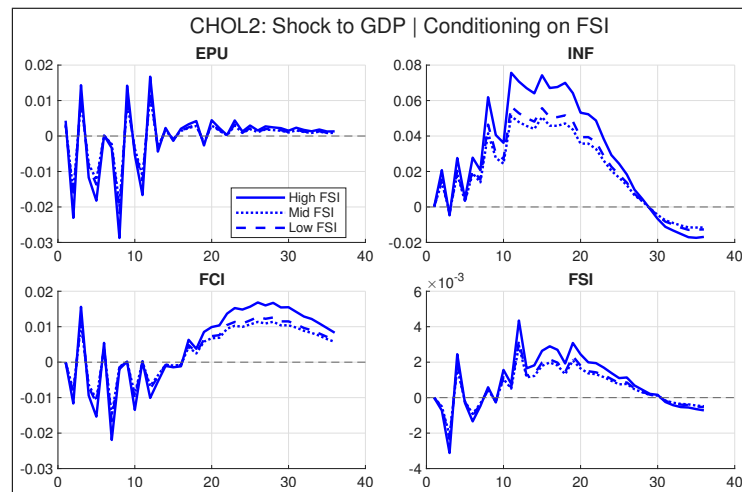


Figure 46: CHOL2 – GIRF of each variable to a one-standard-deviation GDP shock, conditional on EPU regime.

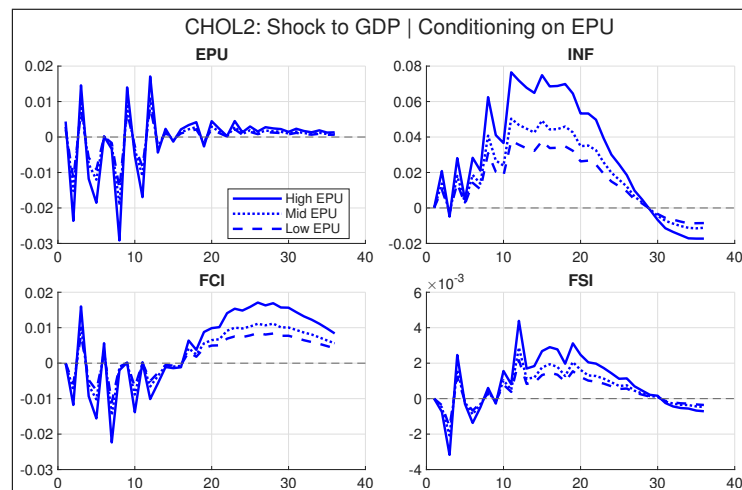


Figure 47: CHOL2 – GIRF of each variable to a one-standard-deviation FCI shock, conditional on FSI regime.

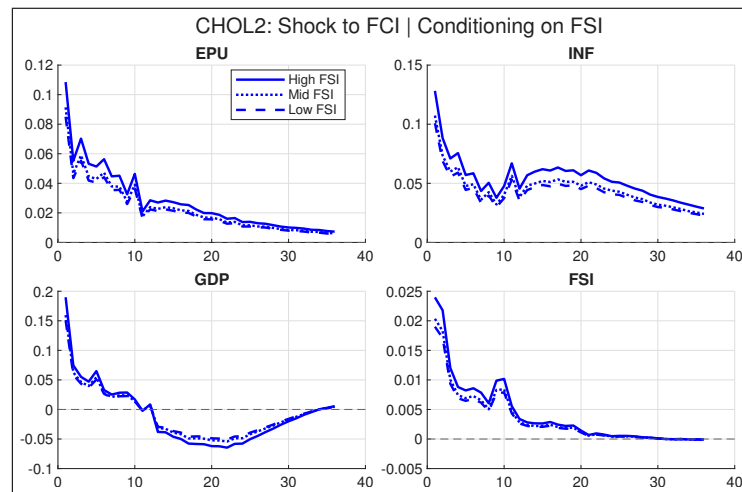


Figure 48: CHOL2 – GIRF of each variable to a one-standard-deviation FCI shock, conditional on EPU regime.

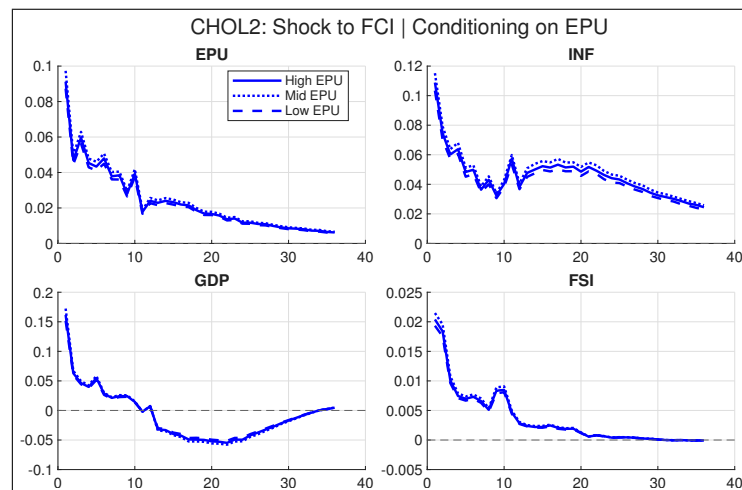


Figure 49: CHOL2 – GIRF of each variable to a one-standard-deviation FSI shock, conditional on FSI regime.

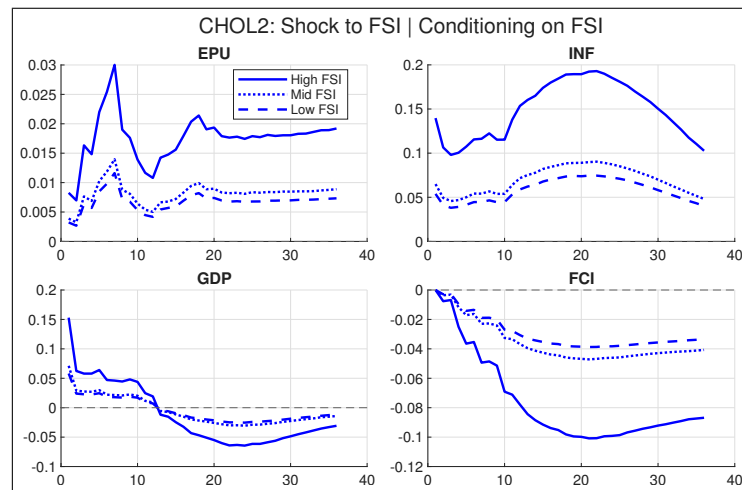
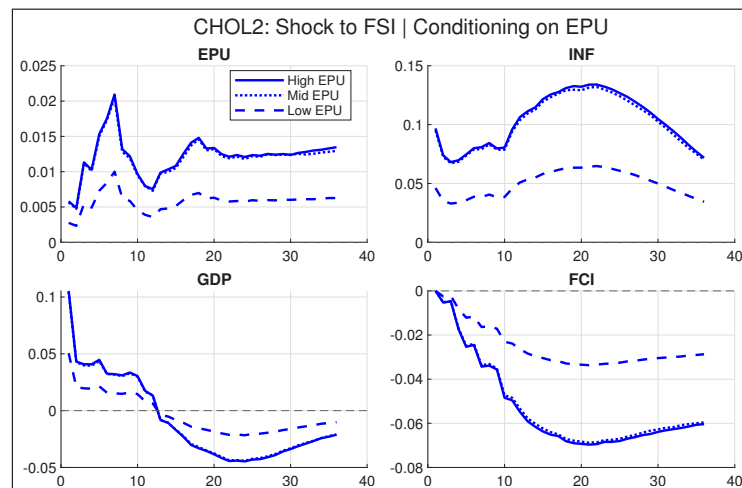


Figure 50: CHOL2 – GIRF of each variable to a one-standard-deviation FSI shock, conditional on EPU regime.



A12 Regime-Conditioned GIRFs under CHOL3

This appendix reports the corresponding regime-conditioned GIRFs under the third Cholesky ordering, CHOL3: (*FCI*, *FSI*, *EPU*, *INF*, *GDP*). This specification also places financial variables first but inserts uncertainty before the macro block, so that *EPU* can respond contemporaneously to financial shocks but not vice versa.

Figure 51: CHOL3 – GIRF of each variable to a one-standard-deviation *EPU* shock, conditional on *FSI* regime.

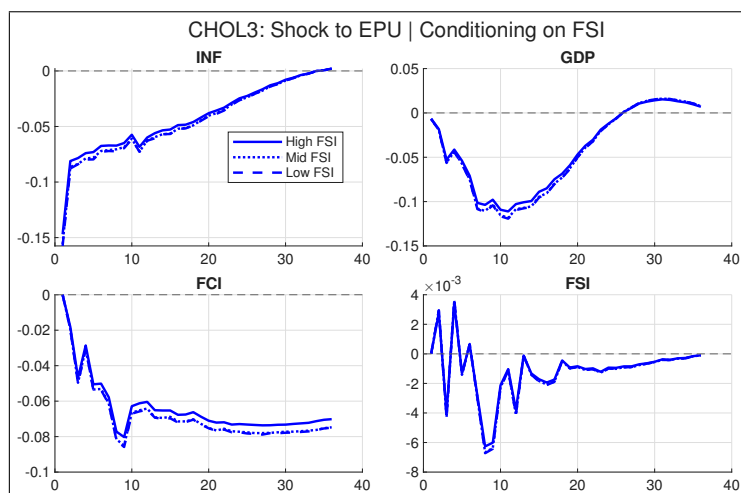


Figure 52: CHOL3 – GIRF of each variable to a one-standard-deviation *EPU* shock, conditional on *EPU* regime.

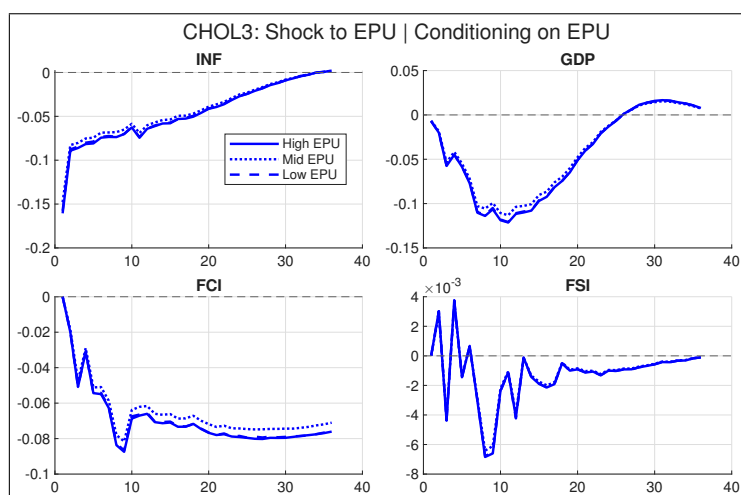


Figure 53: CHOL3 – GIRF of each variable to a one-standard-deviation INF shock, conditional on FSI regime.

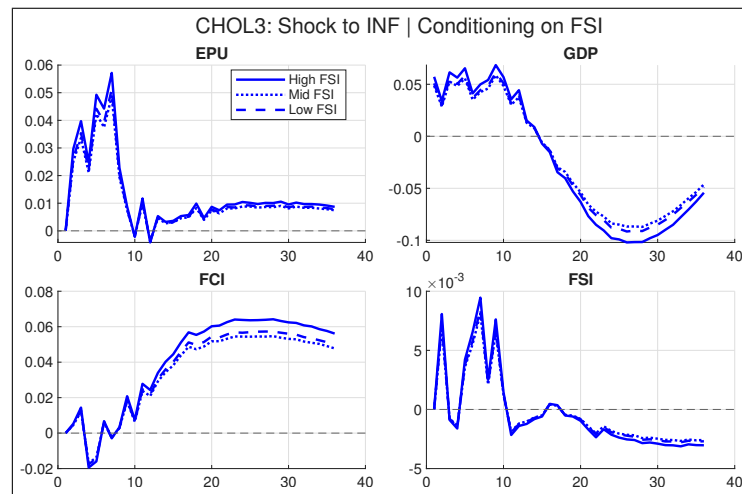


Figure 54: CHOL3 – GIRF of each variable to a one-standard-deviation INF shock, conditional on EPU regime.

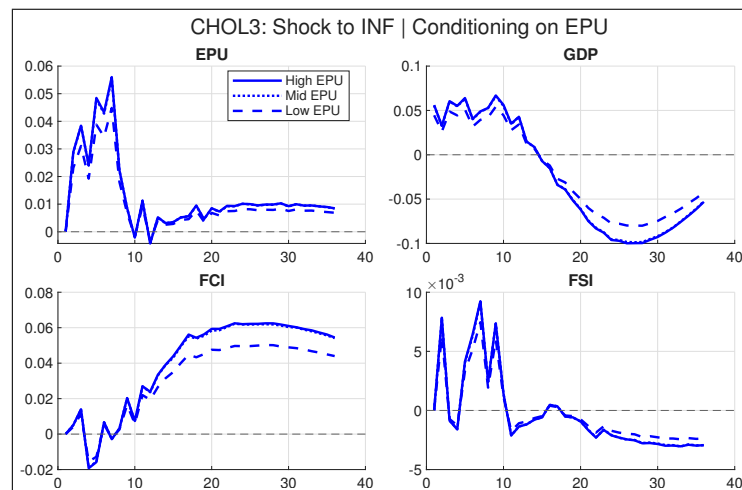


Figure 55: CHOL3 – GIRF of each variable to a one-standard-deviation GDP shock, conditional on FSI regime.

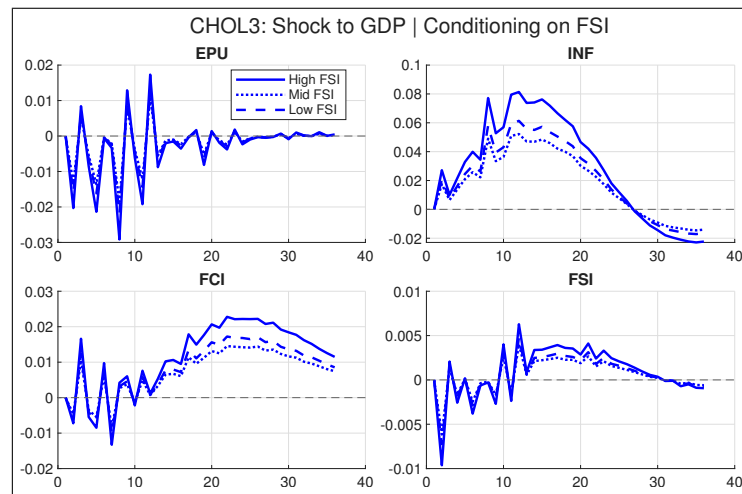


Figure 56: CHOL3 – GIRF of each variable to a one-standard-deviation GDP shock, conditional on EPU regime.

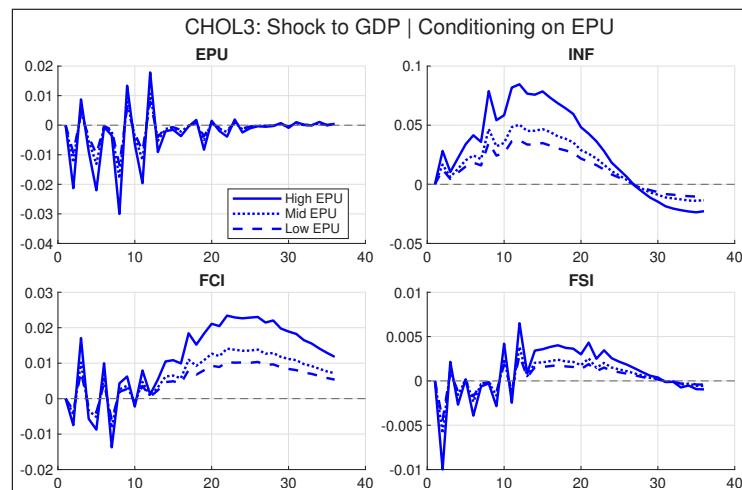


Figure 57: CHOL3 – GIRF of each variable to a one-standard-deviation FCI shock, conditional on FSI regime.

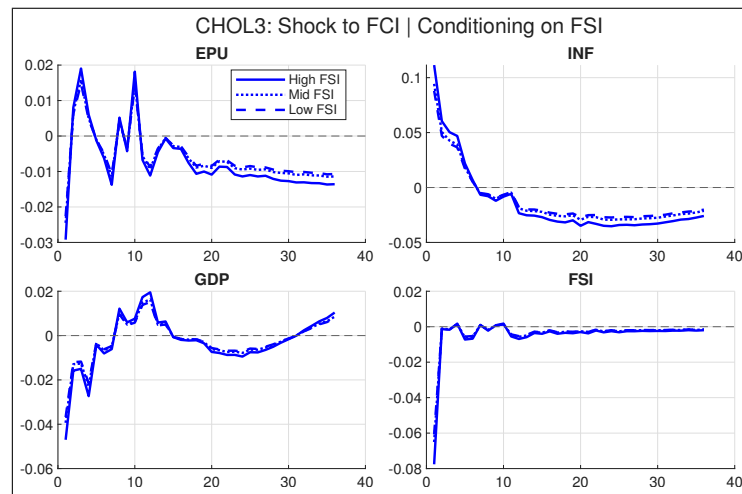


Figure 58: CHOL3 – GIRF of each variable to a one-standard-deviation FCI shock, conditional on EPU regime.

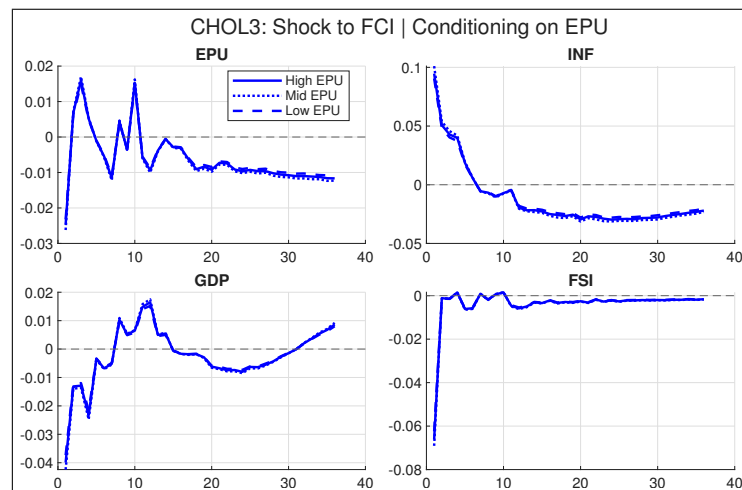


Figure 59: CHOL3 – GIRF of each variable to a one-standard-deviation FSI shock, conditional on FSI regime.

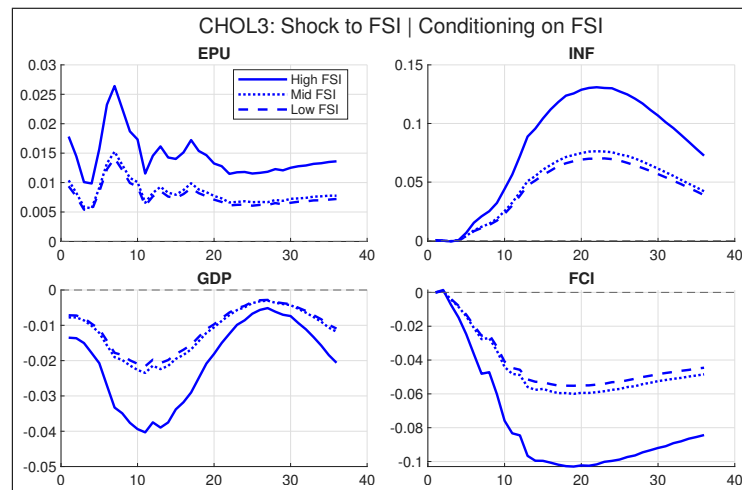
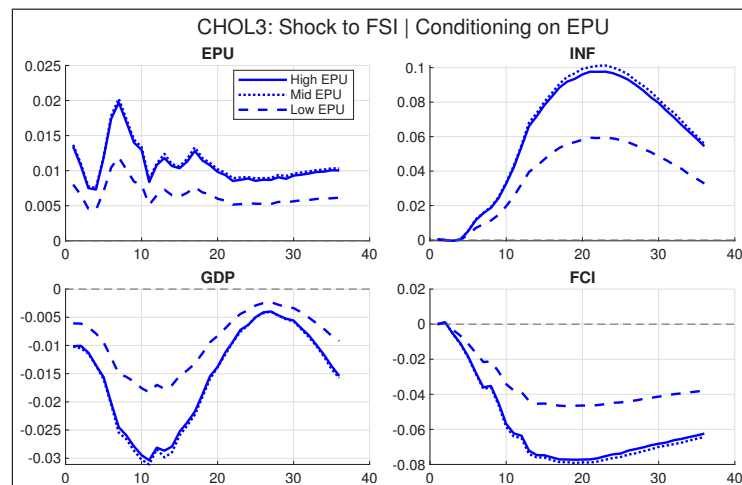


Figure 60: CHOL3 – GIRF of each variable to a one-standard-deviation FSI shock, conditional on EPU regime.



A13 Weak-Dominance Robustness

This appendix reports the full set of high-minus-low difference plots comparing the baseline OINV GIRFs (all thinned draws, blue solid line and shaded band) with the EPU-strong subset (draws where the EPU dominance margin ≥ 0.909 , green dashed line and shaded band). Each panel shows one response variable; columns correspond to the conditioning variable (FSI or EPU). See Section 4.2 for discussion.

Figure 61: Weak-dominance robustness: Δ GIRF to EPU shock, conditioning on FSI regime.

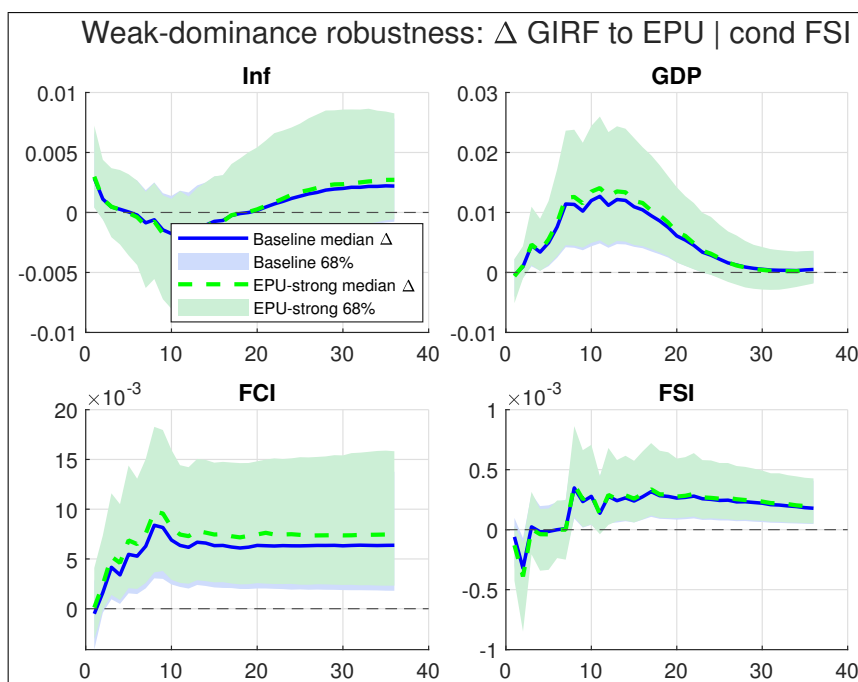


Figure 62: Weak-dominance robustness: Δ GIRF to EPU shock, conditioning on EPU regime.

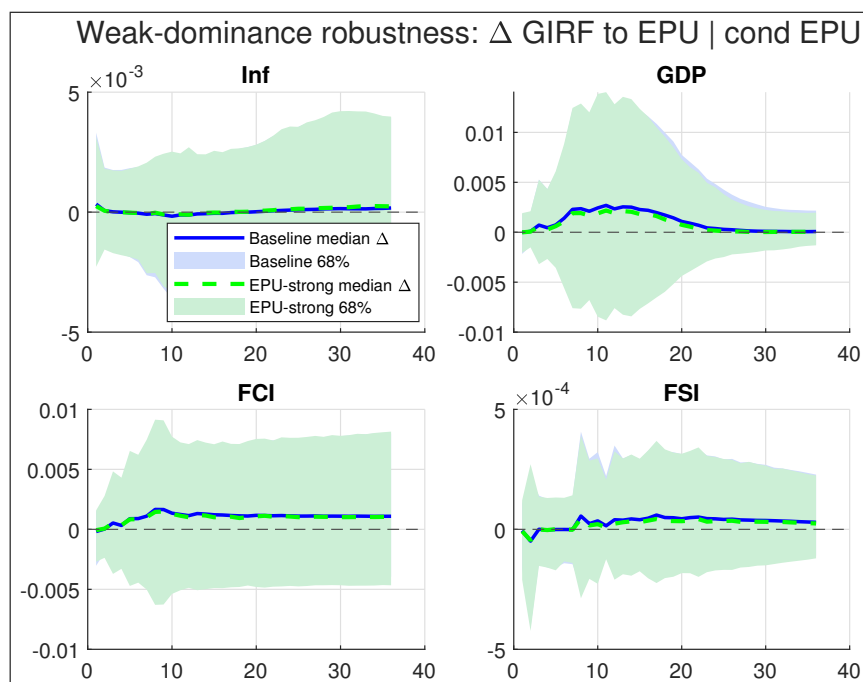


Figure 63: Weak-dominance robustness: Δ GIRF to Inflation shock, conditioning on FSI regime.

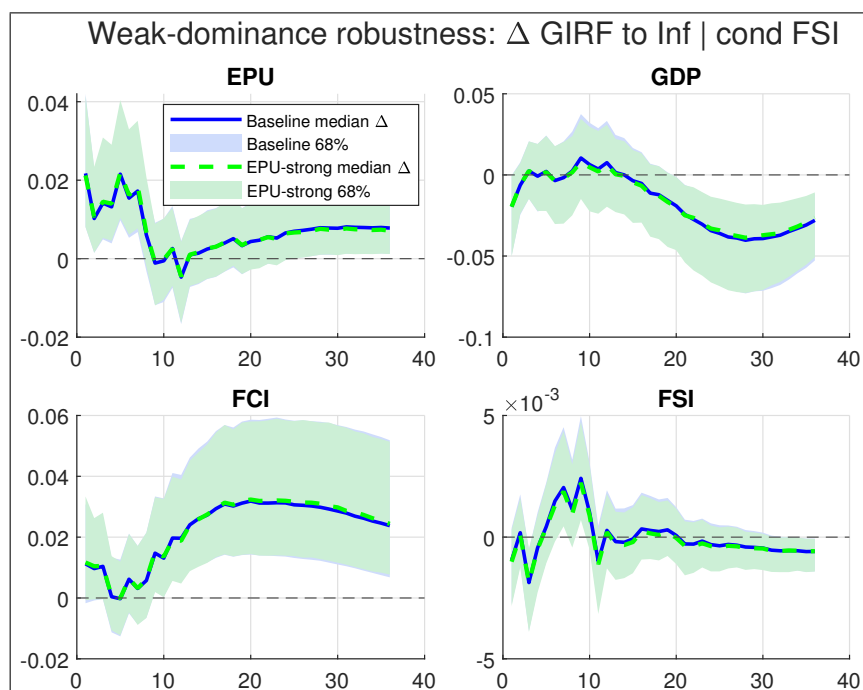


Figure 64: Weak-dominance robustness: Δ GIRF to Inflation shock, conditioning on EPU regime.

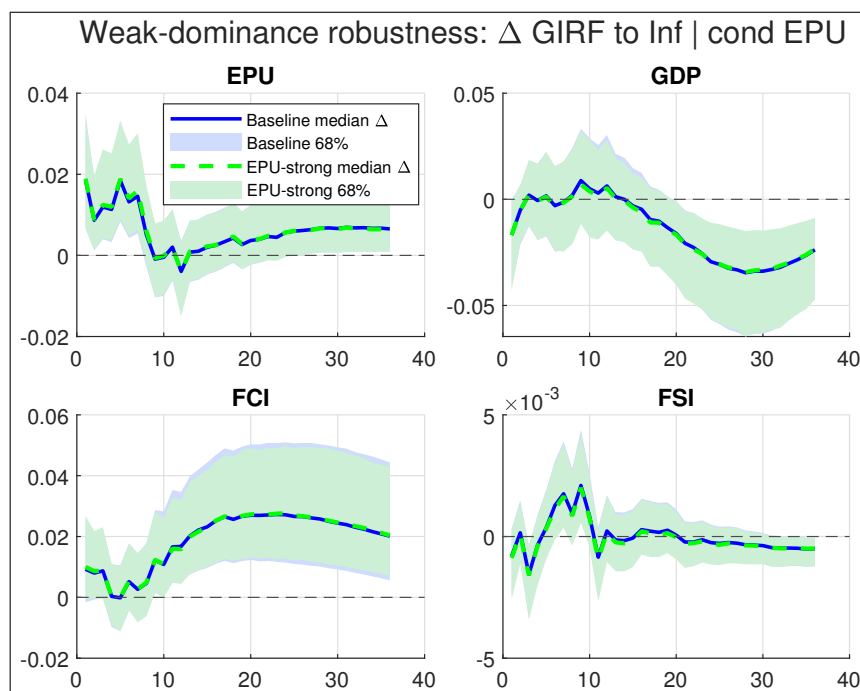


Figure 65: Weak-dominance robustness: Δ GIRF to GDP shock, conditioning on FSI regime.

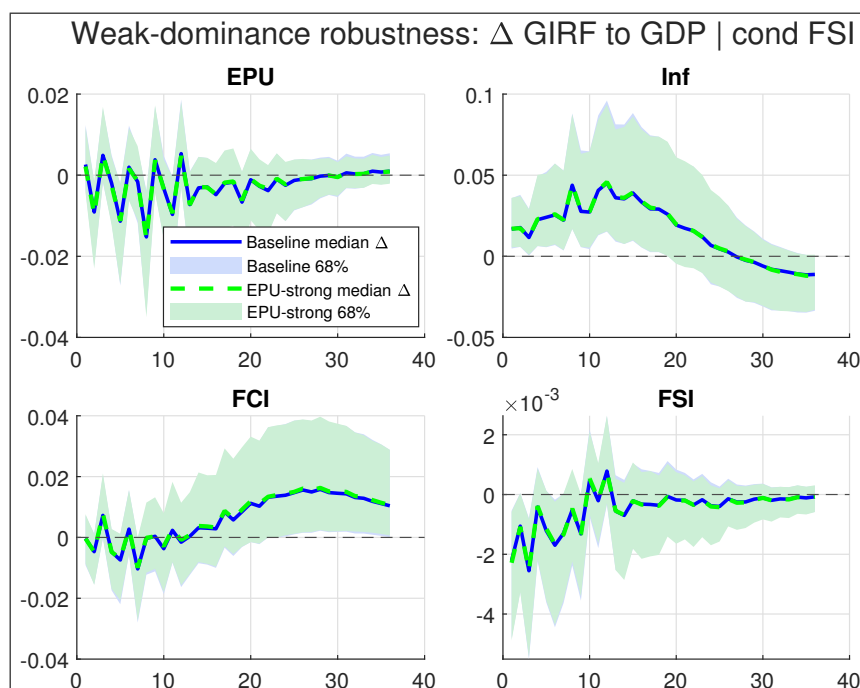


Figure 66: Weak-dominance robustness: Δ GIRF to GDP shock, conditioning on EPU regime.

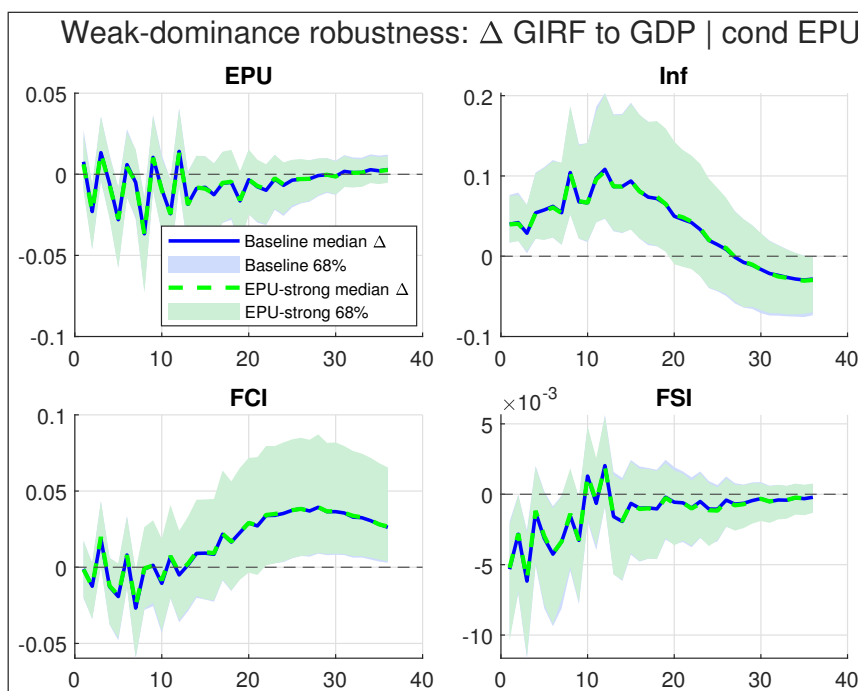


Figure 67: Weak-dominance robustness: Δ GIRF to FCI shock, conditioning on FSI regime.

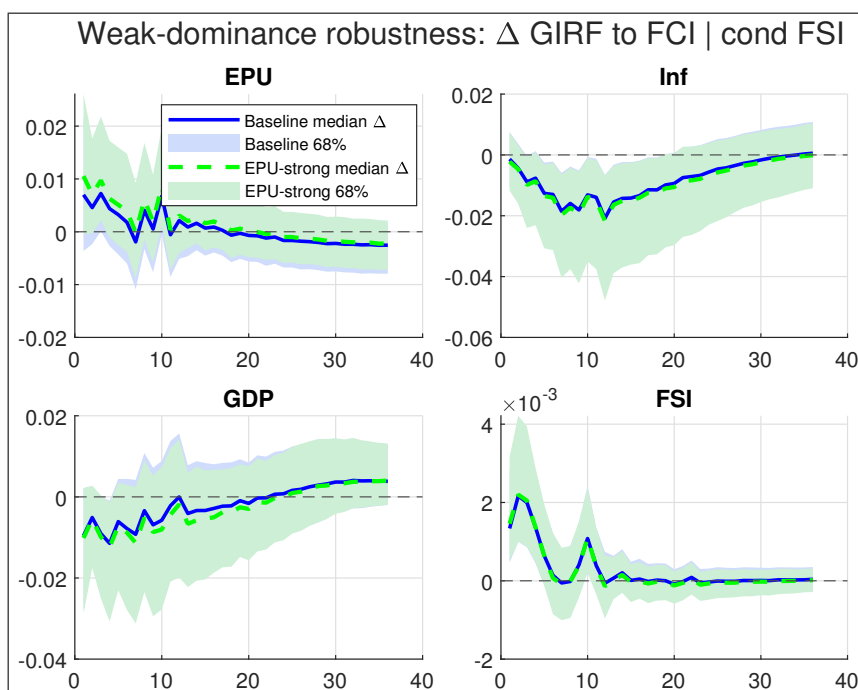


Figure 68: Weak-dominance robustness: Δ GIRF to FCI shock, conditioning on EPU regime.

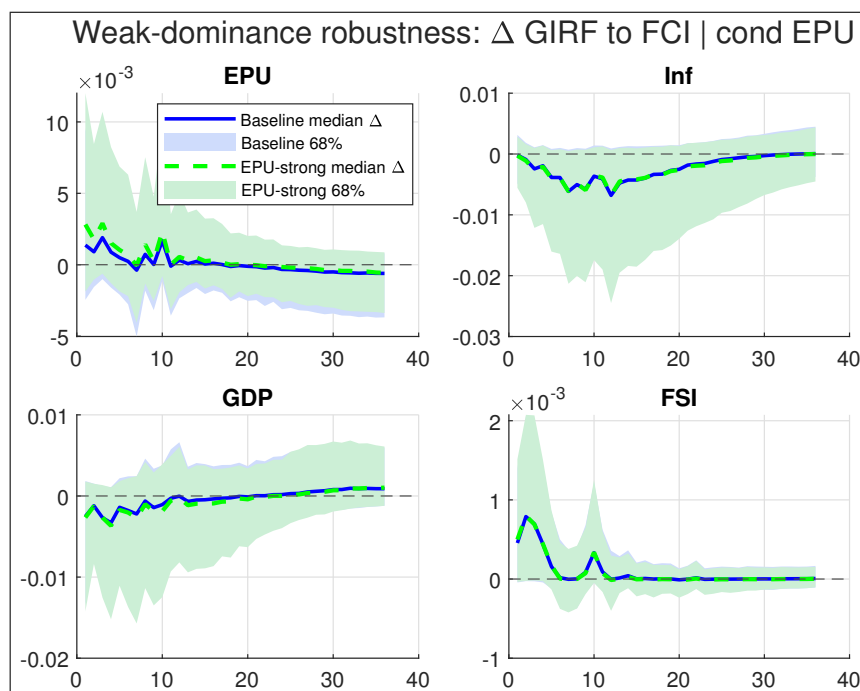


Figure 69: Weak-dominance robustness: Δ GIRF to FSI shock, conditioning on FSI regime.

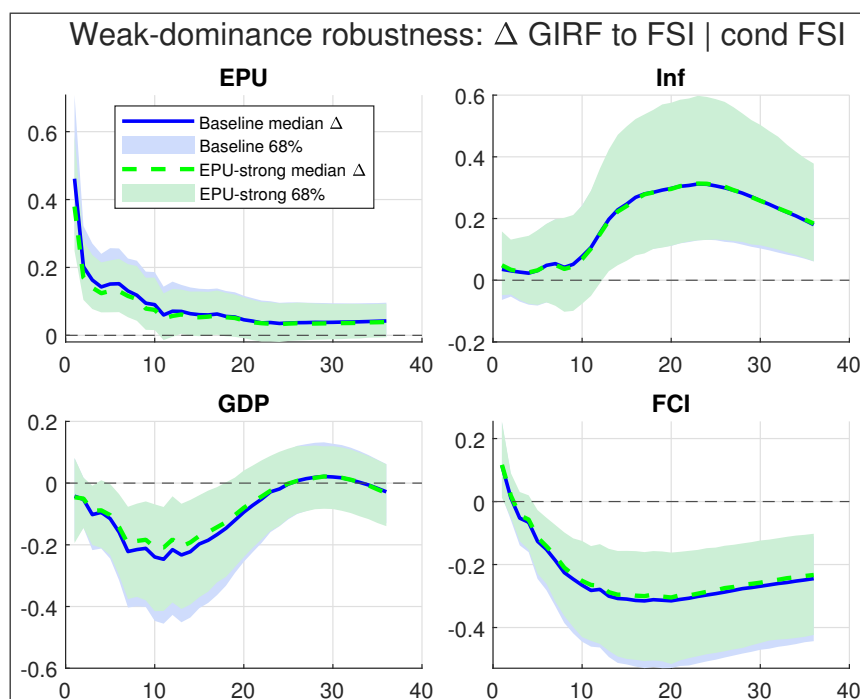
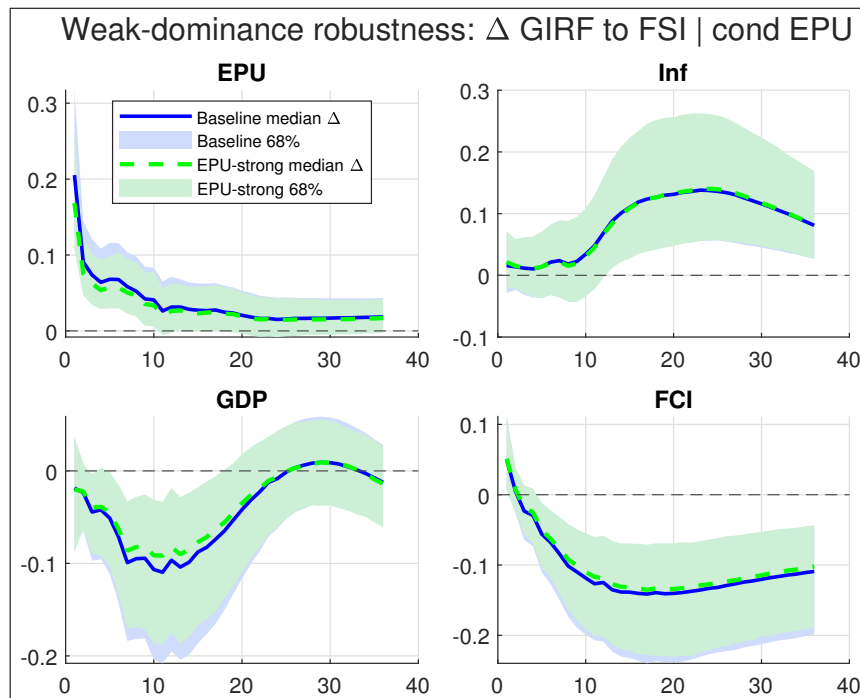


Figure 70: Weak-dominance robustness: Δ GIRF to FSI shock, conditioning on EPU regime.



A14 Joint EPU-GPRI 6-Variable Robustness Check

This appendix reports an additional robustness exercise in which both uncertainty measures are included jointly in the same model: $Y_t = (GPRI_t, EPU_t, INF_t, GDP_t, FCI_t, FSI_t)'$. The objective is to verify that the Section 4 result is not an artifact of replacing one uncertainty proxy with another, but remains visible when both proxies enter simultaneously.

Tables 18, 19, and 20 replicate the three baseline labelling diagnostics in this joint 6-variable setting.

Table 18: Average dominance margin by variable (joint 6-variable model)

	GPRI	EPU	Inflation	GDP	FCI	FSI
Average dominance margin	0.9932	0.7788	0.9980	0.9986	0.9981	1.0000

Table 19: Fraction of draws with weak dominance (joint 6-variable model)

	GPRI	EPU	Inflation	GDP	FCI	FSI
Weak dominance share	0.0226	0.7749	0.0092	0.0059	0.0086	0.0000

Table 20: Dominance violation counts (joint 6-variable model)

	GPRI	EPU	Inflation	GDP	FCI	FSI
GPRI	0	2	0	0	0	493
EPU	10	0	0	0	0	17288
Inflation	0	0	0	0	0	360
GDP	0	0	0	0	0	233
FCI	0	0	0	0	0	333
FSI	0	0	0	0	0	0

The diagnostics are consistent with the baseline interpretation. FSI remains sharply identified, GPRI is strongly identified, and EPU remains the only variable with substantial weak-dominance mass, reflecting overlap between uncertainty and financial-stress innovations. This is quantitatively similar to the baseline concern in Section 3.5, but does not overturn the structural conclusions.

Figures 71 and 72 report the key structural comparison from the same 6-variable model: responses to an FSI shock for the two uncertainty channels (FSI→EPU under EPU regimes, and FSI→GPRI under GPRI regimes), plus the high-minus-low regime gap.

Figure 71: Joint 6-variable model: regime-conditioned responses of EPU and GPRI to an FSI shock. Shaded areas show 68% credible intervals for high and low regimes.

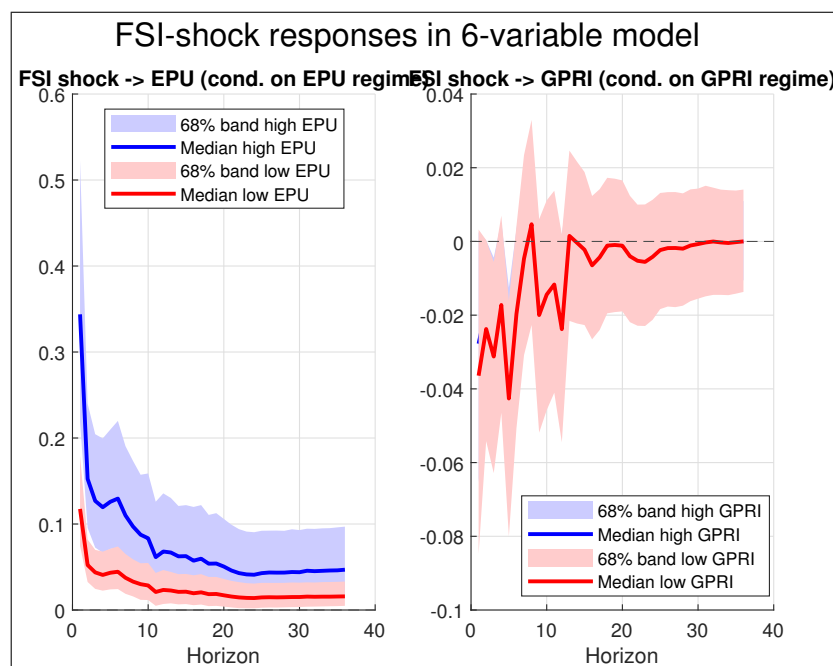
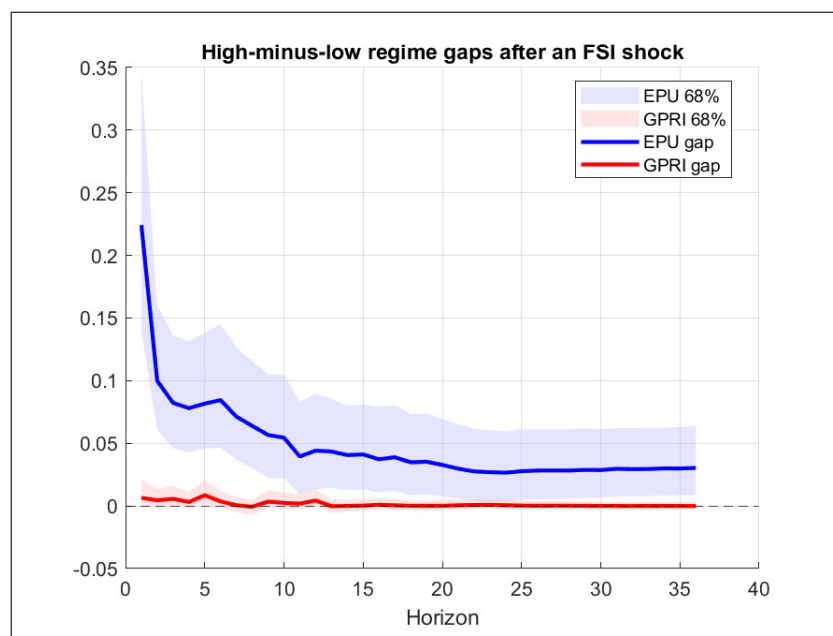


Figure 72: Joint 6-variable model: high-minus-low regime gap for the EPU and GPRI responses to an FSI shock. Shaded areas show 68% credible intervals.



Taken together, the joint-model evidence confirms that the Section 4 result is preserved: the amplification pattern remains concentrated in the policy-uncertainty channel, while the geopolitical-risk channel is comparatively muted. Hence, the main mechanism does not rely on omitting either uncertainty proxy from the system.

A15 Computation of Forecast Simulations

Let Y_t denote the vector of endogenous variables and suppose the model is estimated using a VAR(p) specification with stochastic volatility. For each posterior draw the forecasting procedure proceeds as follows.

1. The VAR coefficients, the structural impact matrix B_0 , and the last estimated stochastic volatility state are extracted from the posterior draw.
2. A new volatility realization is simulated according to the random walk law of motion of the log-volatility process.
3. Given the simulated volatility, the time-varying covariance matrix of the reduced-form innovations is constructed as

$$\Sigma_t = B_0^{-1} \text{diag}(\exp(h_t)) B_0^{-'}.$$

4. A realization of the reduced-form innovation is drawn from a multivariate normal distribution with covariance matrix Σ_t .
5. The VAR equation is then used to generate the next observation of the system, which is appended to the simulated data history.

This procedure is repeated recursively over the forecast horizon so as to generate a complete simulated path of future observations. Repeating the simulation across posterior draws yields the predictive distribution of the variables at each forecast horizon.

A16 Forecast Charts

Figure 73: Average quantile score for the 5th percentile of GDP growth across forecast horizons.

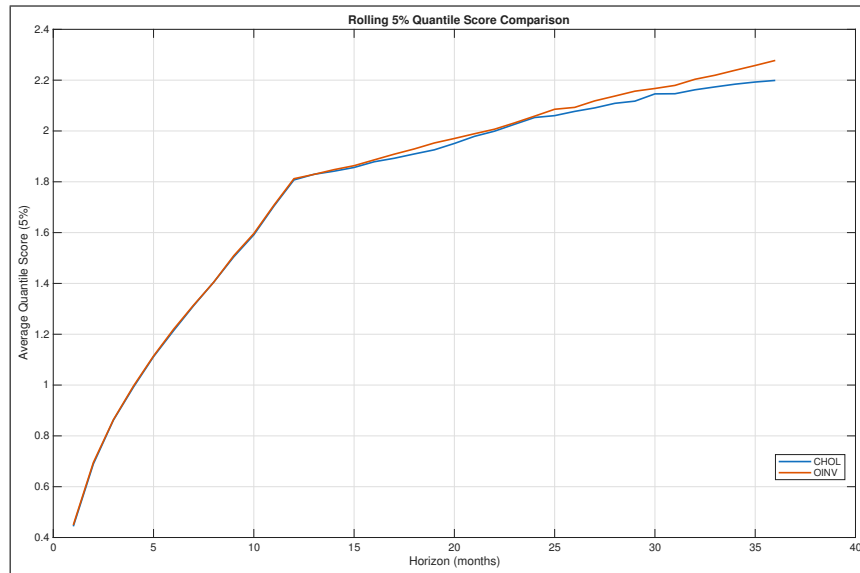


Figure 74: Average Forecast Dispersion across forecast horizons.

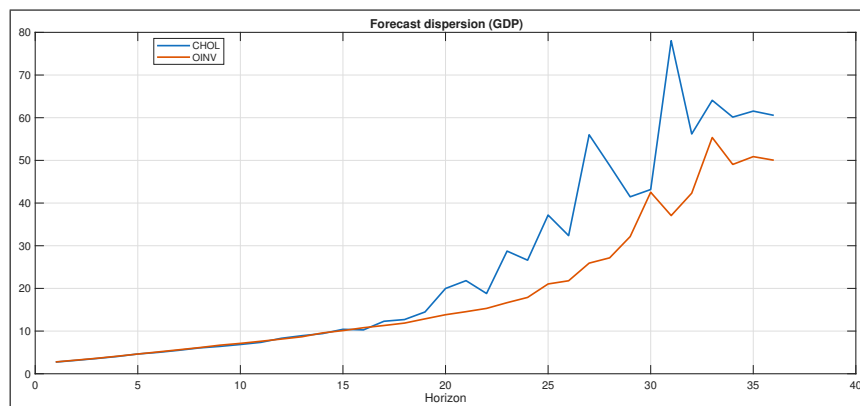


Figure 75: Forecast Skewness across forecast horizons.

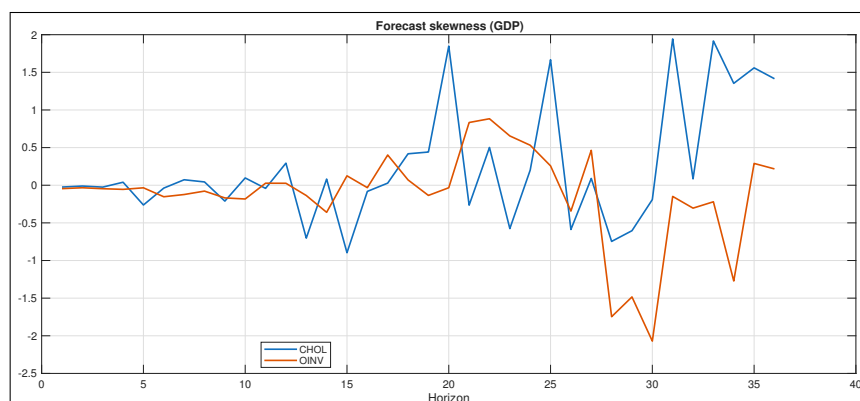


Figure 76: Mean Quantile Forecast across forecast horizons.

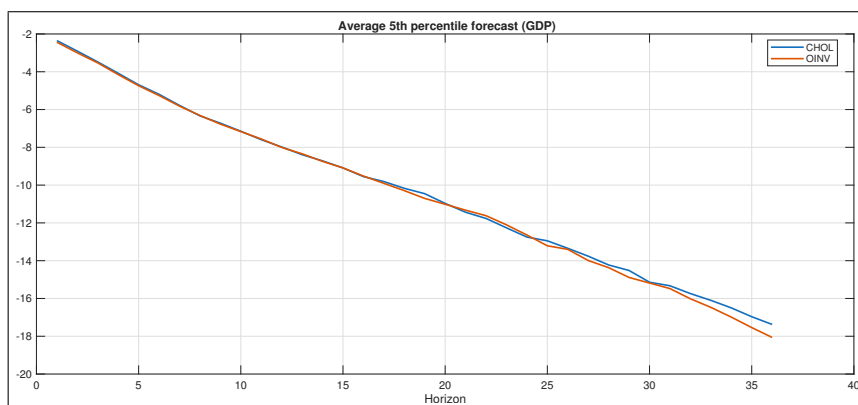


Figure 77: Rolling Forecast Dispersion across forecast horizons.

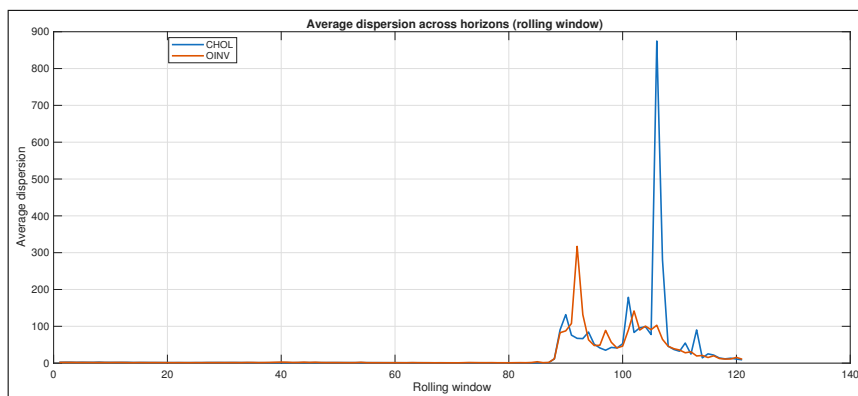


Figure 78: Median v. predicted 5th percentile gap across forecast horizons.

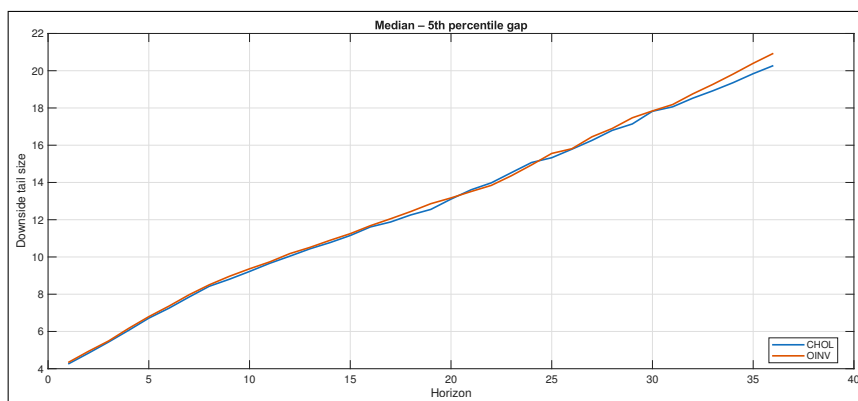


Figure 79: Median forecast across forecast horizons.

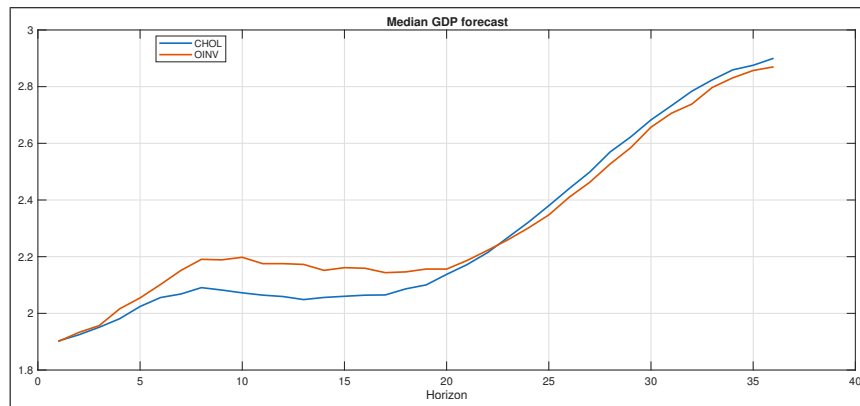


Figure 80: Average quantile scores at 1%, 5%, and 10% across forecast horizons.

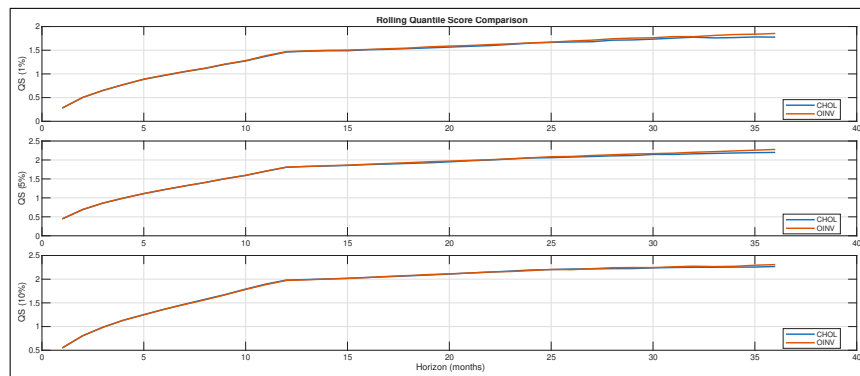
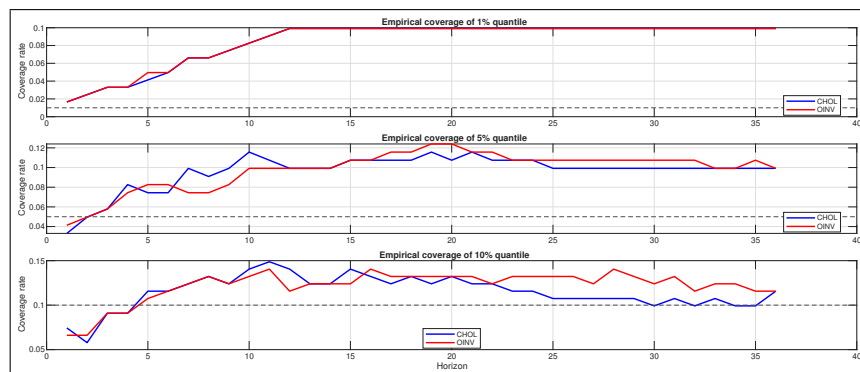


Figure 81: Empirical coverage of lower-tail quantiles at 1%, 5%, and 10%.



A17 Forecast Robustness Diagnostics (Reduced-Origin Four-Model Check)

This appendix documents diagnostics for the reduced-origin four-model forecasting robustness exercise in Section 6.2. The reduced-origin sample includes every sixth recursive origin, and CHOL1/OINV are restricted to the same subset to ensure comparability with CHOL2/CHOL3.

Table 21: CRPS diagnostics in the reduced-origin forecasting robustness exercise

Model	Mean	Median	Std. Dev.	P95	P99	Max
CHOL1	3.3364	0.9382	5.2874	13.7445	28.6166	30.1021
OINV	3.2457	0.8147	5.3210	13.7566	28.9627	30.4751
CHOL2	3.2069	0.8983	5.2595	13.6598	28.7529	30.2640
CHOL3	20.0192	0.8404	246.6137	18.7530	93.6238	5665.0265

Table 21 shows that CHOL3's large mean CRPS is not representative of typical behavior. Its median CRPS remains similar to the other models, while the mean is inflated by extreme outliers.

Table 22: Top outlier origin by average CHOL3 CRPS

Origin Number	CHOL3 mean CRPS	CHOL1 mean CRPS	OINV mean CRPS	CHOL2 mean CRPS
103	358.3746	7.6160	7.5381	7.4773

Table 23: Top horizon outliers for CHOL3 by average CRPS

Horizon	CHOL3 mean CRPS	CHOL1 mean CRPS
36	273.3775	4.0629
35	153.5328	5.0145
34	86.8073	4.9256
33	48.1797	2.9942
32	28.2579	3.3060

These diagnostics show that the CHOL3 mean-CRPS anomaly is concentrated in a narrow origin-horizon cluster (origin 103, long horizons), rather than reflecting broad underperformance across the reduced-origin sample.